

# Kobon Triangle Constructions: Uniform Lower Bounds, Extension Methods, and Formal Verification

Alejandro Zarzuelo Urdiales

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## Abstract

The Kobon triangle problem asks for the largest number of bounded triangular cells determined by a prescribed number of real affine lines. We give a comprehensive account of a construction and verification program begun in March 2026. An all-order simple construction supplies  $G(n) = \lfloor n(n-3)/3 \rfloor + 1 + (n \bmod 2)$  for  $n \geq 4$ , a constant-term refinement of the familiar Füredi–Palásti affine formula. We now also complete an actual geometric iteration of Bartholdi–Blanc–Loisel doubling: for every  $q = 10 \cdot 2^t$ , simple arrangements realize  $(q^2 - 4)/3$  triangles at order  $q + 1$  and  $(q^2 - 4)/3 + q/2$  at order  $q + 2$ . The retained grid, saturation, orientation and visibility invariants justify every finite depth. Combining these families with  $G$  and 138 exact finite certificates gives a proved bound for every natural order and strict improvements over the previous verified repository envelope at infinitely many orders. Formalized classical simple upper bounds place each family count within one triangle of the corresponding simple optimum. Further results include exterior extension and its repetition obstruction, actual edge-incidence and clean-line charging theorems, local multiple-point fan restrictions, and an exact algebraic obstruction to one tangent-grid sign system. We preserve all earlier finite results and distinguish the broader external obstruction census and the uniform 49-line seed from their unfinished Lean integrations. The release checks 375 active Lean sources; native finite checks are explicitly identified. These are construction and formalization advances, not a solution of unrestricted Kobon or a claim of new numerical records for the inherited dyadic families.

**Keywords.** Line arrangements; triangular cells; Kobon problem; constructive lower bounds; projective transformations; exact computation; Lean formalization.

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# 1 Introduction and scope of the results

An arrangement of  $n$  straight lines divides the real plane into bounded and unbounded cells. The Kobon problem asks how many of the bounded cells can be triangles. The difficulty is not creating many triples of intersecting lines: there are  $\binom{n}{3}$  candidate triples. The difficulty is ensuring that their triangles are not cut by any of the remaining lines and that their interiors do not overlap. The distinction between an attractive picture and a valid arrangement is therefore central to both construction and verification.

Let  $K(n)$  denote the classical maximum, allowing multiple intersections and parallel lines, and let  $K_s(n)$  denote the maximum over simple arrangements: no parallel pair and no three concurrent lines. All lower-bound witnesses constructed by the universal theorem below are simple, so they establish lower bounds for both quantities. Some of the strongest retained finite certificates are nonsimple. We never use an upper bound restricted to  $K_s$  as an upper bound for  $K$  without an additional argument.

This paper consolidates the author’s March 2026 extension proposal and the subsequent program of exact construction, comparison with the literature, and formal proof. The purpose is both mathematical and documentary: to state the strongest completed results, to make their proofs and provenance inspectable, and to preserve the failed approaches that determine the next research questions. The original general extension claim is reported as an unresolved claim rather than silently promoted to a theorem.

## 1.1 The main completed all-order theorem

Define  $B(n) = \lceil n(n-3)/3 \rceil$  for  $n \geq 3$ , and set  $B(n) = 0$  for  $n < 3$ . The familiar Füredi–Palásti construction supplies this affine baseline; the underlying construction and its historical priority belong to the earlier literature [6, 10]. The first completed all-order theorem is the following refinement; the recursive envelope below now strengthens its retained finite enhancement.

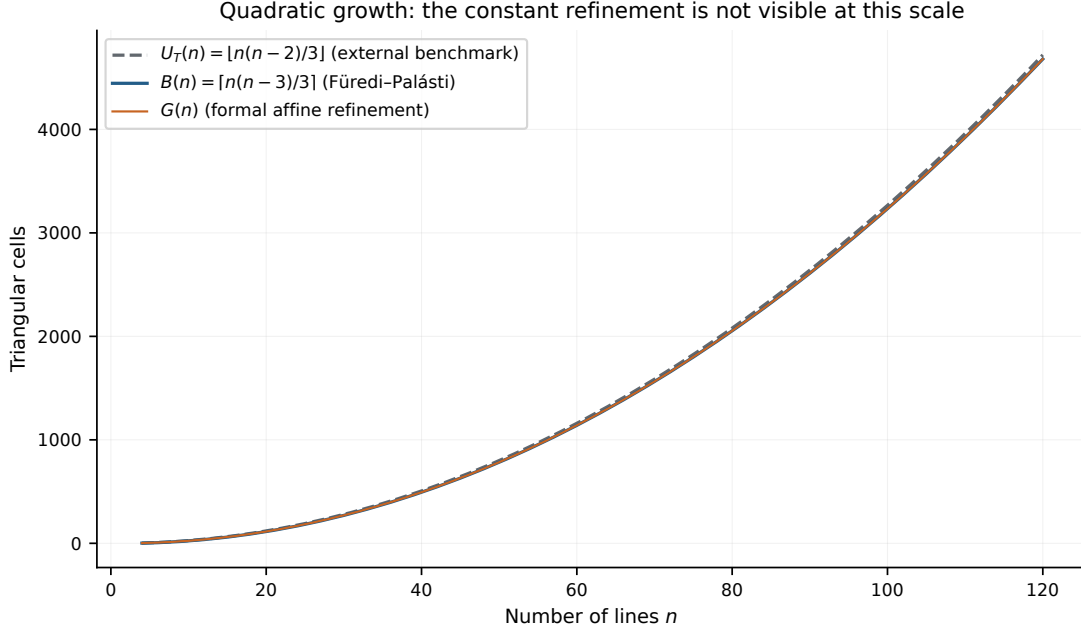
**Theorem 1.1** (Universal affine refinement). *For every  $n \in \mathbb{N}$ , there is a simple arrangement of  $n$  real straight lines with at least  $G(n)$  bounded triangular cells, where*

$$G(n) = \begin{cases} 0, & 0 \leq n < 3, \\ 1, & n = 3, \\ \lfloor n(n-3)/3 \rfloor + 1 + (n \bmod 2), & n \geq 4. \end{cases} \quad (1)$$

Consequently  $K(n) \geq K_s(n) \geq G(n)$ .

The proof, developed in Section 4, combines a shifted trigonometric arrangement with one- and two-cap changes of affine chart. The Lean declaration `Kobon.Universal.baseline_sound` proves the geometric existence statement for arbitrary  $n$ . It has no assumed extension theorem and no dependence on finite experimental verification. Its dependencies are the standard logical axioms used by Lean and Mathlib.

For  $n \geq 4$ , the gains over  $B(n)$  are 1, 1, 0, 2, 0, 1 in residue classes 0, 1, 2, 3, 4, 5 modulo six. Thus the refinement is strict at every odd order at least five and every positive multiple of six. Both formulas have leading behavior  $n^2/3 - n + O(1)$ : this is a constant-term refinement, not an improved quadratic or linear asymptotic coefficient. The original Füredi–Palásti projective count table already contains related constants. The present source audit does not establish first numerical priority for the affine formula.



**Figure 1:** Growth of the classical general baseline, the proved affine refinement, and the classical polynomial upper benchmark. The close overlap on a quadratic scale is mathematically meaningful: the new uniform gain is bounded. Residue and deficit plots below resolve the smaller differences. This figure is computed from the formulas, not copied from earlier publications.

## 1.2 Finite enhancements and other completed components

The universal construction is supplemented by exact coordinates. Let  $C(n)$  be the largest triangle count among the finite certificates in the release at order  $n$ , with  $C(n) = 0$  when no such certificate is saved. Then

$$L(n) = \max\{G(n), C(n)\}, \quad K(n) \geq L(n) \quad (2)$$

is proved for all natural orders. This elementary maximum operation retains published constructions and the author’s earlier values without altering their attribution. It is not a claim that  $L$  is the largest lower bound obtainable from all published constructions by every possible interpolation or extension.

The finite catalog contains 138 distinct coordinate certificates, including 104 simple witnesses. A separate generated table has 64 orders at which a saved finite certificate strictly improves the older baseline  $B$ ; some of these improvements are already supplied by  $G$ . The catalog includes weaker checkpoints and independent reproductions. Its size is not a count of original discoveries.

The October release adds a recursively closed geometric family. Put  $q_t = 10 \cdot 2^t$  and  $A_t = (q_t^2 - 4)/3$ . For every  $t \geq 0$ , simple arrangements realize at least  $A_t$  triangles at  $q_t + 1$  lines and  $A_t + q_t/2$  at  $q_t + 2$  lines. Every next seed retains the sorted grid, saturation, positive central apex and visible-boundary data. Taking the maximum of these candidates and  $L$  gives the current total bound  $H$  of Theorem 7.7. It strictly improves the previous repository envelope at every family pair from 321/322 onward, without claiming a new literature record for the inherited numerical family.

Further completed components are actual exterior addition, its finite boundary-resource obstruction, actual edge incidence and clean-line charging, classical simple upper bounds and local fan restrictions. The simple optimum at each recursive order is between the constructed

| Result                                                                      | Mathematical content                                                                                                                                 | Release status                                         |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Universal $G$<br>Envelope $L$                                               | Simple real-line witnesses at every order<br>Maximum with all saved finite witnesses                                                                 | General Lean proof<br>Lean; finite native checks       |
| 28:238, 30:275, 34:357                                                      | Earlier attributed finite inequalities                                                                                                               | Exact Lean certificates                                |
| Exterior extension<br>Full successor recurrence<br>49-seed numerical target | New line and gain from a visible-pair list<br>$K(n+1) \geq K(n) + \lfloor n/2 \rfloor$<br>Lower function obtained by summing the proposed recurrence | General Lean proof<br>Unproved<br>Independently proved |
| One-step BBL doubling                                                       | $(q+1, T) \mapsto (2q+1, T+q^2)$ for compatible seeds                                                                                                | General Lean proof                                     |
| Infinite hybrid iteration                                                   | Odd and full-gain even geometric families                                                                                                            | Complete Lean; native seed                             |
| Local fan structure                                                         | Incidence, antipodal and run-length restrictions                                                                                                     | General Lean proofs                                    |
| Simple upper windows                                                        | Actual odd/even upper bounds and family existence                                                                                                    | Complete Lean                                          |
| Actual clean-line charging<br>Global multiplicity consequences              | Charge existence and finite fiber bounds<br>Core budgets requiring fan extraction                                                                    | General Lean proof<br>Conditional; bridge open         |
| Tangent-grid obstruction                                                    | Explicit ten-sign algebraic impossibility                                                                                                            | General Lean proof                                     |

**Table 1:** Claim status in the 3 October mathematical release. “General Lean proof” does not imply that every proposed application has been formalized. Published construction inputs retain their original authorship.

count and that count plus one. An explicit ten-sign obstruction on the real 21-line tangent grid is also proved in Lean. Broader exact classification checks and the uniform 49-line compatibility certificate have separate evidence boundaries. Table 1 summarizes the current release.

### 1.3 History, attribution, and the structure of the paper

The March archive records a proposed extension from odd to even orders, associated finite bounds, and an incomplete Lean file. That file contains eight proof placeholders, including its central extension step. Later exact certificates establish the finite inequalities independently, while the current all-order proof uses a different construction. Accordingly, a corrected account must separate the survival of numerical conclusions from the validity of the original proof strategy.

The September work broadened the program to both parities, exterior visibility, boundary-defect estimates, compatible doubling seeds, exact rational realization, and multiplicity-sensitive upper arguments. Several plausible routes failed: direct repetition of exterior gains lost the necessary boundary resources; small perturbations of nonsimple records sometimes reduced the triangle count; and large local searches failed to improve their chosen seeds. These negative findings are recorded with their exact scope rather than converted into impossibility theorems. The October work closes the recursive and clean-line interfaces, adds rigorous grid obstructions, and leaves the unrestricted successor claim and global nonsimple fan extraction open.

Section 3 states the geometric definitions and the certificate-to-cell bridge. Section 4 proves the uniform refinement; Section 5 records finite cases and provenance. The following sections develop extension and family results, local upper-bound ingredients, and computa-

tional searches. The final sections describe the formal trust boundary and future obligations. The appendices contain the complete finite tables and a map from mathematical claims to proof sources. The code and evidence are available at <https://github.com/alejandrozuk/obon-proof> [31].

## 2 Literature, provenance, and the scope of comparison

### 2.1 Four distinctions that affect every comparison

The literature studies several related extremal functions. Their numerical tables cannot be combined without retaining their hypotheses. Throughout this paper,  $K(n)$  concerns straight lines in the real affine plane and permits multiple intersections and parallel pairs, whereas  $K_s(n)$  restricts to pairwise nonparallel lines with no three concurrent. Only bounded triangular cells are counted. A pseudoline arrangement is a topological relaxation; a lower bound obtained there becomes a straight-line lower bound only after realizability has been established. Finally, projective triangular faces need not remain bounded after an affine chart is chosen.

| Distinction           | What transfers immediately                                          | What requires another argument                                              |
|-----------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Simple / unrestricted | A simple example is an unrestricted example.                        | A simple upper bound need not bound a configuration with concurrent lines.  |
| Straight / pseudoline | Every straight arrangement is a pseudoline arrangement.             | A pseudoline construction requires a straight realization.                  |
| Affine / projective   | Projective transformations preserve incidence and projective faces. | Boundedness depends on the line chosen as infinity.                         |
| Finite / infinite     | A certified configuration proves its individual lower bound.        | Iteration requires the next configuration to satisfy every input invariant. |

**Table 2:** Logical distinctions used throughout the source audit.

There are also two different projective-to-affine operations. Choosing a new line at infinity outside an  $n$ -line arrangement keeps  $n$  lines, but may lose triangular faces to infinity. Making one of the arrangement’s own lines the line at infinity and removing it produces an  $(n - 1)$ -line affine arrangement. The latter operation underlies several optimal subsequences; the former is central to the uniform construction formalized here.

### 2.2 From the puzzle to arrangement theory

Fujimura’s puzzle reached English readers through *The Tokyo Puzzles*, edited by Gardner and translated by Adachi [9]; Gardner also discussed it in his collected mathematical recreations [11]. The broader theory of line and pseudoline arrangements is developed in Grünbaum’s monograph [13]. The papers of Strommer and Purdy document an established extremal-triangle literature preceding the construction used in this paper [24, 25, 28]. We cite those early papers as historical context rather than silently identifying all their projective or non-simple questions with the affine function  $K$ .

The familiar comparison polynomial is

$$U_T(n) = \left\lfloor \frac{n(n-2)}{3} \right\rfloor. \quad (3)$$



The attribution to Saburo Tamura is transmitted in the Clément–Bader draft [3]; an original publication by Tamura was not located in this audit. We therefore do not manufacture an independent bibliographic item for that attribution. Clément and Bader report the stronger classical envelope  $U_T(n) - 1$  for  $n \equiv 0, 2 \pmod{6}$ . Section 8 explains both the scope of this reported result and a local proof-reading issue. A source-reported upper bound and an upper bound proved in our Lean development are different kinds of evidence.

### 2.3 The Füredi–Palásti all-order construction

Füredi and Palásti gave highly triangular straight-line arrangements using trigonometric coordinates [10]. Their projective Table 1 has a residue-dependent constant term. The familiar affine consequence is commonly written

$$B(n) = \left\lceil \frac{n(n-3)}{3} \right\rceil \quad (n \geq 3). \quad (4)$$

Erickson states this formula explicitly in Section 7.2 while using arrangements in a computational-geometric lower-bound argument [6]. Felsner and Kriegel place the construction in the study of Euclidean triangular cells [7]. A later discussion by Damásdi, Martínez-Sandoval, Nagy and Nagy quotes the slightly weaker floor version while studying triangle areas [4]. Thus a general lower bound valid at every order existed well before the present work.

Our completed formula is

$$G(n) = \left\lfloor \frac{n(n-3)}{3} \right\rfloor + 1 + (n \bmod 2), \quad n \geq 4. \quad (5)$$

The constructions in later sections prove this for actual simple affine lines. Table 3 separates its arithmetic improvement over (4) from the projective count in the original paper. Here  $P_{\text{FP}}(n)$  denotes the latter construction’s projective count, not the projective maximum.

| $n \bmod 6$ | $G - B$ | $P_{\text{FP}}(n), n \geq 5$ | $P_{\text{FP}} - G$ |
|-------------|---------|------------------------------|---------------------|
| 0           | 1       | $n(n-3)/3$                   | −1                  |
| 1           | 1       | $(n^2 - 3n + 5)/3$           | 0                   |
| 2           | 0       | $(n^2 - 3n + 8)/3$           | 2                   |
| 3           | 2       | $(n^2 - 3n + 9)/3$           | 1                   |
| 4           | 0       | $(n^2 - 3n + 8)/3$           | 2                   |
| 5           | 1       | $(n^2 - 3n + 5)/3$           | 0                   |

**Table 3:** Exact comparison with the quoted affine baseline and with the original projective construction’s face counts. A projective surplus does not itself supply additional bounded affine triangles.

The numerical gain is bounded by two. In particular, the quadratic coefficient  $1/3$  and linear coefficient  $-1$  remain unchanged. What the completed proof adds is a phase refinement, a controlled choice of affine chart for odd orders, and an end-to-end geometric formalization. The audited sources do not provide an explicit statement of (5) with this affine proof. That observation is not an exhaustive numerical-priority theorem: old projective constructions contain related counts, and an unnoticed affine-chart argument could have been available earlier. The defensible claim is the proved improvement over the explicitly quoted formula (4), together with the formal construction; a claim of first numerical discovery is withheld.

## 2.4 Optimal subsequences and the straightening problem

Harborth and Roudneff developed constructions and extremal results for pseudolines [15, 26]. Their scope matters because topological realizability does not imply straight realizability. Forge and Ramírez Alfonsín constructed a projective straight-line family at orders  $2 \cdot 2^t + 2$  [8]. Removing a distinguished line at infinity gives optimal affine orders  $2 \cdot 2^t + 1$ ; in particular, this is the relevant older source for the 33-line order.

Bartholdi, Blanc and Loisel’s preprint appeared in 2007 and its proceedings publication in 2008 [1]. Their compatible doubling step takes  $q + 1$  lines to  $2q + 1$  and adds  $q^2$  triangles. It requires a prescribed tangent grid and a fully used distinguished line. Their straight-line optimal series include  $14 \cdot 2^t + 1$  and  $6 \cdot 2^t + 1$ ; their  $18 \cdot 2^t + 1$  series was then a pseudoline result. The general numerical recurrence is therefore prior work, not a new theorem of this project. Our contribution is the verified real-coordinate specialization, a closed recursive seed invariant and its boundary-visibility refinement. The actual odd and adjacent-even induction is complete in this release; it does not transfer authorship of the published doubling gain.

Parpalak and Utkin supplied a compatible straight 19-line seed and established the straight-line series  $n = 18 \cdot 2^t + 1$  [19]. Its triangle count is  $108 \cdot 4^t - 1$ . Substitution into our formula gives the exact comparison

$$(108 \cdot 4^t - 1) - G(18 \cdot 2^t + 1) = 6 \cdot 2^t - 2. \quad (6)$$

Thus their series is stronger at these orders: the comparisons begin  $107 > 103$  at 19,  $431 > 421$  at 37, and  $1727 > 1705$  at 73. Their separate even-series draft is also relevant prior construction evidence [21]. Its numerical families are not replaced by a weaker universal formula.

It would consequently be misleading to call  $G$  the strongest numerical lower bound for every  $n$ , or an unqualified state of the art among all possible all-order formulas. Taking a maximum with known special constructions already improves it. Padding a configuration by lines sufficiently far away preserves its bounded triangles and extends each special construction to later orders. The retained function in the repository likewise takes a maximum with its finite certificate catalogue. The purpose of  $G$  is a compact uniform guarantee with a complete formal proof, not the suppression of stronger special cases.

## 2.5 Upper-bound refinements retain their simplicity assumptions

The even-order bound

$$\left\lfloor \frac{n(n - 7/3)}{3} \right\rfloor$$

in Bartholdi–Blanc–Loisel applies to *simple affine pseudoline* arrangements. Blanc subsequently proved the sharper common even-order polynomial

$$K_s(n) \leq \left\lfloor \frac{n(n - 5/2)}{3} \right\rfloor, \quad n \text{ even}, \quad (7)$$

also in the simple pseudoline setting [2]. The latter work was posted in 2008 and published in 2011. Its parity argument is the starting point for our clean-line accounting outside a multiple-point core.

The distinction is concrete. Formula (7) equals 53 at  $n = 14$ , whereas a nonsimple configuration with 54 triangles exists. Applying a simple upper theorem to that configuration would be invalid. Our upper-bound work therefore counts shared sides and multiple incidences explicitly rather than treating a small perturbation to general position as automatically harmless.

## 2.6 Recent computational constructions and versioned claims

Savchuk’s 2025 work combines arrangement tables, SAT search, and heuristic straightening [27]. It reports optimal 23- and 27-line configurations and an exclusion of a perfect 33-triangle arrangement at order 11. Its Appendix A.4 records the earlier Honma-based 11-line construction. For our purposes, an arrangement table, a floating-point fit, an exact straight-line certificate, and an exhaustive upper-bound exclusion are four different outputs. The present work independently verifies the coordinate lower bounds it imports; it does not thereby replay the external SAT exclusions.

Maiorana released fifteen exact 14-line, 54-triangle configurations [17]. We preserve their attribution and the source’s CC BY 4.0 license. The current Parpalak–Utkin gallery supplies further exact constructions, including nonsimple even-order examples [22]. The 8-line construction has earlier provenance; citing its current host does not transfer its invention to the host authors. Each retained dataset is tied to a source revision, checked independently, and distinguished from any stronger optimality language used in its original repository.

Versioning also matters for exclusions. The fourth version of Liang, Liu and Zhang’s preprint concerns symmetry restrictions for simple 28-line configurations [16]. It is not treated here as a proof of an unrestricted exact value, and its displayed numerical comparison is not substituted for Blanc’s simple upper theorem. The ProofAtlas research page similarly reports restricted upper results while keeping the unrestricted problem open [23]. We record these as current research sources, not as independently reproduced proofs.

## 2.7 The 2026 enumeration and its use as external input

Parpalak and Utkin’s July 2026 preprint gives a reduced-word enumeration and classification of triangle-maximal pseudoline arrangements, with Euclidean and projective equivalence kept distinct [20]. Its larger first-hit examples are pseudoline constructions; straight realizability is a separate problem. The present tangent-grid obstruction uses their supplied 21-line classification as an external completeness input. Our exact replay checks the algebraic contradiction certificates and all supplied affine-class/support cases, not their full enumeration algorithm inside Lean. Keeping the infinity support and every Euclidean class is essential: a smaller list of projective representatives did not suffice for the affine exclusion.

The compatible 49-line branch also requires careful comparison. Its prospective odd family at  $n = 48 \cdot 2^t + 1$  is the tail of BBL’s older  $n = 6 \cdot 2^s + 1$  series, with  $s = t + 3$ . The counts cannot be presented as new lower-bound records. The uniform explicit slopes, whole-interval sign certificate, boundary data and formal seed interface are the contributions under investigation.

## 2.8 The March result, later verification, and attribution

The author’s March manuscript proposed an even-order extension and recorded the values 238, 275, and 357 at orders 28, 30, and 34[30]. These values are now supported in this project by exact simple-line certificates. The associated original Lean file, however, contained placeholders, including the general extension lemma. The later successful formalizations do not retroactively turn that unfinished universal argument into a proof.

OEIS A006066 indexes numerical data, references, and named contributions [18]. It is not a complete priority database. In particular, the 2007 Bartholdi–Blanc–Loisel table already contains a bold, hence straight-realizable, 30:275 entry. The 34:357 value follows from the older perfect 33-line family and its exterior extension. At 28 lines, Savchuk describes the analogous extension of the 27-line solution. These overlaps require a distinction between an independently supplied construction, a new exact or formal certificate, and the first known

numerical lower bound. We retain the author’s recorded contribution without claiming first numerical priority solely from the OEIS attribution.

The recurrence reproduced by HandWiki [14] is likewise a record of the proposed claim rather than a substitute for its proof. The unrestricted statement

$$K(n+1) \geq K(n) + \lfloor n/2 \rfloor$$

has not been established by the work reported here. What has been established includes a geometric exterior extension under explicit visibility hypotheses, finite successful successor steps, and a separate unconditional all-order construction. Their logical roles remain distinct.

## 2.9 What formal verification contributes

The proof infrastructure uses Lean 4 [5] and mathlib [29]. For the universal result, it connects analytic line equations, determinant signs, triangular cells, affine-chart control, and finite counting, rather than checking the final arithmetic formula in isolation. For finite results, it turns attributed coordinates into explicit certificates. For the upper-bound investigation, it now proves actual segment incidence, clean-line charging and the classical simple polynomial bounds, as well as local geometric restrictions. Extraction and summation of cyclic fans across a nonsimple core remain unfinished.

The accompanying repository [31] records source revisions, negative searches, axiom reports, and the boundary between proved statements and proposed research. This separation is part of the result: a complete formal proof of a modest uniform improvement is stronger evidence than an ambitious recurrence with an unproved geometric step. It does not by itself confer numerical priority or solve the unrestricted Kobon problem.

## 3 Real-line geometry and exact certificates

### 3.1 The two extremal problems

**Definition 3.1.** An affine line is a set  $\{(x, y) \in \mathbb{R}^2 : ax + by = c\}$  with  $(a, b) \neq (0, 0)$ . An arrangement is a finite set of distinct lines. A triangular cell is a bounded connected component of the complement whose closure is a nondegenerate triangle. The maximum number of such cells among arrangements of  $n$  lines is  $K(n)$ . The maximum among simple arrangements, with neither parallel lines nor triple concurrence, is  $K_s(n)$ .

The classical problem permits degeneracies that the simple problem excludes. Consequently  $K_s(n) \leq K(n)$ , and a simple lower-bound construction is valid for both problems. The reverse use of an upper bound is invalid without a separate reduction. Our coordinate validator permits several lines through a triangular vertex, but excludes every line that cuts the triangle’s interior.

The formal lower-bound predicate uses witnesses with no parallel pairs. This is a sufficient class of witnesses, not a claim that every maximizer of the classical problem has that property. For nontrivial orders, pairwise nonzero determinants also imply that every line is valid and that no two supporting lines coincide. The cases with fewer than three lines are handled separately.

### 3.2 A polynomial sign certificate

Represent a line  $\ell$  by  $(a_\ell, b_\ell, c_\ell)$ , with affine evaluation  $f_\ell(x, y) = a_\ell x + b_\ell y - c_\ell$ . For two lines set

$$D(\ell, m) = a_\ell b_m - b_\ell a_m, \quad (8)$$

$$\begin{aligned} V(\ell, m) &= (X_{\ell m}, Y_{\ell m}, D(\ell, m)) \\ &= (c_\ell b_m - b_\ell c_m, a_\ell c_m - c_\ell a_m, D(\ell, m)). \end{aligned} \quad (9)$$

When  $D(\ell, m) \neq 0$ , their actual affine intersection is  $p_{\ell m} = (X_{\ell m}/D(\ell, m), Y_{\ell m}/D(\ell, m))$ . For a third line  $r$  define

$$E(r; \ell, m) = a_r X_{\ell m} + b_r Y_{\ell m} - c_r D(\ell, m), \quad (10)$$

$$O(r; \ell, m) = E(r; \ell, m) D(\ell, m). \quad (11)$$

The useful identity is

$$O(r; \ell, m) = f_r(p_{\ell m}) D(\ell, m)^2. \quad (12)$$

Thus  $O$  has the sign of the affine evaluation without requiring division in the certificate. All operations are polynomial over integer or rational coordinates.

**Definition 3.2** (Certified supporting triple). For a pairwise nonparallel arrangement  $(\ell_0, \dots, \ell_{n-1})$ , an increasing triple  $i < j < k < n$  is certified if

1.  $E(\ell_k; \ell_i, \ell_j) \neq 0$ ;
2. for each line  $r$  of the arrangement, the three numbers

$$O(r; \ell_i, \ell_j), \quad O(r; \ell_i, \ell_k), \quad O(r; \ell_j, \ell_k)$$

are either all nonnegative or all nonpositive.

The first condition excludes concurrence of the three supports; the second permits zeros at vertices but forbids a strict sign change.

**Theorem 3.3** (From a sign certificate to a triangular cell). *Every certified supporting triple determines a nondegenerate bounded triangular cell. Its interior is nonempty. Two distinct increasing certified triples determine disjoint interiors.*

*Proof.* The three pair intersections exist because their denominators in (8) are nonzero. The nonvanishing evaluation excludes concurrence and gives three noncollinear points. Their positive barycentric combinations form a nonempty open triangle  $\Delta$ .

By (12), each arrangement line has a common weak sign at the three vertices. Affine linearity gives the same sign throughout the closed triangle. It is strict in  $\Delta$ : otherwise all three vertex evaluations would be zero, forcing a valid line to contain three noncollinear points. Hence no arrangement line enters  $\Delta$ .

Fix an interior point  $z$ . Its strict sign cell is

$$S_z = \{x \in \mathbb{R}^2 : f_r(x) f_r(z) > 0 \text{ for every arrangement line } r\}.$$

The preceding argument gives  $\Delta \subseteq S_z$ . Conversely, the three support inequalities alone force all three barycentric coordinates of  $x$  to be positive, so  $S_z \subseteq \Delta$ . The sign cell is an intersection of open half-planes and is convex; it is exactly a component of the complement. It is bounded by the three finite segments of the triangle.

If two certified interiors meet, they have the same strict sign vector and hence the same sign cell. Its three supporting lines are determined by its three open sides. Since distinct arrangement lines cannot share an open side, the sets of supports agree, and sorting their indices makes the triples equal. The contrapositive proves disjointness. Q.E.D.

The modules `Kobon.Euclidean` and `Kobon.Cells` prove the sign-cell and nonoverlap content through `interior_nonempty`, `interior_eq_signCell`, and `distinct_interiors_disjoint`. The connected-component interpretation is the elementary convexity argument above; it is not separately encoded through a topological component API. Thus the certificates have an explicit interpretation as actual Kobon cells.

### 3.3 What a finite lower-bound certificate establishes

**Proposition 3.4** (Soundness of exact finite validation). *Let  $A$  be an array of  $n$  integer line coefficients, and let  $\mathcal{T}$  be a list of at least  $T$  certified increasing triples with no duplicates. If all pair determinants are nonzero, then  $K(n) \geq T$ . If all increasing triples of lines additionally have nonzero concurrence determinant, then  $K_s(n) \geq T$ .*

*Proof.* The coefficient inclusion  $\mathbb{Z} \hookrightarrow \mathbb{R}$  preserves polynomial equalities, strict inequalities, and weak signs. The resulting real arrangement therefore satisfies Definition 3.2. Apply Theorem 3.3 to the distinct triples. The additional concurrence test is exactly the simplicity condition. Q.E.D.

Rational coordinates are cleared of denominators line by line before the integer check. The mathematical lower bound needs only a list of  $T$  valid triangles; it need not prove that no further triangle exists in the same arrangement. For the catalog, however, two independent exact counters also enumerate the triangle count. One works from consecutive intersections along each line, while the other directly tests triples against the arrangement. Agreement is an independent check on extraction and counting, not a substitute for the general soundness theorem.

These distinctions explain the three evidential levels used later. A proved symbolic theorem applies at arbitrary order. A finite Lean certificate establishes its listed order. An exact experimental count outside the catalog remains a computationally verified witness until promoted through the same formal pipeline. No finite list, however large, proves an infinite family.

## 4 A parity-sensitive all-order construction

This section gives the mathematical ingredients of Theorem 1.1. The underlying trigonometric family is a version of the classical Füredi–Palásti family. Its attribution is independent of the present formalization and choice of affine chart.

### 4.1 Trigonometric geometry and the classical count

For  $0 < \theta < \pi$ , define

$$\ell(\theta) : \quad \sin \theta x + \cos \theta y = \sin(3\theta). \quad (13)$$

The normal has length one. Direct trigonometric expansion gives

$$D(\ell(x), \ell(y)) = \sin(x - y), \quad (14)$$

$$E(\ell(z); \ell(x), \ell(y)) = 4 \sin(x - y) \sin(z - y) \sin(z - x) \sin(x + y + z). \quad (15)$$

In particular, distinct angles in  $(0, \pi)$  give nonparallel lines. Multiplying (15) by (14) yields

$$O(\ell(z); \ell(x), \ell(y)) = 4 \sin^2(x - y) \sin(z - y) \sin(z - x) \sin(x + y + z). \quad (16)$$

Only three factors determine the sign on the right.

For the half-phase choice  $\theta_i = (i + 1/2)\pi/n$ , select increasing triples with

$$i + j + k \equiv n - 1 \text{ or } n - 2 \pmod{n}. \quad (17)$$

For the shifted choice  $\theta_i = (i + 1/6)\pi/n$ , select instead

$$i + j + k \equiv 0 \text{ or } n - 1 \pmod{n}. \quad (18)$$

**Lemma 4.1** (Triangle selection). *For  $n \geq 3$ , the triples in (17) certify triangles of the half-phase arrangement. The triples in (18) certify triangles of the shifted arrangement. Both arrangements are simple.*

*Sign proof.* The angle sum of any three lines is  $(i + j + k + 3/2)\pi/n$  in the first arrangement and  $(i + j + k + 1/2)\pi/n$  in the second. It is never an integer multiple of  $\pi$ , which, together with (15), excludes triple concurrence.

For a selected  $i < j < k$ , fix a fourth index  $r$ . The factors  $\sin(\theta_r - \theta_j)$  and  $\sin(\theta_r - \theta_i)$  change sign only when  $r$  passes the corresponding support index. For each of the pairs  $(i, j)$ ,  $(i, k)$ ,  $(j, k)$ , the remaining factor is  $\sin((i + j + r + 3a)\pi/n)$  with  $a = 1/2$  or  $1/6$ . Substituting the selected residue of  $i + j + k$  fixes on which side of the next multiple of  $\pi$  this sum lies when  $r < k$  and when  $r > k$ . Splitting the index range at  $i, j, k$  shows that the three expressions in (16) have one weak sign in each range. At a support index the appropriate evaluations are zero. Their common sign need not be the same for different  $r$ ; that is precisely the permitted sign choice in Definition 3.2.

Equivalently, this check uses only the sign of  $\sin t$  on consecutive intervals  $(s\pi, (s + 1)\pi)$  and the inequalities  $0 \leq r < n$ . No numerical trigonometric approximation is used. The complete index splits are proved in `FurediPalasti.lean` and `ShiftedFurediPalasti.lean`. Definition 3.2 and Theorem 3.3 then give the claimed cells. Q.E.D.

For any residue  $b \in \mathbb{Z}/n\mathbb{Z}$ , the equation  $i + j + k = b$  has exactly  $n^2$  ordered solutions. At most  $3n$  have a repeated index, because each of  $i = j$ ,  $j = k$  and  $i = k$  leaves at most  $n$  choices. Two distinct residue classes therefore give at least  $2n^2 - 6n$  ordered distinct-index triples. Each increasing triple accounts for six permutations, so the half-phase arrangement has at least

$$\left\lceil \frac{2n^2 - 6n}{6} \right\rceil = B(n) \quad (19)$$

certified triangles. This is the classical all-order estimate, now proved directly in the real-coordinate formalization.

## 4.2 The repeated-index correction

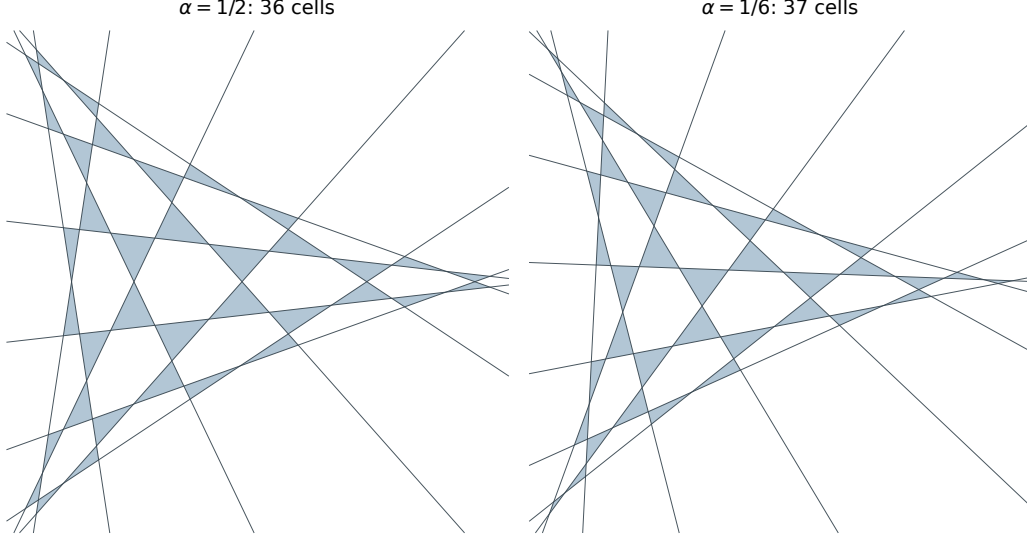
**Lemma 4.2** (Shifted count). *For every  $n \geq 3$ , the shifted arrangement has at least*

$$\left\lfloor \frac{n(n - 3)}{3} \right\rfloor + 1$$

*certified triangular cells.*

*Proof.* Use the two residues 0 and  $n - 1$  in (18). In the zero residue, the three repeated-index sets all contain  $(0, 0, 0)$ . Their union therefore has at most  $3n - 2$  members, not merely  $3n$ .





**Figure 2:** New rational-coordinate illustrations at  $n = 12$ , obtained from the half-phase and shifted trigonometric families. Two independent exact counters certify 36 and 37 triangles in these rational proxies. The axes of each panel are rescaled independently. The real-coordinate universal theorem follows from the symbolic sign and modular arguments, independently of this illustration. The underlying family is attributed to Füredi and Palásti.

In the other residue retain the bound  $3n$ . The number of ordered distinct-index solutions is at least  $2n^2 - 6n + 2$ . If  $T$  is the number of increasing selected triples, then

$$6T \geq 2n^2 - 6n + 2, \quad T \geq \left\lceil \frac{n(n-3)+1}{3} \right\rceil = \left\lfloor \frac{n(n-3)}{3} \right\rfloor + 1.$$

Lemma 4.1 supplies their actual geometry. This argument uses the overlap of bad-index sets, not a numerical rounding conjecture. Q.E.D.

### 4.3 Projective changes of affine chart

The shifted count is enough for the even part of (1). To obtain the additional odd-order gain we use exposed vertices of the half-phase arrangement. The exact coordinate map is

$$(x, y) \mapsto \left( \frac{x}{h-x}, \frac{y}{h-x} \right), \quad h \neq 0. \quad (20)$$

It replaces the line  $x = h$  by the line at infinity. The transformed coefficients of  $ax + by = c$  are

$$\Phi_h(a, b, c) = (ha - c, hb, c). \quad (21)$$

Unlike an affine transformation, this change can alter which projective triangular faces appear bounded. Thus counting triangles before and after the change requires an explicit sign argument.

**Lemma 4.3** (Exact covariance). *If  $D(\ell, m) \neq 0$ , then*

$$D(\Phi_h \ell, \Phi_h m) = h D(\ell, m) (h - x_{\ell m}), \quad (22)$$

$$E(\Phi_h r; \Phi_h \ell, \Phi_h m) = h^2 E(r; \ell, m), \quad (23)$$

$$O(\Phi_h r; \Phi_h \ell, \Phi_h m) = h^3 (h - x_{\ell m}) O(r; \ell, m), \quad (24)$$



where  $x_{\ell m}$  is the abscissa of the old intersection. If the cut avoids every old vertex, the transformed arrangement has no parallel pairs. Triple nonconcurrency is preserved.

*Proof.* Substitute (21) into the polynomial expressions in Section 3, expand, and cancel. Nonzero  $h$  and avoidance of all  $x_{\ell m}$  give the pair determinant claim, while multiplication by  $h^2$  preserves a nonzero triple evaluation. Q.E.D.

The sign multiplier in (24) also gives a direct triangle-preservation rule. If  $h > 0$  and all vertices of a certified triangle lie to the left of  $x = h$ , all three multipliers are positive. If  $h < 0$  and its vertices lie to the right, they are again positive. At selected exposed vertices on the opposite side, the sign reverses. Carefully choosing which vertices are separated by the cut creates the extra triangles.

**Lemma 4.4** (One exposed vertex). *For odd  $n = 2m + 1 \geq 3$ , a change of affine chart of the half-phase arrangement has at least  $B(n) + 1$  bounded triangles.*

*Proof.* The intersection of  $\ell(x)$  and  $\ell(y)$  has abscissa

$$X(x, y) = \cos(2x) + \cos(2y) + \cos(2x + 2y). \quad (25)$$

For the half-phase angles, the vertex of lines  $0, n - 1$  is uniquely rightmost, with abscissa  $c_n = 1 + 2\cos(\pi/n) > 0$ . Indeed, each of the first two cosine terms in (25) is at most  $\cos(\pi/n)$  and the last is at most one; if the pair is not  $(0, n - 1)$ , at least one necessary inequality is strict. Choose  $h > 0$  between all other abscissae and  $c_n$ .

No selected triangle of (17) uses the pair  $(0, n - 1)$ : such a triple would require its middle index to be 0 or  $n - 1$  modulo  $n$ . Hence all counted triangles are preserved by the positive multiplier rule. The additional triple  $(0, m, 2m)$  has, for each test line, nonnegative evaluation at the exposed vertex and nonpositive evaluations at its other two vertices before the transformation. The cut reverses exactly the exposed-vertex sign, so the transformed triple satisfies the certificate. These signs follow from (16) by splitting the test index at  $m$ . The triple is absent from the old selected list, and Lemma 4.3 preserves simplicity. This gives one additional bounded triangle. Q.E.D.

**Lemma 4.5** (Two exposed vertices). *For  $n = 6k + 3$  with  $k \geq 1$ , a change of affine chart of the half-phase arrangement has at least  $B(n) + 2$  bounded triangles.*

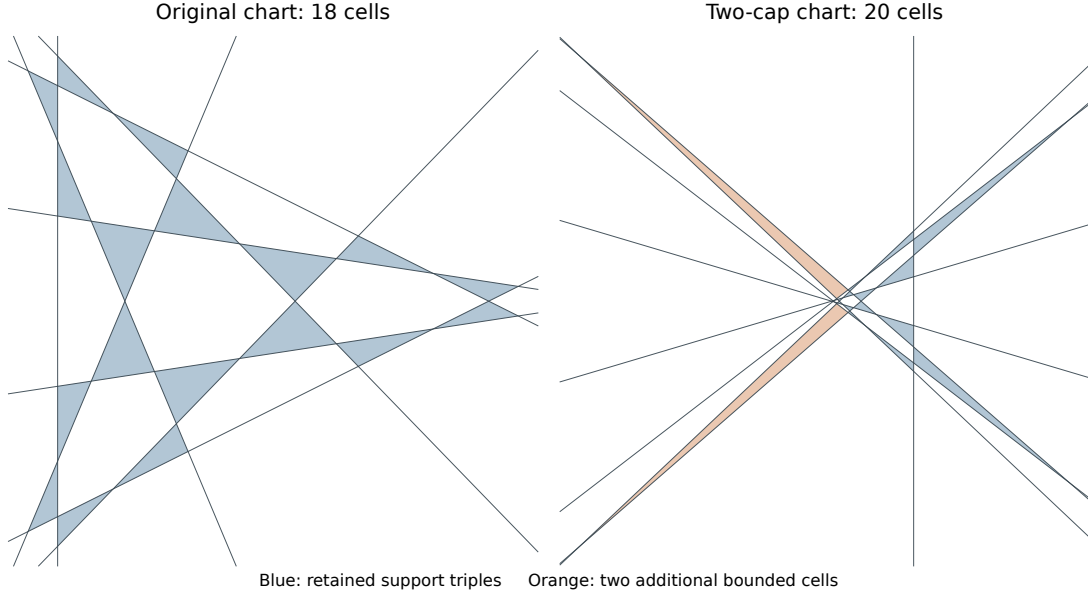
*Proof.* Write  $n = 3m$ , where  $m = 2k + 1$ . The two intersections supported by  $(m - 1, m)$  and  $(2m - 1, 2m)$  have common abscissa  $-c_n/2$ . Every other intersection has strictly larger abscissa. To verify the latter assertion, rewrite (25) as

$$X(u, v) = 2zw + 2z^2 - 1, \quad z = \cos(u + v), \quad w = \cos(u - v).$$

For nonadjacent angle indices, the cosine bounds on the discrete grid, together with completing the square in  $z$ , give  $X > -c_n/2$ . For adjacent indices,  $w = \cos(\pi/n)$  and the same quadratic attains equality precisely at  $u + v = 2\pi/3$  or  $4\pi/3$ , giving the two listed pairs. The exceptional wrapping pair  $(0, n - 1)$  is the rightmost vertex. These exposure inequalities are proved uniformly in `FurediPalastiCaps.lean`.

Choose  $h < 0$  strictly above  $-c_n/2$  and below all other abscissae. The modular conditions (17) show that no counted triangle uses either exposed pair. Its vertices are therefore all to the right of the cut and its signs are preserved. The two additional support triples are

$$(2k, 2k + 1, 5k + 2), \quad (k, 4k + 1, 4k + 2).$$



**Figure 3:** New exact rational illustration at  $n = 9$ : the change of affine chart retains all 18 old triangular cells and creates two more. The resulting 20 is below the known optimum 21; this figure illustrates the uniform mechanism. Panels are rescaled independently. Both counts are checked exactly. The general proof tracks the determinant multiplier at every vertex, independently of the drawing.

For each test index, substitution into (16) gives one sign at the exposed pair and its opposite weak sign at the other two vertices. The negative cut reverses exactly the exposed-vertex multiplier in each case, establishing the transformed certificates. The two triples are distinct and neither satisfies the old selection residues. Simplicity follows from Lemma 4.3. Appending both triples to the retained list proves the claim. Q.E.D.

#### 4.4 Assembly, exact gain, and asymptotic scale

*Proof of Theorem 1.1.* Use Lemma 4.2 for even  $n \geq 4$ . For odd  $n$  not divisible by three,  $n(n-3) \equiv 1 \pmod{3}$ , so  $B(n) + 1 = \lfloor n(n-3)/3 \rfloor + 2$ ; apply Lemma 4.4. For odd multiples of three at least nine,  $B(n) = n(n-3)/3$  and Lemma 4.5 gives the same target. Three nonconcurrent lines give  $G(3) = 1$ , while the smaller cases have zero triangles. Each construction used is simple. Q.E.D.

**Corollary 4.6** (Exact improvement over the quoted baseline). *For  $n \geq 4$ ,*

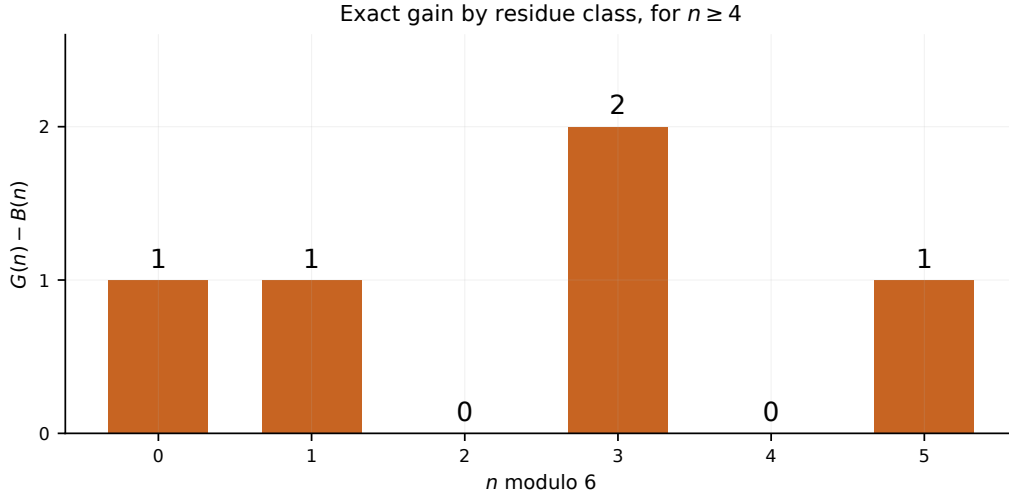
$$G(n) - B(n) = \begin{cases} 1 + (n \bmod 2), & 3 \mid n, \\ n \bmod 2, & 3 \nmid n. \end{cases}$$

*The gains in classes  $0, 1, 2, 3, 4, 5 \pmod{6}$  are respectively  $1, 1, 0, 2, 0, 1$ .*

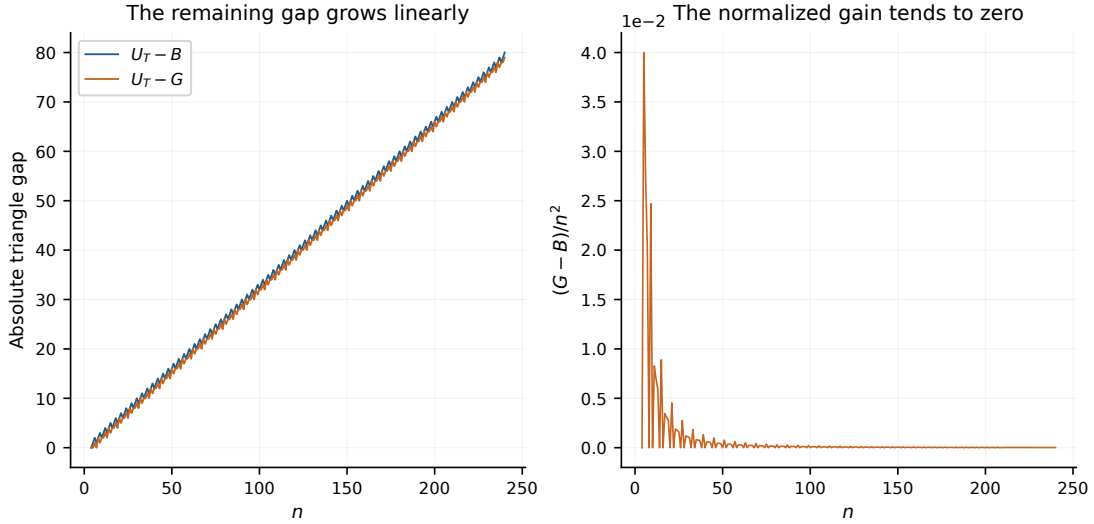
*Proof.* The product  $n(n-3)$  is zero modulo three when  $3 \mid n$ , and one modulo three otherwise. Substitute these remainders in the floor and ceiling formulas. Q.E.D.

For comparison with the classical polynomial benchmark  $U(n) = \lfloor n(n-2)/3 \rfloor$ , direct integer arithmetic gives, for  $n \geq 4$ ,

$$U(n) - G(n) = \left\lfloor \frac{n}{3} \right\rfloor + \mathbf{1}_{n \equiv 2 \pmod{3}} - 1 - (n \bmod 2). \quad (26)$$



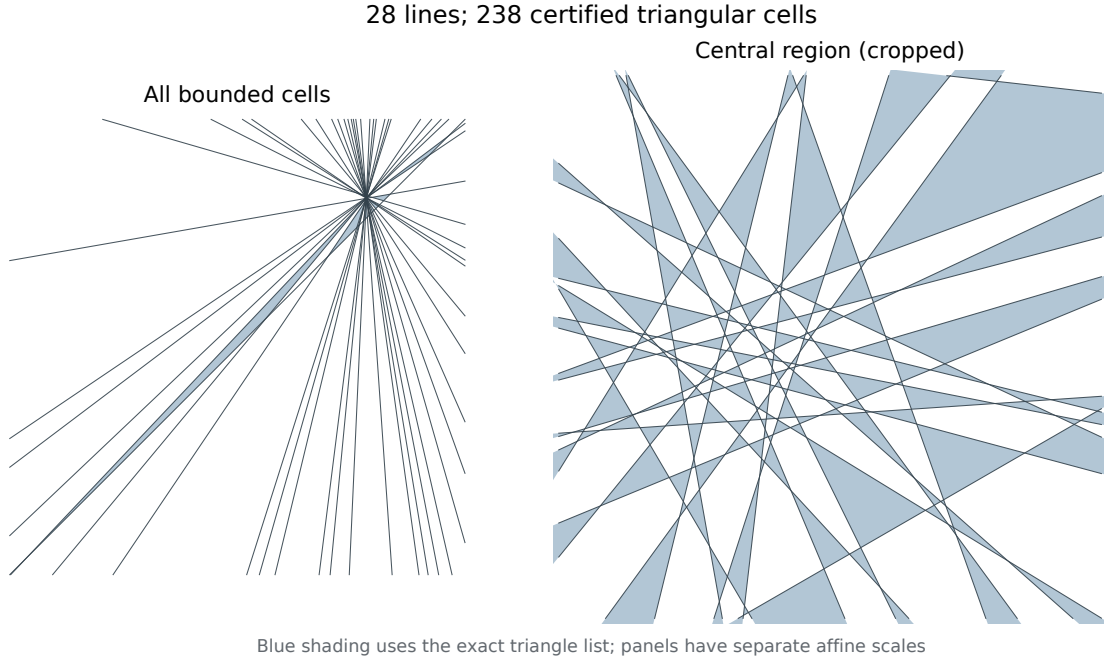
**Figure 4:** The exact proved gain  $G - B$ , resolved by residue class modulo six. The gain is strict on four classes; the universal statement includes arbitrarily large orders in each class.



**Figure 5:** Deficit and normalized growth computed from the proved formulas. The constant improvement is visible in the deficit, while both general lower formulas have the same leading quadratic and linear terms. The polynomial is the classical benchmark, now also formalized for simple arrangements; its numerical upper bound is not new.

This comparison is arithmetic. Section 8.7 now formalizes the classical simple upper theorem; unrestricted attribution remains external. The deficit is still of order  $n$ , and sharper bounds or special constructions apply at some orders.

The outcome is an unconditional affine theorem, stronger than the quoted baseline at infinitely many orders. Its numerical first priority remains a separate literature question. The construction gives a witness for each  $n$ ; it does not provide an operation that adds one line to every preceding optimal witness.



**Figure 6:** The retained exact 28-line witness with 238 certified triangles. This is a new rendering of the saved coordinates associated with the earlier extension program. The marked zoom, where present, is an enlargement of the same arrangement; it does not add or remove lines from the certificate. The picture illustrates a lower bound, while the exact sign tests establish it.

## 5 Individual cases, finite enhancements, and provenance

### 5.1 The March values and their present verification

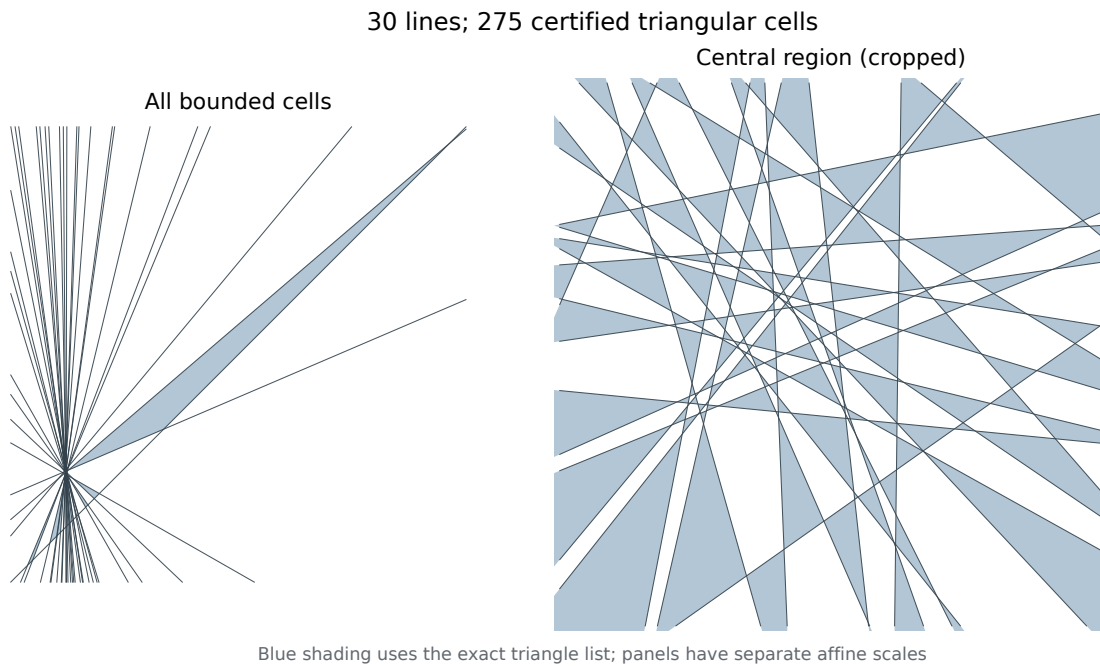
The three finite inequalities associated with the author’s early extension work are

$$K(28) \geq 238, \quad K(30) \geq 275, \quad K(34) \geq 357. \quad (27)$$

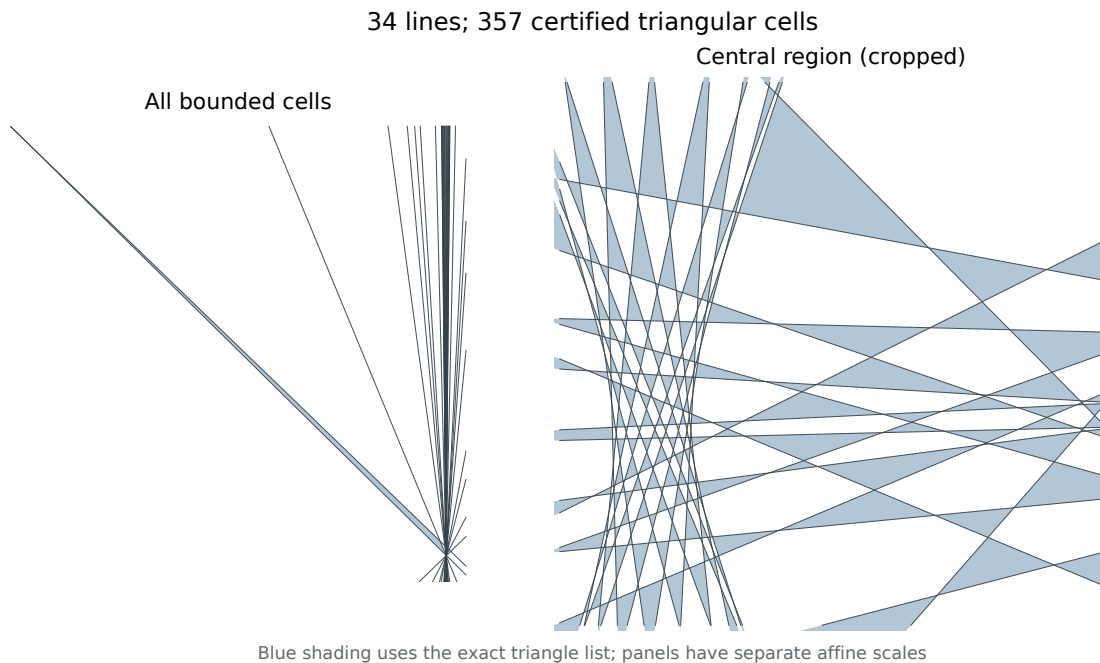
Each now has an exact simple-arrangement certificate, so the same inequalities hold for  $K_s$ . Their present validity is independent of the unproved recurrence in the original March manuscript. The certificates give actual lines and complete checked triangle lists, not only numerical consequences of an extension hypothesis.

The current OEIS table associates the name Zarzuelo with these entries [18, 30]. Attribution in an index and first discovery are different questions. The earlier literature already contains matching constructions or consequences for some of these counts; the literature review records these overlaps. The claims made here are the documented attribution trail, the independently verified witnesses, and their place in the author’s extension program. No new priority claim is inferred from the absence of a value from a particular online table.

The original inventory also listed outputs for every order from 3 through 60, together with larger cases at 99 and 195. Some were direct certificates, some were numerical consequences of a boundary-defect argument, and many were weaker than an already available construction. Appendix C preserves that inventory as history. The current certificate aliases independently retain its numerical inequalities; a label inherited from the old table is not used as a present formal-proof status.



**Figure 7:** The retained 30-line witness with 275 certified triangles, rendered from its exact coordinates. The count has prior literature overlap and is not asserted to be a new numerical record. The complete witness remains in the catalog even though the historical general extension proof was incomplete.



**Figure 8:** The retained 34-line witness with 357 certified triangles. Original plotting of the certificate makes the finite conclusion inspectable independently of the claimed universal extension. The construction input and earlier matching literature are credited separately.

| $n$ | Earlier retained count | Current simple count | Increase |
|-----|------------------------|----------------------|----------|
| 39  | 469                    | 470                  | 1        |
| 47  | 690                    | 691                  | 1        |
| 48  | 720                    | 721                  | 1        |
| 51  | 817                    | 818                  | 1        |
| 53  | 884                    | 885                  | 1        |
| 54  | 918                    | 919                  | 1        |
| 55  | 954                    | 955                  | 1        |
| 59  | 1102                   | 1103                 | 1        |
| 60  | 1140                   | 1141                 | 1        |
| 99  | 3169                   | 3170                 | 1        |
| 195 | 12481                  | 12482                | 1        |

**Table 4:** Changes relative to this project’s previous retained witnesses, not a table of first discoveries or global record improvements. “Simple” is certified for every current coordinate witness in this table.

## 5.2 Materialized improvements from the phase and chart construction

Table 4 records simple rational witnesses promoted during the last construction phase. Each listed new count is supplied by the universal theorem and also realized by an independently checked finite coordinate file. This double presentation is useful: the theorem proves all orders, while the coordinates support visualization and further exact searches.

In particular, the earlier additional-line values 39 : 469, 51 : 817, 99 : 3169 and 195 : 12481 are now exceeded by  $G$  itself. Their finite correctness survives, but those numbers are not distinctive of the additional-line family. Historical evidence is retained rather than deleted when a stronger construction is found.

## 5.3 The 44-line simple case

The exact exterior extension of the retained 43-line arrangement creates 21 triangles while preserving its 587 old triangles. Both exact counters and the Lean certificate give a simple 44-line arrangement with 608 triangles. Blanc’s classical simple upper bound at this order is also 608 [2]. Both inequalities are now formally supported, giving

$$K_s(44) = 608. \quad (28)$$

The finite Lean certificate supplies the lower inequality, and `UpperEvenSimpleOptimality` supplies the simple upper inequality. Equation (28) is their immediate numerical corollary, not a separate maximality wrapper. It neither establishes  $K(44) = 608$  for unrestricted arrangements nor claims first discovery of this numerical value.

This example shows why upper-bound scope belongs next to the numerical statement. A simple witness can establish a classical lower bound, but a simple upper theorem cannot by itself close the classical problem.

## 5.4 External records and independently checked inputs

The updated catalog incorporates exact gallery witnesses from Parpalak and Utkin at

8 : 15, 14 : 54, 20 : 117, 26 : 204, 32 : 315, 36 : 402, 38 : 450, 42 : 553, 46 : 667, 50 : 792.

These are constructions or coordinate sources of the cited researchers, not newly discovered values in this project. The eight-line value has older priority; the modern gallery is the source of the checked coordinates. The public source snapshot and the formal certificate index preserve the provenance [18, 19].

All fifteen of Maiorana’s supplied 14-line arrangements with 54 triangles were independently checked and promoted as finite certificates [17]. Fourteen have two triple points and one has four. Their source license and immutable commit are retained. These witnesses are a particularly useful warning against conflating simple and nonsimple arrangements: removing a multiple intersection by a small perturbation need not preserve the full triangle count. Section 9 reports the exact resolution experiments.

The two source snapshots used for these imports are Maiorana commit `e47c7cfd9661e54d29befb83118bf83d5f804028` and gallery commit `feae4f5571a988a3ed45bd26a36259ebb7d7d3d2`. Their mathematical attribution is attached to each source record even when a new Lean wrapper or new figure is generated here.

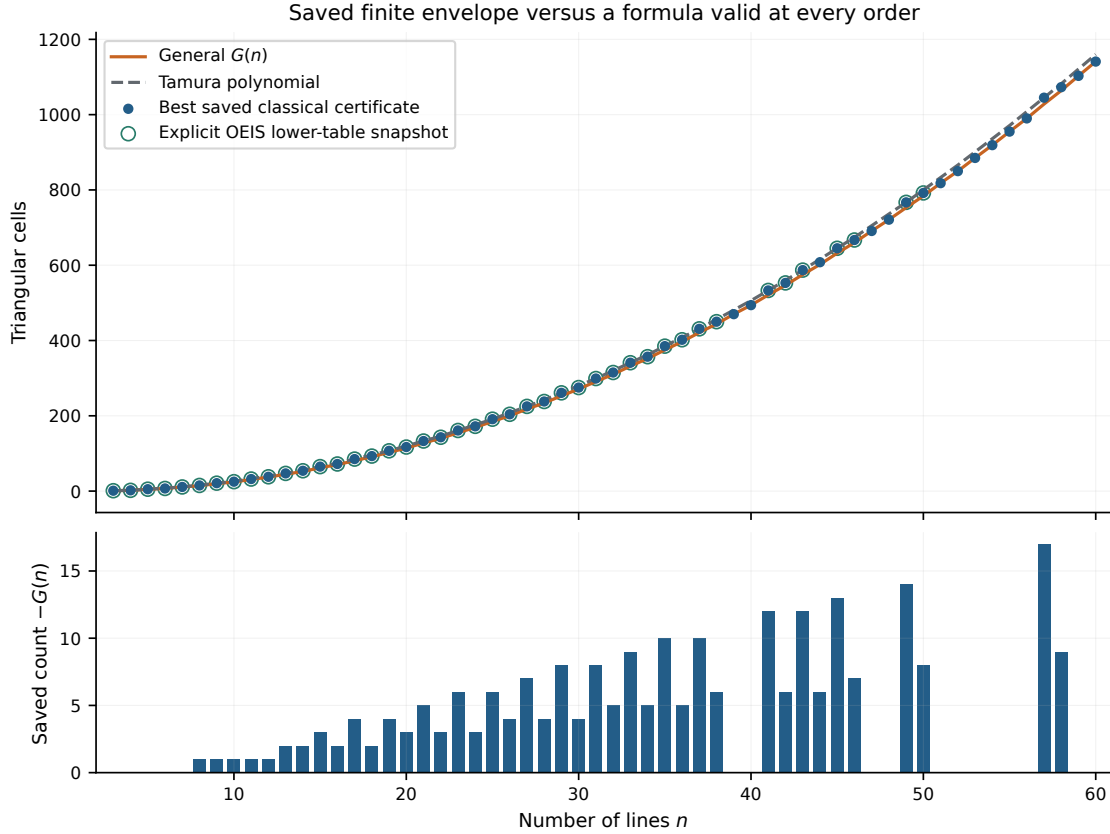
## 5.5 The finite envelope and the complete tables

**Proposition 5.1** (Retained all-order envelope). *Let  $C(n)$  be the maximum count in the finite catalog at order  $n$ , or zero if that order is absent. Then  $K(n) \geq L(n) = \max\{G(n), C(n)\}$  for every  $n$ . In the frozen release,  $L(n) = G(n)$  for every  $n > 195$ .*

*Proof.* When  $C(n) \leq G(n)$ , use Theorem 1.1. Otherwise choose a certificate attaining the finite maximum and apply Proposition 3.4. No coordinate certificate in this release has order greater than 195. Larger computational witnesses discussed later have not been silently added to the Lean catalog. Q.E.D.

The formal definition is `Universal.bound = max Universal.baseline AllN.bound`. Since  $G \geq B$ , it equals the expression in Proposition 5.1; the older AllN layer preserves the previous finite-envelope theorem. This makes comparisons reproducible without rewriting the historical construction history.

Appendix A gives the best saved simple and classical values from 3 through 60. Appendix B lists all 138 certificate identities, including repeated orders and weaker checkpoints. Appendix C gives the older author-output inventory. Together these tables prevent three common confusions: a retained output is not necessarily the best available value; a missing OEIS row is not proof of novelty; and an exact count for one arrangement is not automatically the extremal value  $K(n)$ .



**Figure 9:** The strongest saved simple and classical coordinate counts, compared with the uniform formula and the polynomial upper benchmark. External witnesses retain their attribution. A marker above  $G$  is a finite enhancement of the general formula, not automatically an original record.

## 6 Exterior extensions: the March proposal and its precise replacement

The starting point of this project was the March 2026 proposal that an odd Kobon arrangement could always be extended with the gain

$$K(2m + 2) \geq K(2m + 1) + m. \quad (29)$$

The proposal was presented in [30] and subsequently described as an iterative rule in [14]. The preserved March source is important evidence of the development of the idea, but is not a completed proof of (29): its Lean file contains eight proof placeholders, including the extension assertion. The present work replaces the informal alternating-segment argument by an actual geometric extension theorem and identifies exactly which additional counting hypothesis yields the desired gain. Independently verified finite bounds are retained regardless of the status of that original argument.

The later OEIS attribution at orders 28, 30, and 34 is evidence of the proposal’s reception, rather than a proof of numerical priority [18]. In particular, the BBL table already records the straight-line value  $30 : 275$ . The perfect 33-line input belongs to the family of Forge and Ramírez Alfonsín [8]; applying the exterior extension stated by BBL gives  $34 : 357$  [1]. The historical attribution, the validity of a finite count, and the novelty of a general extension theorem are therefore assessed separately.



Two distinctions are essential. A recurrence for maxima need not extend *every* arrangement attaining a given count, and a construction for one odd-to-even step need not admit repeated full-gain extensions. Throughout this section the geometric hypotheses concern simple affine arrangements; consequently their conclusions give lower bounds for both  $K_s$  and  $K$ , but nonsimple witnesses for  $K$  cannot automatically be substituted into the hypotheses.

### 6.1 A geometric theorem from explicitly visible pairs

Write an old line as  $L_i = \{p : \ell_i(p) = 0\}$ , with  $\ell_i$  affine, and let  $A = \{L_0, \dots, L_{n-1}\}$  be simple. A linear functional  $w$  is *admissible* if it is nonzero on every old line direction. At  $p_{ij} = L_i \cap L_j$ , choose the direction vectors  $u_i, u_j$  on the two supports so that  $w(u_i) = w(u_j) = 1$ . The following criterion uses only the old arrangement.

**Definition 6.1** (Visible pair). The pair  $\{i, j\}$  is visible from  $w$  if, for every old affine equation  $\ell_k$ , the three real numbers

$$\ell_k(p_{ij}), \quad D\ell_k(u_i), \quad D\ell_k(u_j)$$

have one common weak sign: either all are nonnegative or all are nonpositive. Here  $D\ell_k$  denotes the linear part of  $\ell_k$ .

Geometrically, the two increasing- $w$  rays bound an unbounded empty wedge. The algebraic definition is convenient because it gives the required half-plane inequalities directly, without assuming that a future triangle already exists.

**Theorem 6.2** (Verified exterior realization). *Let  $A$  be a simple arrangement of  $n$  real affine lines, let  $\mathcal{T}$  be a set of  $T$  distinct triangular cells of  $A$ , and let  $\mathcal{V}$  be a set of  $c$  distinct pairs visible from an admissible  $w$ . If*

$$H > \max_{i < j} w(p_{ij}),$$

*then adjoining  $L = \{p : w(p) = H\}$  produces a simple arrangement containing all triangles in  $\mathcal{T}$  and at least  $c$  further triangles. In particular,  $K_s(n+1) \geq T + c$ .*

*Proof.* Admissibility excludes a parallel old line, and the strict height inequality excludes every old vertex from  $L$ . Thus simplicity is preserved. Every old triangle lies in the convex hull of its vertices, strictly below the level  $H$ , and survives unchanged.

For a visible pair, the new vertices on its two supports are

$$p_{ij} + (H - w(p_{ij}))u_i, \quad p_{ij} + (H - w(p_{ij}))u_j.$$

At either vertex an old affine equation has the form

$$\ell_k(p_{ij}) + (H - w(p_{ij}))D\ell_k(u).$$

The second coefficient is positive. Definition 6.1 therefore puts all three vertices in one closed half-plane of each old line. The new line is itself a supporting side, and the triangle is nondegenerate because the old intersection lies strictly below it. No arrangement line cuts its interior. Different old pairs give different support triples, and every new triple uses  $L$ , whereas no old one does. The two lists are consequently disjoint. Q.E.D.

This is the geometric content of **Exterior.extension**. The theorem is proved for arbitrary real coefficients and either parity, with no assumption about a future triangle count. An explicit formal height is

$$H = 1 + \sum_{\substack{i, j < n \\ i \neq j}} |w(p_{ij})|,$$

so existence of a sufficiently distant line is part of the construction. The proof uses the standard Lean logical axioms and no finite native evaluation. What it does *not* supply is a universal lower bound of  $\lfloor n/2 \rfloor$  on the number of visible pairs.

## 6.2 Defect, boundary wedges, and a rule for both parities

Let  $t(A) = T$  and define the bounded-segment defect

$$d(A) = n(n - 2) - 3T. \quad (30)$$

There are  $n(n - 2)$  elementary bounded segments. A segment can support at most one triangular cell: its endpoints determine the two other supports, and their intersection lies on at most one side of the segment. Thus  $d(A)$  counts the unused bounded segments. Let  $b(A)$  count the unbounded two-sided cells, called wedges below.

**Lemma 6.3** (Boundary charging). *For a simple arrangement of  $n \geq 3$  lines,*

$$3 \leq b(A) \leq n, \quad b(A) + d(A) \geq n.$$

*Proof.* The arrangement has  $2n$  free rays. A wedge pairs two such rays at their common endpoint, and each free ray belongs to at most one wedge. An unpaired ray on  $L$ , ending at  $p = L \cap M$ , meets an interior vertex of  $M$ . The two bounded elementary segments of  $M$  incident to  $p$  cannot both support triangles: both triangles would also use the unique bounded segment of  $L$  starting at  $p$ . Charge the unpaired ray to an unused segment of  $M$  at  $p$ . Each unused segment receives at most two charges, one at each endpoint. Hence  $2n - 2b \leq 2d$ . This also proves  $b \leq n$ . Finally, the convex hull of all finite vertices has at least three vertices. Both supports at a hull vertex have free outgoing rays, which give a wedge. Distinct hull vertices give distinct wedges. Q.E.D.

Order the  $2n$  oriented line directions cyclically. The two directions of a wedge are consecutive: a line direction strictly inside that sector would force a line to enter the wedge and cross a free boundary ray. Write  $B \subset \mathbb{Z}/(2n)$  for the wedge-start indices. Define

$$c_j = |B \cap \{j, j + 1, \dots, j + n - 2\}|, \quad C(A) = \max_j c_j. \quad (31)$$

The  $n$  directions on which an admissible  $w$  is positive form one consecutive block, and every such block occurs for some normal sector. An exterior line closes precisely the wedges whose two rays lie in this block. Consequently  $C(A)$  is the exact largest exterior gain and

$$\sum_{j=0}^{2n-1} c_j = (n - 1)b(A), \quad \left\lceil \frac{(n - 1)b(A)}{2n} \right\rceil \leq C(A) \leq \left\lfloor \frac{n}{2} \right\rfloor. \quad (32)$$

The sum follows by counting the  $n - 1$  blocks containing a fixed wedge. The upper bound follows because complete wedge pairs use disjoint rays in a block of  $n$  rays.

**Theorem 6.4** (Boundary-defect extension). *For a simple  $n$ -line arrangement with  $n \geq 3$  and  $T$  triangles, an exterior line preserves the old triangles and gives at least*

$$T + g(n, T), \quad g(n, T) = \left\lceil \frac{n - 1}{2n} \max\{3, 3T - n(n - 3)\} \right\rceil \quad (33)$$

*triangles. Thus, for either parity,*

$$K_s(n) \geq T \implies K_s(n + 1) \geq T + g(n, T).$$

*Proof.* Lemma 6.3 gives  $b(A) \geq \max\{3, n-d(A)\}$ . Substitute into (32) and apply Theorem 6.2. If  $T$  is only a lower bound, the witness has some count  $T' \geq T$ ; the right side of (33) is nondecreasing in  $T$ , which gives the stated maximum-function implication. Q.E.D.

*Remark 6.5* (Formal scope and attribution). Theorem 6.4 is an ordinary geometric theorem in this manuscript. The cyclic sum, averaging argument, and defect arithmetic are Lean checked, and Theorem 6.2 supplies actual geometric realization. The universal charging map and the correspondence between Euclidean boundary rays and the finite cyclic data have not yet been joined into an end-to-end Lean proof. Exterior addition and unused-segment arguments have precedents in [1, 2]. The contribution under investigation is the explicit quantitative defect formulation and its certificate interface; first numerical or historical priority is not asserted.

**Corollary 6.6** (Full gain for an odd saturated input). *If a simple arrangement of  $2m+1$  lines has defect at most two, an exterior line creates exactly  $m$  additional triangles. In particular, this applies to every simple odd arrangement attaining  $\lfloor n(n-2)/3 \rfloor$ .*

*Proof.* The wedge count is at least  $2m-1$ . If all  $4m+2$  direction sectors had gain at most  $m-1$ , then

$$2m(2m-1) \leq 2mb(A) = \sum_j c_j \leq (4m+2)(m-1) = 2m(2m-1) - 2,$$

a contradiction. Equation (32) bounds the gain above by  $m$ . The segment-count upper bound at odd  $n$  leaves defect zero or two. Q.E.D.

For an even input  $n = 2m$ , the sufficient condition  $b(A) \geq 2m-1$  likewise forces gain  $m$ , because  $(2m-1)^2 > 4m(m-1)$ . In either parity the exact criterion is the finite condition  $C(A) = \lfloor n/2 \rfloor$ . There is no claim that every even or odd maximum has this boundary profile. Iterating (33) is nevertheless legitimate ordinary geometry and supplies a weaker lower-bound rule from any simple seed.

### 6.3 What fails in indefinite full-gain exterior iteration

The March argument made the number of new triangles follow from an alternation assertion after placing a line far from all old vertices. Distance proves preservation; it does not prove emptiness for half the new segments. For example, the five lines  $y = ix + i^2$ ,  $i = 0, \dots, 4$ , have three triangles, and adding  $y = x/2 + 100$  creates only one further triangle. These counts are checked by exact certificates. This refutes that unqualified step of the argument, without refuting a better choice of line or an inequality about maxima.

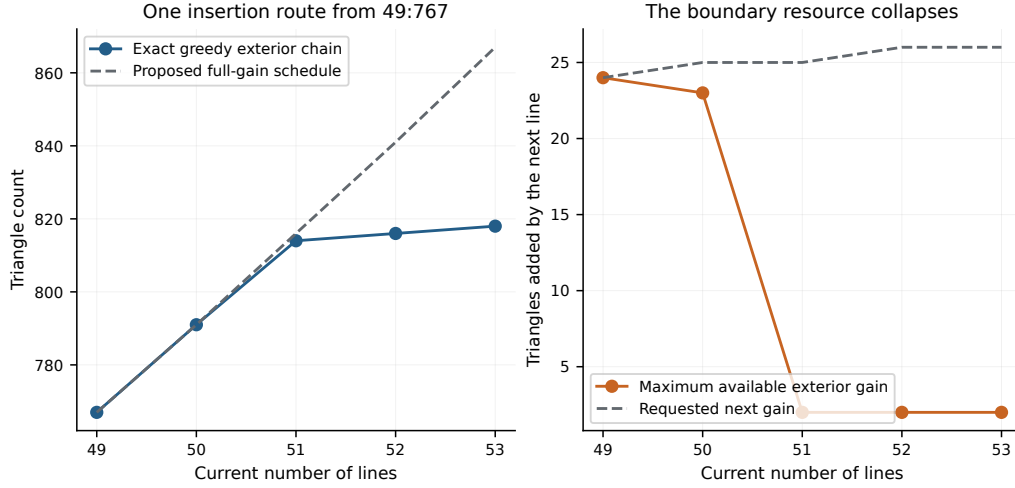
A stronger obstruction concerns every chain of exterior additions. Write  $b_r$  for its wedge count and  $c_r$  for the gain at step  $r$ . Closing  $c_r$  wedges consumes those wedges. No new free ray can appear at an old vertex, and a wedge at a new vertex must use one of the two free rays of the newly added line. Therefore

$$b_{r+1} + c_r \leq b_r + 2, \quad \sum_{j < r} c_j \leq b_0 + 2r - b_r \leq b_0 + 2r - 3. \quad (34)$$

The accumulated exterior gain is at most linear in the number of steps. The accumulated gains  $\lfloor n/2 \rfloor$  are quadratic, so an infinite chain of such full-gain exterior steps is impossible. The numerical contradiction is formalized in `Iteration.no_infinite_full_gain_budget`; as in Remark 6.5, the universal wedge-budget interpretation is an ordinary geometric argument.

| Input order | Triangles | Wedges | Best exterior gain | Requested gain |
|-------------|-----------|--------|--------------------|----------------|
| 49          | 767       | 47     | 24                 | 24             |
| 50          | 791       | 24     | 23                 | 25             |
| 51          | 814       | 3      | 2                  | 25             |
| 52          | 816       | 3      | 2                  | 26             |
| 53          | 818       | 3      | 2                  | 26             |

**Table 5:** One exact greedy exterior chain from the attributed 49-line input. Every direction sector was counted with rational arithmetic. These are capacities of the displayed arrangements, not upper bounds for  $K$  or  $K_s$  at the corresponding orders.



**Figure 10:** The boundary resource in the particular exact 49-line chain. The sharp drop in available wedges explains why its first full-gain extension cannot be repeated with the requested next gain. The plotted capacities concern this chain only.

Table 5 records the motivating test from 49 : 767. The obstruction is independent of the greedy tie break: two exterior gains 24 and 25 from this seed would imply

$$b_2 \leq 47 + 4 - 24 - 25 = 2,$$

contrary to  $b_2 \geq 3$ . The five finite profile maxima have a checked Lean arithmetic certificate; their identification with the saved coordinates is supplied by the exact geometry program. A reconstruction between steps or an interior insertion could evade this boundary budget, so it does not disprove the proposed recurrence for maxima.

#### 6.4 A conditional recurrence and an unconditional numerical bound

For any integer-valued function  $F$ , assume

$$F(n+1) \geq F(n) + \lfloor n/2 \rfloor \quad (n \geq s). \quad (35)$$

Summation, including both parity cases, gives the fully checked arithmetic implication

$$F(n) \geq F(s) + \left\lfloor \frac{(n-1)^2}{4} \right\rfloor - \left\lfloor \frac{(s-1)^2}{4} \right\rfloor \quad (n \geq s). \quad (36)$$

The hypothesis (35) remains explicit in the formal theorem. Substituting  $s = 49$  and  $F(49) = 767$  yields the numerical target

$$Q_{49}(n) = \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 191. \quad (37)$$

There is now a separate unconditional Lean theorem

$$K(n) \geq Q_{49}(n) \quad (n \geq 49), \quad (38)$$

proved using the saved witnesses at 49 and 50 and the classical all-order baseline thereafter. Indeed, for  $n \geq 51$ ,

$$\frac{n(n-3)}{3} - \frac{(n-1)^2}{4} - 191 = \frac{n^2 - 6n - 2295}{12} \geq 0,$$

with equality at  $n = 51$  and an increasing numerator beyond it. Rounding gives  $B(n) \geq Q_{49}(n)$ . This proves the desired numerical formula without constructing a nested full-gain chain. The stronger all-order bound  $G$  and the finite envelope elsewhere in this paper also dominate it.

For comparison, direct iteration of the defect rule  $T_{n+1} = T_n + g(n, T_n)$  from 49 : 767 guarantees 50 : 791, 51 : 803, 52 : 805, and 60 : 821. The stronger target (37) gives 51 : 816, 52 : 841, and 60 : 1061. These are different constructions and different assertions. The completed numerical theorem does not retrospectively prove the March extension claim, and the failure of its proposed exterior iteration does not invalidate the completed numerical theorem.

## 7 Compatible seeds, geometric doubling, and sparse families

The second extension mechanism inserts a whole collection of lines near a distinguished support. It belongs to Bartholdi, Blanc, and Loisel (BBL) [1]; it is fundamentally different from the one-line exterior operation of Section 6. The present work supplies explicit compatible seeds, a recursively closed geometric Lean proof, and finite certificates at considerably larger orders. The October release completes both the odd iteration and the stronger adjacent-even family. The numerical doubling gain retains its published attribution.

### 7.1 The exact one-step theorem

Let  $q = 4r$  with  $r \geq 5$ , put  $\alpha = \pi/q$ , and take

$$0 < \varepsilon < \tan(\alpha/2). \quad (39)$$

Define  $q$  ordered intercepts, indexed from zero, by

$$a_j = \begin{cases} \tan((j - q/2 + 1)\alpha), & 0 \leq j < q/2 - 1, \\ -\varepsilon, & j = q/2 - 1, \\ \varepsilon, & j = q/2, \\ \tan((j - q/2)\alpha), & q/2 < j < q. \end{cases} \quad (40)$$

In particular, the extreme intercepts are  $-\cot \alpha$  and  $\cot \alpha$ . The two central intercepts replace the repeated zero that would otherwise occur in this grid.

**Definition 7.1** (Compatible tangent-grid seed). A seed at  $(q, \varepsilon)$  consists of  $Y_0 : y = 0$  and the  $q$  lines  $L_j : y = m_j(x - a_j)$ , with the following properties: the arrangement is simple; every interval  $[a_j, a_{j+1}]$  on  $Y_0$  supports the triangular cell with supports  $Y_0, L_j, L_{j+1}$ ; and the central apex  $L_{q/2-1} \cap L_{q/2}$  lies above  $Y_0$ . A counted seed additionally carries a list of distinct actual triangular cells whose length is at least  $T$ .

Simplicity already requires all  $m_j$  to be nonzero and pairwise distinct. The distinguished-line condition is a substantial geometric restriction, not a consequence of having a large total count. It is precisely why an arbitrary optimal arrangement cannot simply be used as a repeatable BBL seed.

**Theorem 7.2** (Fully verified geometric doubling). *Suppose  $q = 4r \geq 20$  and (39) holds. A compatible tangent-grid seed with at least  $T$  certified triangles gives a simple real straight-line arrangement of  $2q + 1$  lines with at least*

$$T + q^2 \quad (41)$$

*triangles.*

The precise formal theorem is `BBLDoubling.doubling`. It assumes the actual old line equations, simplicity, each distinguished triangle, the positive central height, and a duplicate-free triangle list. It does not assume a future crossing order, a list of future empty triangles, or a triangle-count recurrence. Its conclusion is an actual `SimpleLowerBound`; the generic proof uses only the standard Lean logical axioms. The numerical doubling principle is BBL's. The contribution here is its completed geometric proof for these explicit hypotheses, using the following two-parameter realization.

*Geometric proof and its formal decomposition.* For  $0 \leq i < q$  set

$$\beta_i = -\frac{\pi}{2} + (i + \frac{1}{2})\alpha, \quad b_i = \tan \beta_i.$$

Adjoin the lines

$$M_i : \quad y = \kappa g_i(\delta)(x - b_i), \quad g_i(\delta) = \frac{2b_i}{1 + b_i^2} + \frac{\delta}{b_i} = \sin(2\beta_i) + \delta \cot \beta_i. \quad (42)$$

Choose  $\delta > 0$  sufficiently small and then  $\kappa > 0$  sufficiently small relative to the fixed  $\delta$  and the seed. This order of choices is part of the proof. BBL's published construction uses explicit scales; the qualitative two-scale statement here is established independently by the formal crossing analysis, rather than inferred from those fixed scales.

For two new lines put  $t = b_i$  and  $u = b_k$ . Their intersection has  $x$ -coordinate independent of  $\kappa$ :

$$X_{ik}(\delta) = \frac{2tu(t + u)}{2tu(1 - tu) - \delta(1 + t^2)(1 + u^2)}. \quad (43)$$

When  $tu \neq 1$ , its limit is the reduced tangent  $(t + u)/(1 - tu) = \tan(\beta_i + \beta_k)$ , and the exact displacement is

$$X_{ik}(\delta) - \frac{t + u}{1 - tu} = \frac{\delta(t + u)(1 + t^2)(1 + u^2)}{[2tu(1 - tu) - \delta(1 + t^2)(1 + u^2)](1 - tu)}. \quad (44)$$

For sufficiently small positive  $\delta$  its nonzero sign is that of  $(t + u)tu$ . If  $tu = 1$ , the crossing diverges along the specified exterior ray as  $\delta \downarrow 0$ ; if  $t + u = 0$ , its coordinate is exactly zero. Thus neither exceptional case is hidden inside a continuity argument with a vanishing denominator.

The reduced tangents, the old intercepts, the new half-grid intercepts, and  $-\varepsilon, 0, \varepsilon$  determine a strictly ordered finite system of cuts. Equations (43)–(44) place every new–new crossing in its prescribed cut interval. Taking  $\delta$  in the intersection of the finitely many resulting positive neighborhoods also makes the new slope factors nonzero and distinct. For fixed such  $\delta$ , each old–new intersection tends to the corresponding old intercept as  $\kappa \rightarrow 0$ . Finite

continuity then realizes all the prescribed strict crossing comparisons simultaneously. Small  $\kappa$  separates old and new slopes. Injectivity along each new crossing row excludes new triple intersections; old simplicity excludes the remaining ones. This proves actual simplicity and crossing order.

The count can now be separated from trigonometry. Treat the  $q$  new lines and  $Y_0$  as  $q + 1$  auxiliary supports. The integer row model places old support  $j$  at key  $4j + 2$ . A pair of auxiliary supports with crossing key  $z$  yields a mixed triangle with old support  $j$  when

$$|z - (4j + 2)| < 4. \quad (45)$$

Every auxiliary pair has two such old neighbors, except for at most  $q$  exterior pairs, each of which has one. Hence the number is at least

$$2 \binom{q+1}{2} - q = q^2. \quad (46)$$

The row inequalities make two sides consecutive on their supports. The two-side emptiness lemma proves that the corresponding triangle is an actual empty cell. The supporting triples are distinct. Thus (46) counts genuine triangles, not merely local crossing patterns.

It remains to preserve the seed's contribution. An old triangle not using  $Y_0$  has a strict half-plane sign against  $Y_0$ ; all new lines converge to  $Y_0$ , so its three vertex signs persist for sufficiently small  $\kappa$ . The old triangles with a side on  $Y_0$  require a different argument. Saturation forces consecutive old apices to have opposite heights. The positive central apex fixes the alternating parity. The crossing order identifies a new cap support for each noncentral distinguished triangle. On both cap endpoints, every new line has the parity-correct weak sign; strict signs against other old supports persist by continuity. This replaces each such old triangle by one empty triangle with the same two old supports. The central triangle itself is retained. The finite cap and noncap requirements can again be satisfied by one common positive  $\kappa$ .

Finally, these  $T$  retained or replaced triangles have at least two old nonhorizontal supports, while the  $q^2$  mixed triangles have two auxiliary supports. The supporting triples cannot coincide. Their union is a duplicate-free witness of length at least  $T + q^2$ . Q.E.D.

The recurrence preserves the certified segment deficit:

$$\Delta(q, T) = q^2 - 1 - 3T, \quad \Delta(2q, T + q^2) = \Delta(q, T). \quad (47)$$

Here the equality concerns the transported lower count; if the output has additional uncounted triangles, its actual defect can be smaller. Thus doubling amplifies a compatible seed to larger orders while preserving its certified constant deficit. A smaller-defect seed or a new source of additional triangles is needed to improve that deficit.

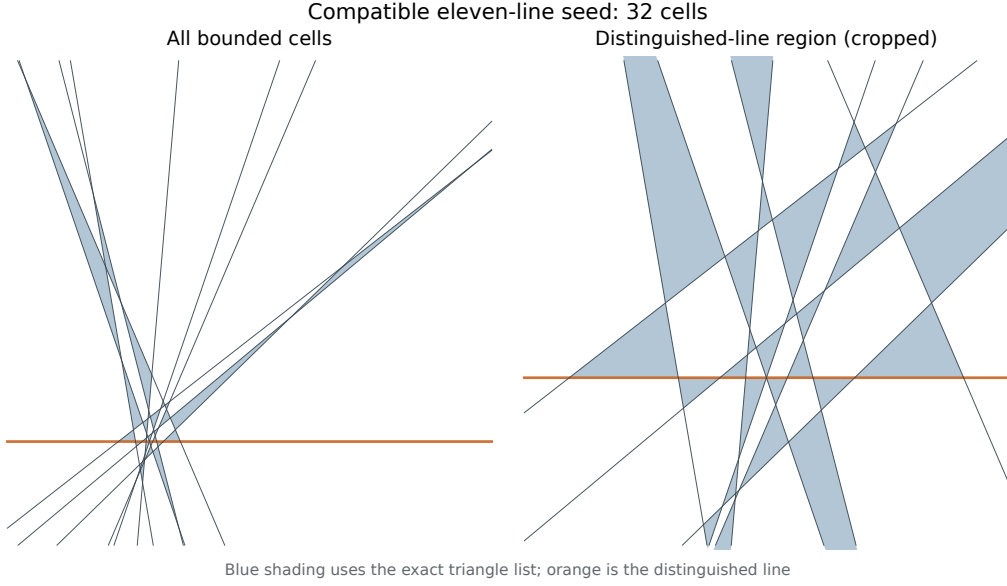
## 7.2 The compatible eleven-line realization

The useful eleven-line count is 32, an existing result; the search started from the Honma-based construction displayed by Savchuk [27]. The additional work here is an explicit realization compatible with the BBL grid for arbitrarily small positive central separation. Take  $Y_0 : y = 0$  and

$$x - v_j y = a_j, \quad (v_0, \dots, v_9) = \frac{1}{10}(15, -2, 14, 1, 4, -4, 5, -3, 12, -5), \quad (48)$$

where

$$(a_0, \dots, a_9) = (-\tan \frac{4\pi}{10}, -\tan \frac{3\pi}{10}, -\tan \frac{2\pi}{10}, -\tan \frac{\pi}{10}, -\varepsilon, \varepsilon, \tan \frac{\pi}{10}, \tan \frac{2\pi}{10}, \tan \frac{3\pi}{10}, \tan \frac{4\pi}{10}). \quad (49)$$



**Figure 11:** A finite rational realization of the compatible eleven-line seed, with global and distinguished-line views. The figure illustrates the arrangement; the all-small-parameter claim is certified by exact determinant and half-plane inequalities, not by the rendering. The count 32 is inherited from earlier work; the compatible realization and parameter certificate are the contribution recorded here.

For every  $0 < \varepsilon \leq 10^{-5}$  the seed is simple, has at least 32 certified triangles, and has all nine distinguished triangles. Its central slopes are  $5/2$  and  $-5/2$  and its central height is  $5\varepsilon/2 > 0$ .

The stability check is finite but uniform in  $\varepsilon$ . The determinant for a triple of nonhorizontal lines is affine in the intercepts and therefore affine in  $\varepsilon$ :

$$D_{ijk} = (a_j - a_k)v_i + (a_k - a_i)v_j + (a_i - a_j)v_k.$$

Exact rational enclosures for the four tangent values and endpoint sign checks certify the whole interval. The first certificate used outward rational interval arithmetic; the subsequent Lean theorem uses proved trigonometric enclosures and the parametric certificate soundness theorem. The limit  $\varepsilon = 0$  is used only to bound expressions; it is not presented as a simple arrangement. Five visible pairs for  $w = (10, -13)$  are also proved uniformly, giving the actual exterior bound  $12 : 37$  by Theorem 6.2.

### 7.3 Closing the geometric invariant

A numerical inequality after one step is insufficient for induction. The output must again satisfy every geometric hypothesis in Definition 7.1. The October construction returns precisely that information. The [seed record](#) contains actual slopes, a duplicate-free valid triangle list, simplicity, every consecutive distinguished triangle, and the positive central apex. The line labels are normalized by explicit inverse permutations; the transport theorem preserves actual support triples and their triangle predicates, rather than merely transporting a count.

**Theorem 7.3** (Closed geometric iteration). *Fix  $r \geq 5$  and a counted seed size  $T$ . Suppose that for every sufficiently small positive  $\varepsilon$  a compatible tangent-grid seed with  $4r + 1$  lines and at least  $T$  triangles exists. Then for every integer  $t \geq 0$  there is a simple arrangement of  $4r 2^t + 1$  lines with at least  $T_t$  triangles, where*

$$T_0 = T, \quad T_{t+1} = T_t + (4r 2^t)^2, \quad 3T_t + (4r)^2 = 3T + (4r 2^t)^2. \quad (50)$$



*At every finite depth the full seed invariant is available on some nonempty positive parameter interval.*

*Proof.* The geometric construction in Theorem 7.2 is performed while retaining the next distinguished triangles and the positive central triangle. Its output intercepts, after the proved reindexing, are exactly the tangent grid for  $2r$ . Simplicity and the counted list are carried in the same witness. This is `seed_step`; its coefficient identities are proved in `BBLRecursiveGrid` and its unreindexed geometric witness in `BBLRecursiveWitness`.

More explicitly, let  $\eta > 0$  be a valid input threshold. The next seed exists for every

$$0 < \varepsilon < \min\{\eta, \tan(\pi/(8r))\}.$$

The minimum is positive. For each such  $\varepsilon$ , the small pencil parameters are chosen as in the one-step proof. No uniform lower bound on these auxiliary parameters is needed. Thus the whole seed property, including its nonempty interval, is preserved by  $r \mapsto 2r$ . Induction proves the first assertion and the recurrence. Multiplying the recurrence by three and using  $(2q)^2 = q^2 + 3q^2$  proves the displayed identity by induction. The formal results are `uniform_iterate`, `infinite_family`, and `triangleCount_identity`. Q.E.D.

The quantifier is  $\forall t \exists \eta_t > 0 \forall \varepsilon \in (0, \eta_t)$ , with a seed witness for each such parameter. It is not  $\exists \varepsilon > 0 \forall t$ . In particular, the theorem does not claim that a fixed drawn arrangement can be iterated with one fixed positive gap forever. This distinction closes the earlier proof gap without inserting an unproved compactness assertion.

The normalized 21-line seed supplies  $r = 5$ ,  $T = 132$  throughout  $0 < \varepsilon \leq 10^{-5}$ . Its nineteen distinguished triangles, actual coefficient identity, and central height  $5\varepsilon/2$  are proved in `BBLSeed21Normalized`. The generic theorem starts at  $r \geq 5$ ; the separately verified eleven-line seed supplies the initial term in the following family.

**Theorem 7.4** (Verified odd family). *For every integer  $t \geq 0$ , set  $q_t = 10 \cdot 2^t$ . Then*

$$K_s(q_t + 1) \geq T_t = \frac{100 \cdot 4^t - 4}{3} = \frac{q_t^2 - 4}{3}. \quad (51)$$

*Proof.* For  $t = 0$  use the eleven-line certificate. For  $t \geq 1$  apply Theorem 7.3 to the normalized 21-line seed at depth  $t - 1$ . Equation (50) gives  $3T_t + 4 = q_t^2$ . The exact formal declaration is `eleven_odd_family`. Q.E.D.

The generic iteration uses only standard logical axioms. The concrete family additionally inherits four existing native certificate checks from its 21-line base. This is an unconditional geometric existence theorem with an explicitly reported computation trust boundary, not a conditional recurrence on unknown arrangements.

## 7.4 Even companions and the retained boundary invariant

The defect argument of Theorem 6.4 gives the weaker ordinary mathematical consequence

$$K_s(q + 2) \geq \frac{q^2 - 4}{3} + \frac{q}{2} - 1, \quad q = 10 \cdot 2^t. \quad (52)$$

Indeed its gain is  $\lceil q(q-2)/(2(q+1)) \rceil = q/2 - 1$ . The October proof establishes the additional visible pair at every depth, rather than extrapolating from finite checkpoints.

A visible seed extends the preceding seed record by an admissible normal  $w = (w_x, w_y)$ , a duplicate-free list of at least  $V$  visible pairs, and a rightmost old line  $R$  satisfying

$$m_R > 0, \quad w_x + w_y m_R > 0, \quad x(R \cap L_i) \leq a_R \quad (L_i \neq R).$$

These are concrete line-intersection conditions. The record is `VisibleSeed`. The generic iteration additionally assumes  $w_x > 0$ .

**Theorem 7.5** (Closed visible-boundary iteration). *Under these uniform visible-seed hypotheses with  $r \geq 5$ , doubling preserves the normal and the whole visible-seed record, with*

$$(r, T, V) \mapsto (2r, T + (4r)^2, V + 2r). \quad (53)$$

Consequently, at depth  $t$  the visible count satisfies  $V_t + 2r = V + 2r 2^t$ . If  $V = 2r$ , the exterior extension yields at least  $T_t + 2r 2^t$  triangles at order  $4r 2^t + 2$ .

*Proof.* Old visible pairs avoiding  $Y_0$  persist under a sufficiently small pencil; at most one visible old pair uses  $Y_0$ . The new construction supplies  $r$  negative complementary pairs,  $r - 1$  positive almost-complementary pairs, the pair of  $Y_0$  and the rightmost new support, and the pair of  $R$  with the support at  $\beta_* = \pi/4 - \alpha/2$ . This supplies  $2r + 1$  pairs, compensating for the possible old loss and leaving a gain of  $2r$ . Pair supports and new/old tags prove that the retained list has no duplicates.

The last pair is controlled by

$$\cot \alpha \sin(2\beta) + \cos(2\beta) - 1 = \frac{\sin(2\beta + \alpha)}{\sin \alpha} - 1.$$

On the half-grid its unique maximum occurs at  $\beta_*$ , with a gap at least  $2 \sin \alpha$ . First shrink  $\delta$  to preserve this strict maximum, then shrink  $\kappa$  to compare the actual new crossing with all old vertices on  $R$ . The positivity of projected graph directions transfers horizontal crossing order to  $w$ -order. The corresponding rightmost clean ray is then established on the next grid. The finite requirements are intersected with the eventual one-step witness, so the same output has the old seed invariant and the new boundary invariant.

This is the geometric content of `canonical_visible_step` and `BBLEvenRecursive.visible_seed_step`; `BBLProjection`, `BBLRightmost` and `BBLNextBoundary` supply the actual order and intersection lemmas. Reindexing transports visibility and admissibility. The positive parameter-interval induction is the same as for odd seeds. Summing  $V_{t+1} - V_t = 2r 2^t$  gives  $V_t = V + 2r(2^t - 1)$ . Finally apply the proved exterior theorem to this actual visible-pair list. Q.E.D.

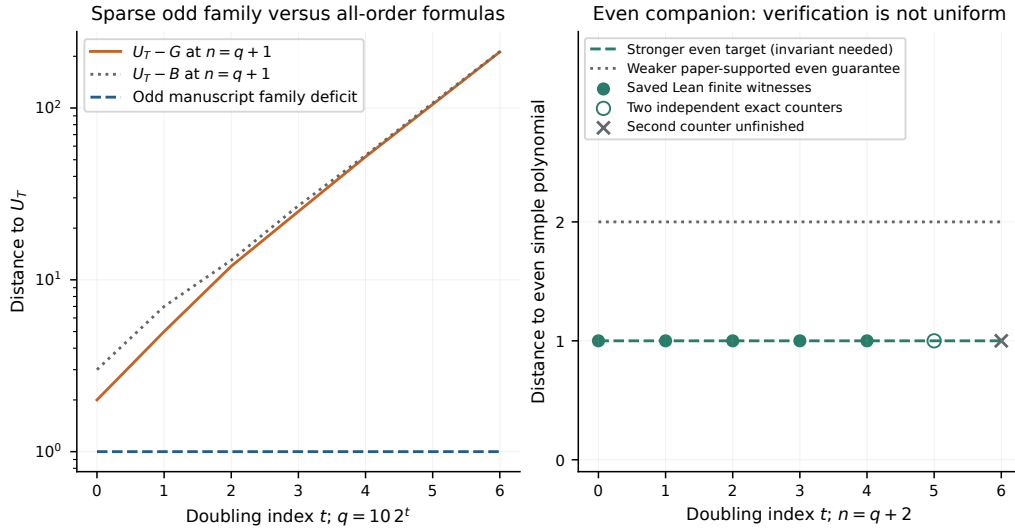
**Theorem 7.6** (Verified full-gain even family). *For every integer  $t \geq 0$  and  $q = 10 \cdot 2^t$ ,*

$$K_s(q + 2) \geq \frac{q^2 - 4}{3} + \frac{q}{2}. \quad (54)$$

*Proof.* The normalized 21-line visible seed has  $r = 5$ ,  $T = 132$ ,  $V = 10$  and  $w = (10, -13)$ . Its full rightmost-boundary and visibility hypotheses hold uniformly for  $0 < \varepsilon \leq 10^{-5}$ . Apply Theorem 7.5; the independently verified 12 : 37 witness supplies  $t = 0$ . The unconditional declaration is `eleven_even_family`. Its dependencies include the four odd-seed native checks and two further finite admissibility/visibility checks. No unresolved visibility hypothesis remains in this concrete theorem. Q.E.D.

| $q$ | Odd order | Odd count | Even order | Even count | Family proof   |
|-----|-----------|-----------|------------|------------|----------------|
| 10  | 11        | 32        | 12         | 37         | Lean existence |
| 20  | 21        | 132       | 22         | 142        | Lean existence |
| 40  | 41        | 532       | 42         | 552        | Lean existence |
| 80  | 81        | 2132      | 82         | 2172       | Lean existence |
| 160 | 161       | 8532      | 162        | 8612       | Lean existence |
| 320 | 321       | 34132     | 322        | 34292      | Lean existence |
| 640 | 641       | 136532    | 642        | 136852     | Lean existence |

**Table 6:** The completed odd and even family. Every displayed existential bound now follows from one general Lean theorem. This does not change the separate verification status of previously exported coordinate files, nor assert that these numbers exceed all literature constructions.



**Figure 12:** The original sparse-family deficit plot, retained as a historical numerical figure. Its arithmetic is unchanged. The family formulas now have completed Lean existence proofs; any finite-check status encoded in the original plot describes the September coordinate artifacts. The new figure below compares the completed envelope.

The independent adjacency and half-plane counters agree on the archived 321-, 322- and 641-line coordinate files, with 34132, 34292 and 136532 triangles. Their largest rational components have 916, 985 and 1103 decimal digits. The 642-line file passed adjacency and simplicity checks but its separate direct recount was interrupted. That coordinate artifact still has partial computational status. In contrast, existence of some 642-line simple arrangement with at least 136852 triangles is now proved by Theorem 7.6. An existential proof does not retrospectively certify the exact bytes of a saved coordinate file.

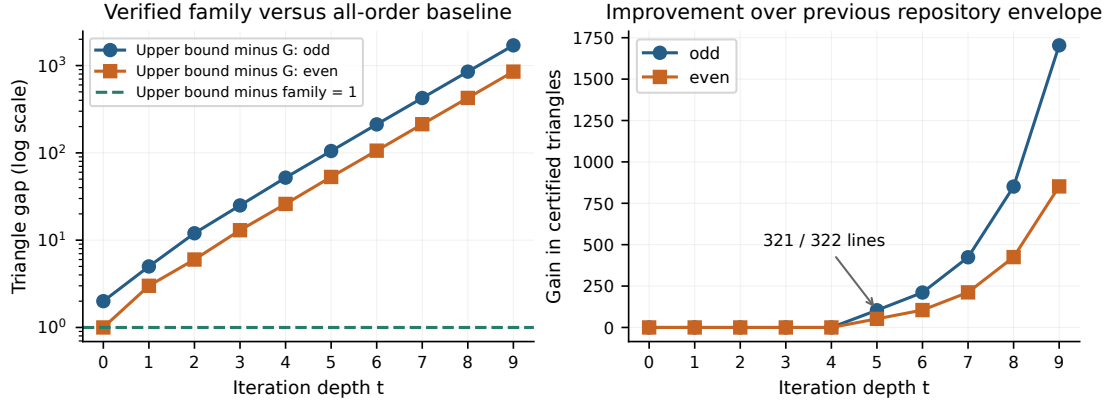
## 7.5 An improved verified bound at every natural order

Recall the previous retained bound  $L(n) = \max\{G(n), C(n)\}$  in (2). Define the sparse family contribution

$$R(n) = \begin{cases} (q_t^2 - 4)/3, & n = q_t + 1, \\ (q_t^2 - 4)/3 + q_t/2, & n = q_t + 2, \\ 0, & \text{otherwise,} \end{cases} \quad q_t = 10 \cdot 2^t, \quad (55)$$

and put

$$H(n) = \max\{L(n), R(n)\}. \quad (56)$$



**Figure 13:** The completed recursive family and the retained all-order bound. The improvement is over the previous repository envelope; it is not a numerical-priority comparison against all published constructions. The upper comparisons refer to simple arrangements. Formula values are proved in Lean; the plot is newly generated from those formulas.

The two cases have different parity and each determines  $t$  uniquely. The executable Lean definition uses the maximum of the candidates for  $0 \leq t \leq n$ , rather than an infinite search.

**Theorem 7.7** (Total retained and recursive envelope). *For every  $n \in \mathbb{N}$ ,  $K(n) \geq H(n) \geq L(n)$ . For every  $t \geq 5$ ,*

$$H(q_t + 1) - L(q_t + 1) = \lfloor q_t/3 \rfloor - 2, \quad (57)$$

$$H(q_t + 2) - L(q_t + 2) = q_t/2 - \lfloor q_t/3 \rfloor - 2, \quad (58)$$

and both differences are positive.

*Proof.* Each nonzero candidate in (55) has a simple witness by the preceding two theorems, hence an unrestricted witness. The finite maximum chooses one such witness or a retained finite/baseline witness; it is not an addition of incompatible triangle counts. If a candidate occurs at order  $n$ , then  $t < 2^t \leq n$ , so the executable finite maximum includes it. This is `all_n`.

Since  $q_t^2 \equiv 1 \pmod{3}$  and  $q_t$  is even, substitution in  $G$  gives

$$T_t - G(q_t + 1) = \lfloor q_t/3 \rfloor - 2, \quad T_t + q_t/2 - G(q_t + 2) = q_t/2 - \lfloor q_t/3 \rfloor - 2.$$

All saved finite enhancements end at order 195. Thus  $L(n) = G(n)$  for  $n > 195$ , as proved by `BBLFamilyBenchmarks.previous_envelope`. For  $t \geq 5$ ,  $q_t \geq 320$  and both differences are positive. The formal strict-tail statement is `RecursiveEnvelope.strict_previous_tail`; uniqueness of the sparse candidate identifies  $H$  with that candidate here. Q.E.D.

At 321 and 322 lines the gains over the old verified envelope are respectively 104 and 52; at 641 and 642 they are 211 and 105. The first members do not displace stronger retained certificates: 21 : 133, 22 : 143, 41 : 533 and 42 : 553 remain in  $H$ . All 138 coordinate identities and their published attribution are retained. This theorem improves the previous verified repository envelope at infinitely many orders. It does not prove a numerical record against all published families, and its off-family values retain the preceding all-order linear deficit.

## 7.6 Actual one-triangle optimality windows

The October release also closes the geometric upper-bound bridge for simple arrangements, as proved in Section 8.7. This upgrades a comparison with a polynomial to a statement about the actual extremal quantity.

**Corollary 7.8** (Simple optimality windows). *For  $q = 10 \cdot 2^t$  and  $t \geq 0$ , write  $A = (q^2 - 4)/3$ . Then*

$$A \leq K_s(q+1) \leq A+1, \quad A+q/2 \leq K_s(q+2) \leq A+q/2+1. \quad (59)$$

*Proof.* The lower halves are the constructed witnesses. For every simple arrangement, the geometric upper results give  $3T \leq n(n-2)$  and, for even  $n \geq 4$ ,  $6T \leq n(2n-5)$ . Substitution gives respectively

$$\lfloor (q+1)(q-1)/3 \rfloor = A+1, \quad \lfloor (q+2)(2q-1)/6 \rfloor = A+q/2+1.$$

The exact combined declarations are `odd_window` and `even_window`.

Q.E.D.

The upper halves use standard logical axioms only; the lower halves retain their seed's native dependencies. This one-triangle interval need not be sharp at the smaller orders: the literature settles some of them more precisely. In particular it is not an upper theorem for arrangements with multiple concurrence, and it does not close the unrestricted Kobon problem.

## 7.7 Known optimal families and independent reproduction

The classical dyadic five-line family must be kept separate from the new compatible eleven-line realization. A convenient normalized seed uses intercepts  $(-1, -\varepsilon, \varepsilon, 1)$  and

$$x - v_j y = a_j, \quad (v_0, v_1, v_2, v_3) = (-15, 10, -10, 30),$$

together with  $Y_0$ . The complete interval  $0 < \varepsilon \leq 10^{-5}$  has five certified triangles, all three distinguished triangles, and two visible pairs. Its central slopes are  $1/10$  and  $-1/10$ ; the rightmost slope is  $1/30$ . The normalization is deliberate: an earlier affine-equivalent numerical realization had both central slopes negative, which was inadequate as an unchecked premise for the sign convention in the published iterative proof. The corrected whole-parameter seed is verified in `TamuraSeed5`.

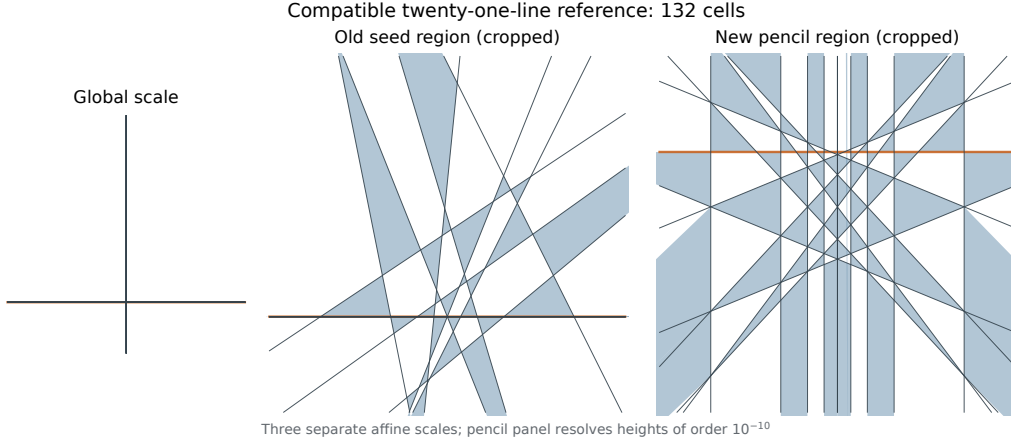
The known optimal odd family and its simple even companions have

$$q = 4 \cdot 2^t, \quad K_s(q+1) \geq \frac{q^2 - 1}{3}, \quad K_s(q+2) \geq \frac{q^2 - 1}{3} + \frac{q}{2}. \quad (60)$$

The current finite certificates include  $5 : 5$ ,  $6 : 7$ ,  $9 : 21$ ,  $10 : 25$ ,  $17 : 85$ ,  $18 : 93$ ,  $33 : 341$ ,  $34 : 357$ ,  $65 : 1365$ ,  $66 : 1397$ ,  $129 : 5461$ , and  $130 : 5525$ . These reproduce known numerical consequences of the affine family of Forge and Ramírez Alfonsín and BBL's odd-to-even discussion [1, 8]; they are not claimed as newly discovered bounds. Theorem 7.2 as currently formalized does not cover the small bases  $q = 4, 8, 16$ , and the unfinished large parametric 33-line draft is not counted as a completed theorem.

Parpalak and Utkin supply a different straight-line family at  $q = 18 \cdot 2^t$  [19]. Its odd count is

$$K_s(q+1) \geq \frac{q^2}{3} - 1, \quad (61)$$



**Figure 14:** A 21-line hybrid checkpoint with 132 triangles. The global, core and thin-pencil views illustrate scale separation. The finite rendering is distinct from the true-tangent-grid parametric theorem in `BBLSeed21`. The normalized seed now starts the fully verified recursion; the picture is not used as its proof.

including  $19 : 107$  and  $37 : 431$ . The present work independently checked their compatible 19-line seed on  $0 < \varepsilon \leq 10^{-3}$  with 1632 exact endpoint determinant checks, a rational 107-triangle witness, and all 17 distinguished triangles. The interval reproduction is separate from an end-to-end Lean proof of their infinite construction. Both the seed and the family retain Parpalak–Utkin attribution. The earlier BBL statement at  $18 \cdot 2^t + 1$  concerned pseudolines; it cannot be used as a straight-line result without a realization argument. Applying Corollary 6.6 to the published straight-line family gives the attributed simple even corollary

$$K_s(q+2) \geq \frac{q^2}{3} + \frac{q}{2} - 1,$$

with  $20 : 116$ ,  $38 : 449$ ,  $74 : 1763$ , and  $146 : 6983$ . Stronger nonsimple published values at some of these orders concern  $K$ , and do not invalidate this simple-arrangement distinction.

## 7.8 The former integration gap and the remaining seed branches

The verified 21-line base has the true tangent intercepts (40) with  $q = 20$ , fixed rational slopes, 132 triangles, 19 distinguished triangles, ten visible pairs, and the required rightmost-line property for every  $0 < \varepsilon \leq 10^{-5}$ . Its finite interval checks use native evaluation; their analytic and geometric soundness proofs are proved separately.

The September manuscript correctly identified four missing interfaces: retained geometric data, whole-list transport under the next-grid permutation, normalization of the 21-line grouped labels, and induction on a positive parameter interval. All four are now complete. The central source labels 5 and 6 have apex height  $5\varepsilon/2$ ; normalization preserves that actual point. The next central triangle is retained independently of whether it appears in the counted list. The invariant used in the successful induction is

$$\exists \eta > 0 \quad \forall \varepsilon \in (0, \eta) \quad \exists \text{ a compatible counted seed at } (q, \varepsilon). \quad (62)$$

The former target

$$q_t = 20 \cdot 2^t, \quad K_s(q_t + 1) \geq \frac{q_t^2 - 4}{3} \quad (63)$$

is therefore a theorem, the tail of (51). The visible-boundary interfaces are also closed by the same release.

This completion does not discharge every possible concrete seed. For the Forge–Ramírez Alfonsín branch, the active `BBLForgeConditional` proves the generic formulas

$$K_s(32 \cdot 2^t + 1) \geq \frac{1024 \cdot 4^t - 1}{3}, \quad K_s(32 \cdot 2^t + 2) \geq \frac{1024 \cdot 4^t - 1}{3} + 16 \cdot 2^t$$

under the explicit premises `UniformSeed 8 341` and its visible analogue. The large concrete 33-line uniform base and its dependent sources remain excluded drafts: the full native check did not finish. Finite certificates at 33, 34, 65, 66, 129 and 130 remain valid. A finite base at one parameter is not the uniform base premise above.

## 7.9 A uniform 49-line compatibility certificate

The October search found a stronger compatible seed at order 49 using fixed rational reciprocal slopes of denominator  $10^{10}$ . Its 48 intercepts are

$$-\tan \frac{23\pi}{48}, \dots, -\tan \frac{\pi}{48}, -\varepsilon, \varepsilon, \tan \frac{\pi}{48}, \dots, \tan \frac{23\pi}{48}.$$

Exact rational interval arithmetic verifies, throughout  $0 < \varepsilon \leq 1/100$ , a simple arrangement with 767 certified triangles, all 47 distinguished-line caps and a positive central apex. The fixed normal  $(1, -2)$  has 24 visible pairs, including rightmost pair  $(0, 48, 49)$ ; the last slope is  $1/3$  and the required boundary inequalities hold uniformly. These are exact external certificate claims, not completed finite Lean base theorems.

The analytic part is now inside Lean: `BBLTangent48Bounds` proves rational enclosures for each of the 23 positive tangent values, using half-angle and reciprocal identities beginning with  $\tan(\pi/3) = \sqrt{3}$ . The endpoint widths are at most  $5 \cdot 10^{-36}$ . The exported finite predicates were independently replayed: 1,176 direction checks, 18,424 simplicity checks, 37,583 triangle/support checks and 1,176 visibility/support checks passed. Two independent exact cell counters also find 767 triangles at the rational midpoint witness. None of these facts is a substitute for the unfinished Lean evaluation of the full parametric base.

If this uniform seed is promoted into the proved recursive interface, its counts will be

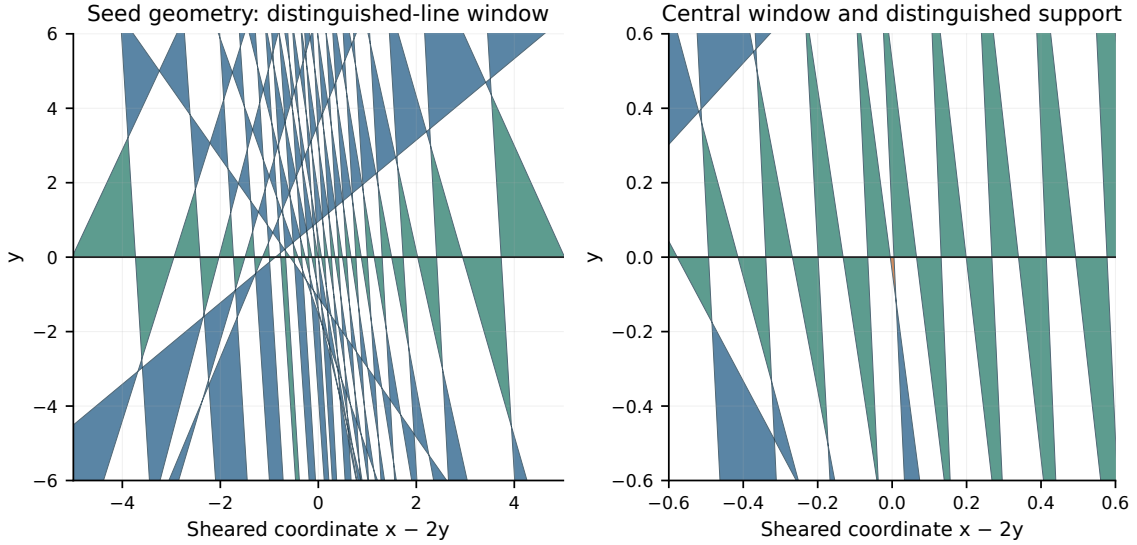
$$q = 48 \cdot 2^t, \quad T_{\text{odd}} = 768 \cdot 4^t - 1, \quad T_{\text{even}} = 768 \cdot 4^t + 24 \cdot 2^t - 1 \quad (64)$$

at orders  $q + 1$  and  $q + 2$ . The odd formula is exactly the tail  $s = t + 3$  of BBL’s Theorem 1.3 for  $n = 6 \cdot 2^s + 1$  [1]; these are not new numerical records. The contribution of this branch is an explicit, well-conditioned uniform compatibility certificate and a partially completed formal linkage. Five dependent seed/family files are visibly excluded as unverified drafts. Isolated tangent, graph and arithmetic proofs do not make the full base proof complete.

Earlier positive-margin fits had moved triangles across infinity. The successful fit additionally preserves the original affine sign system. This illustrates why incidence or a projective order type alone cannot certify the desired affine cell count.

A preceding search for a perfect compatible 21-line seed tested 11,547,910 floating proposals and 3,234 linear-feasibility programs without improving 132. Those bounded failures were not impossibility proofs. The exact obstruction in Section 9.5 now proves a much more precise limitation for a prescribed grid and supplied classification; it still does not exclude other compatible families or a different geometric construction.





**Figure 15:** Two clipped windows of the rational midpoint witness for the uniform 49-line seed, with  $\varepsilon = 1/200$ . The invertible shear  $x' = x - 2y$  makes the distinguished structure easier to see. The left window has  $x' \in [-5, 5]$ ,  $y \in [-6, 6]$ ; the right has both coordinates in  $[-0.6, 0.6]$ . Ordinary certified cells are blue, distinguished caps teal and the central cell orange. Distant cells lie outside these windows: the panels do not display all 767 triangles. The uniform certificate is exact external evidence; full Lean seed integration remains incomplete.

## 8 Multiplicity-sensitive upper-bound arguments

This section develops upper estimates by separating the ordinary part of an arrangement from its finite multiple points. The October release completes extraction of actual elementary segments, their incidence identity and clean-line charging for pairwise nonparallel arrangements. It also proves the classical simple upper bounds from actual witnesses. The remaining global cyclic-fan extraction is stated explicitly: conditional core-budget arithmetic is not an unrestricted Lean upper theorem. None of the restricted estimates below is claimed as a new numerical upper bound for the full function  $K$ .

The distinction is necessary because the even-order estimates of Bartholdi–Blanc–Loisel and Blanc require simplicity [1, 2]. The classical Clément–Bader envelope is treated as a reported external theorem [3]; the local example in Section 8.6 explains why we do not silently replace its proof by a bound on all shared edges incident to a multiple point.

### 8.1 An exact elementary-segment identity

**Definition 8.1.** For an arrangement of  $n$  distinct affine lines, let  $t_r$  be the number of finite intersection points incident to exactly  $r$  lines, and let  $P$  be the number of unordered parallel pairs. A bounded *elementary segment* is the portion of an arrangement line between two consecutive finite intersection points. Let  $E$  count these segments,  $U$  count those incident to no triangular cell, and  $D$  count those incident to two triangular cells. Let  $a$  be the number of lines containing at least one finite intersection point, and put

$$S = \sum_{r \geq 3} r(r-2)t_r, \quad \delta = n(n-2) - 3T,$$

where  $T$  is the number of bounded triangular cells.



**Proposition 8.2** (Segment and defect identities). *For every such arrangement,*

$$E = n(n-1) - 2P - S - a, \quad (65)$$

$$3T = E - U + D, \quad (66)$$

$$\delta = 2P + S + a - n + U - D. \quad (67)$$

*If every line meets another line, then  $a = n$  and  $\delta = 2P + S + U - D$ .*

*Proof.* Each nonparallel unordered pair contributes to exactly one finite point, so

$$n(n-1) - 2P = \sum_{r \geq 2} r(r-1)t_r.$$

If a line has  $s > 0$  distinct finite intersection points, it contributes  $s - 1$  bounded elementary segments; an isolated line contributes none. Summing point-line incidences gives

$$E = \sum_{r \geq 2} rt_r - a.$$

Subtracting these expressions yields the first identity. Each triangular cell uses three elementary segments. A segment is incident to zero, one, or two such cells, so the total number of triangle-side incidences is  $E - U + D$ . Substitution into the definition of  $\delta$  gives the last identity. Q.E.D.

The term  $D$  cannot be dropped in a nonsimple arrangement. Multiple points reduce the available segment count by  $S$ , but sharing can compensate for some of that loss. This is precisely the interaction that a classical upper proof must control.

**Lemma 8.3** (A shared side has a multiple endpoint). *An elementary segment incident to two bounded triangular cells cannot have both endpoints ordinary, where an ordinary point is incident to exactly two arrangement lines.*

*Proof.* At each ordinary endpoint there is a unique supporting line other than the line containing the segment. Both adjacent triangles would therefore have the same three supporting lines. Three distinct lines have at most one bounded triangular region, whereas the two incident cells lie on opposite sides of the shared segment. This is impossible. Q.E.D.

For the remainder of the section assume that  $n \geq 4$  is even and that all pairs of lines are nonparallel. Let the *core* consist of the finite points of multiplicity at least three. Write

$$q = \sum_{r \geq 3} t_r, \quad I = \sum_{r \geq 3} rt_r,$$

and let  $h$  count the lines meeting the core. Let  $D_1$  count shared elementary segments with exactly one core endpoint and  $D_2$  those with two core endpoints. By Lemma 8.3,  $D = D_1 + D_2$ , and hence

$$\delta = S + U - D_1 - D_2. \quad (68)$$

## 8.2 Parity on a line outside the core

A line is *clean* if it contains no core point. The following argument adapts the local parity method in Blanc's Section 2 [2]. Its proof is included because the transverse lines may have multiple intersections away from the clean line.

**Lemma 8.4** (Clean-line charging). *For even  $n \geq 4$  and pairwise nonparallel lines,*

$$n - h \leq 2U + D_1. \quad (69)$$

*Proof.* First fix a clean line  $L$ , place it horizontally, and number its  $n - 1$  crossings from left to right. Let  $R_i$  be the transverse line at crossing  $i$ . Denote the bounded segment of  $L$  between crossings  $i$  and  $i + 1$  by  $\ell_i$ , and its two exterior rays by  $\ell_0$  and  $\ell_{n-1}$ . Write  $p_i^+$  and  $p_i^-$  for the faces above and below  $\ell_i$ , and  $r_i^+, r_i^-$  for the elementary transverse pieces of  $R_i$  immediately above and below  $L$ . A transverse piece may be a bounded segment or an unbounded ray.

We show that some bounded  $r_i^\pm$  is unused or shared. Suppose otherwise: every bounded transverse piece is incident to exactly one triangle. Each bounded  $\ell_i$  is unshared by Lemma 8.3.

If both  $p_1^+$  and  $p_1^-$  are nontriangular, then  $r_1^+$  and  $r_1^-$  are unused: their other adjacent faces meet the exterior ray  $\ell_0$ . At least one of these transverse pieces is bounded, because  $R_1$  meets another transverse line. This contradicts the supposition. Reflecting across  $L$  if necessary, assume  $p_1^+$  is triangular;  $p_1^-$  is then not.

For  $1 \leq i \leq n - 2$ , prove the following assertions together:

- (i) if  $i$  is even,  $p_i^+$  is nontriangular; if  $i$  is odd,  $p_i^-$  is nontriangular;
- (ii) if both faces at  $\ell_{i-1}$  are nontriangular, then  $p_i^-$  is triangular for even  $i$ , and  $p_i^+$  is triangular for odd  $i$ .

The base case follows from the choice of  $p_1^+$  and the unbounded faces at  $\ell_0$ . Consider an even step; the odd step exchanges signs. If  $p_{i-1}^+$  is triangular, then  $p_i^+$  cannot be triangular, since otherwise the two triangles share the bounded piece  $r_i^+$ , contrary to the supposition. Assertion (ii) has a false antecedent in this case.

Otherwise both  $p_{i-1}^+$  and  $p_{i-1}^-$  are nontriangular by the preceding odd step. This cannot happen for  $i = 2$ , so  $i \geq 4$ . The earlier even step makes  $p_{i-2}^+$  nontriangular. Hence  $r_{i-1}^+$  is unused, and must be unbounded. The intersection of  $R_{i-1}$  and  $R_i$  must consequently lie below  $L$ . It follows that  $r_i^-$  is bounded. Its left adjacent face is nontriangular, so its right adjacent face  $p_i^-$  must be triangular. Since  $\ell_i$  is unshared,  $p_i^+$  is nontriangular. This proves both assertions.

Now  $n - 2$  is even, so  $p_{n-2}^+$  is nontriangular. If  $p_{n-2}^-$  is also nontriangular, both pieces at the last crossing are unused and at least one is bounded, again a contradiction. Otherwise  $p_1^+$  and  $p_{n-2}^-$  are triangular, while  $r_1^-$  and  $r_{n-1}^+$  are unused. The lines  $R_1$  and  $R_{n-1}$  intersect. If their intersection is below  $L$ , then  $r_1^-$  is bounded; if above, then  $r_{n-1}^+$  is bounded. This is the final contradiction.

Charge each clean line to one bounded unused or shared transverse segment supplied by this argument. An unused segment can receive at most two charges, one at each ordinary endpoint. A shared segment must have a multiple endpoint, so it can receive at most one charge if counted by  $D_1$ , and none if counted by  $D_2$ . There are  $n - h$  clean lines, giving (69). Q.E.D.

*Remark 8.5* (Why parallel pairs have been excluded). Three parallel vertical lines and one horizontal line give a clean horizontal line whose transverse pieces are all unbounded. Thus simply removing parallel-participating lines from the set to be charged does not prove the lemma. The identities of Proposition 8.2 allow parallels, but the charging argument above does not. Every subsequent upper estimate in this section retains the pairwise nonparallel hypothesis.

Combining (68) and (69) gives the basic budget

$$2\delta \geq n + 2S - h - 3D_1 - 2D_2. \quad (70)$$

The remaining problem is to bound the shared-side contribution at the core.

### 8.3 The geometric obstruction in a cyclic fan

At an  $r$ -fold point  $v$ , the incident lines determine  $2r$  rays in cyclic order, with opposite rays separated by  $r$  positions. Let  $d_1(v)$  count shared elementary segments starting at  $v$  and ending at an ordinary point, and let  $d_2(v)$  count those ending at another core point. A ray supports at most one such incident elementary segment.

**Lemma 8.6** (No long run of ordinary-ended shared rays). *For  $r \geq 3$ , no  $r - 1$  consecutive rays at an  $r$ -fold point can all support segments counted by  $d_1(v)$ .*

*Proof.* Such a run would border  $r$  consecutive triangular sectors. At the ordinary endpoint of each selected shared segment, the opposite supporting lines of the two neighboring triangles must coincide: besides the radial line through  $v$ , only one arrangement line passes through that endpoint. Propagating this equality makes the opposite sides of all  $r$  triangles lie on one line  $M$ .

The line  $M$  would meet  $r + 1$  consecutive rays from  $v$ , which include an opposite pair. A line meeting both open opposite rays must contain their connecting segment and hence  $v$ . But the opposite side of a nondegenerate triangle with vertex  $v$  cannot lie on a line through  $v$ . This contradiction proves the lemma. Q.E.D.

This is an actual geometric statement, not an assumed combinatorial prohibition. The Lean module `FanGeometry` proves uniqueness and propagation of opposite supports for real lines and points. The module `CyclicFan` supplies the cyclic-to-geometric bridge from triangular sectors, antipodal rays and ordinary selected endpoints.

**Proposition 8.7** (Local fan budget). *At every finite point of multiplicity  $r \geq 3$ ,*

$$d_1(v) \leq 2r - 3, \quad 3d_1(v) + d_2(v) \leq 6r - 6. \quad (71)$$

*Proof.* Count selected rays in all  $2r$  cyclic windows of length  $r - 1$ . Each window contains at most  $r - 2$  selected rays by Lemma 8.6, and each selected ray belongs to exactly  $r - 1$  windows. Therefore

$$(r - 1)d_1(v) \leq 2r(r - 2).$$

If  $d_1(v) \geq 2r - 2$ , the left side is at least  $2(r - 1)^2 = 2r(r - 2) + 2$ , a contradiction. Finally  $d_1(v) + d_2(v) \leq 2r$ , so

$$3d_1(v) + d_2(v) = 2d_1(v) + (d_1(v) + d_2(v)) \leq 6r - 6.$$

Q.E.D.

The cyclic double count is proved in `FanCount`. The theorem `CyclicFan.selected_card_le` combines it with the real geometric obstruction, without assuming the no-long-run conclusion as an unexplained premise.

### 8.4 A weighted bound for an arbitrary finite core

**Proposition 8.8** (Multiplicity-weighted defect estimate; manuscript geometry). *Let  $n \geq 4$  be even and let the arrangement be pairwise nonparallel. Then*

$$2\delta \geq n + \sum_{r \geq 3} (2r^2 - 11r + 6)t_r. \quad (72)$$

*Proof.* Each  $D_1$  segment has exactly one core endpoint and each  $D_2$  segment has two. Summing (71) gives

$$3D_1 + 2D_2 \leq 6I - 6q.$$

Also  $h \leq I$ , since every core line contributes at least one point-line incidence. Inserting both inequalities into (70) yields

$$2\delta \geq n + 2S - 7I + 6q,$$

which is (72) after expanding  $S, I, q$ .

Q.E.D.

The local weights are  $-9, -6, 1, 12, 27$  at multiplicities  $3, 4, 5, 6, 7$ , respectively. Consequently the obstruction to recovering a simple-style parity defect is concentrated at triple and quadruple points; higher multiplicity has a positive weight in this accounting.

**Corollary 8.9** (Nonnegative core weight; manuscript geometry). *Under the hypotheses of Proposition 8.8, if*

$$\sum_{r \geq 3} (2r^2 - 11r + 6)t_r \geq 0,$$

*then*

$$T \leq \left\lfloor \frac{n(n-5/2)}{3} \right\rfloor.$$

*In particular, this holds if every finite multiple point has multiplicity at least five.*

*Proof.* The weighted estimate implies  $\delta \geq n/2$ , giving the displayed upper bound by the definition of  $\delta$ . For  $r \geq 5$ , the exact factorization

$$2r^2 - 11r + 6 = 1 + (r-5)(2r-1) \geq 1$$

proves the stated sufficient condition.

Q.E.D.

This extends the same polynomial to a restricted nonsimple class. It says nothing comparable for a general core consisting of many triple points. Numerical first priority for this restricted consequence has not been established.

## 8.5 A sharper estimate with at most two core points

**Lemma 8.10** (A fan with at most one core-to-core ray). *If  $d_2(v) \leq 1$  at a point of multiplicity  $r \geq 3$ , then  $d_1(v) \leq 2r - 4$ .*

*Proof.* Mark which of the  $2r$  sectors at  $v$  are triangular. A ray supports a shared elementary segment exactly when both neighboring sectors are triangular. If every sector is triangular, every ray is shared; deleting at most one  $d_2$  ray leaves a run of at least  $2r - 1$  selected rays, contradicting Lemma 8.6. Hence a nontriangular sector exists.

Suppose  $d_1(v) \geq 2r - 3$ . Two separated nontriangular sectors exclude at least four shared rays, as do three consecutive nontriangular sectors. Both situations are impossible. If exactly one sector is nontriangular, the shared rays form a path of length  $2r - 2$ . Deleting at most one  $d_2$  ray leaves at most two selected runs. Their total capacity is  $2(r - 2)$ , less than the required  $2r - 3$ . If exactly two adjacent sectors are nontriangular, at most  $2r - 3$  shared rays remain. All would have to be selected, forming a run of length  $2r - 3 > r - 2$ . These cases exhaust the possibilities and prove the lemma.

Q.E.D.

**Theorem 8.11** (At most two finite multiple points; manuscript geometry). *An arrangement of an even number  $n \geq 4$  of pairwise nonparallel affine lines with at most two finite multiple points satisfies*

$$T \leq \left\lfloor \frac{n(n-5/2)}{3} \right\rfloor + 1. \quad (73)$$

More precisely, the defect bounds are  $\delta \geq n/2$  if  $q = 0$ ,  $\delta \geq n/2 - 1$  if  $q = 1$ , and  $\delta \geq n/2 - 3$  if  $q \leq 2$ .

*Proof.* With at most two core points, at most one ray at a core point can terminate at another core point. Lemma 8.10 gives  $D_1 \leq 2I - 4q$ . There is at most one core-to-core shared segment, so  $D_2 \leq q(q-1)/2 \leq 1$ . Moreover,

$$h \leq I - D_2,$$

because any such segment lies on a line counted twice in the core-incidence total and once among the core lines. Substitution in (70) yields

$$2\delta \geq n + \sum_{r \geq 3} (2r^2 - 11r + 12)t_r - D_2.$$

For  $r \geq 3$ ,

$$2r^2 - 11r + 12 = -3 + (r-3)(2r-5) \geq -3.$$

Thus  $2\delta \geq n - 3q - D_2 \geq n - 7$ . Since  $n$  and  $2\delta$  are even,  $2\delta \geq n - 6$ . Rearranging gives (73). If  $q = 1$ , then  $D_2 = 0$  and the inequality  $2\delta \geq n - 3$  rounds to  $2\delta \geq n - 2$ ; if  $q = 0$ , it gives  $2\delta \geq n$  directly. Q.E.D.

The hypotheses are attained by externally attributed exact constructions. Table 7 records examples whose pairwise nonparallel status, triangle count, and two triple points were checked exactly. Their equality with (73) proves sharpness within this restricted class, not optimality over all classical arrangements.

| $n$ | $T$ | Restricted upper | Coordinate provenance                                 |
|-----|-----|------------------|-------------------------------------------------------|
| 8   | 15  | 15               | Earlier construction, retained in the current gallery |
| 14  | 54  | 54               | Maiorana, two-triple-point witnesses                  |
| 26  | 204 | 204              | Parpalak–Utkin gallery                                |
| 50  | 792 | 792              | Parpalak–Utkin gallery                                |

**Table 7:** Attainment in the class covered by Theorem 8.11. Sources: Maiorana [17], Parpalak and Utkin [22]. These are not new numerical records claimed by this paper.

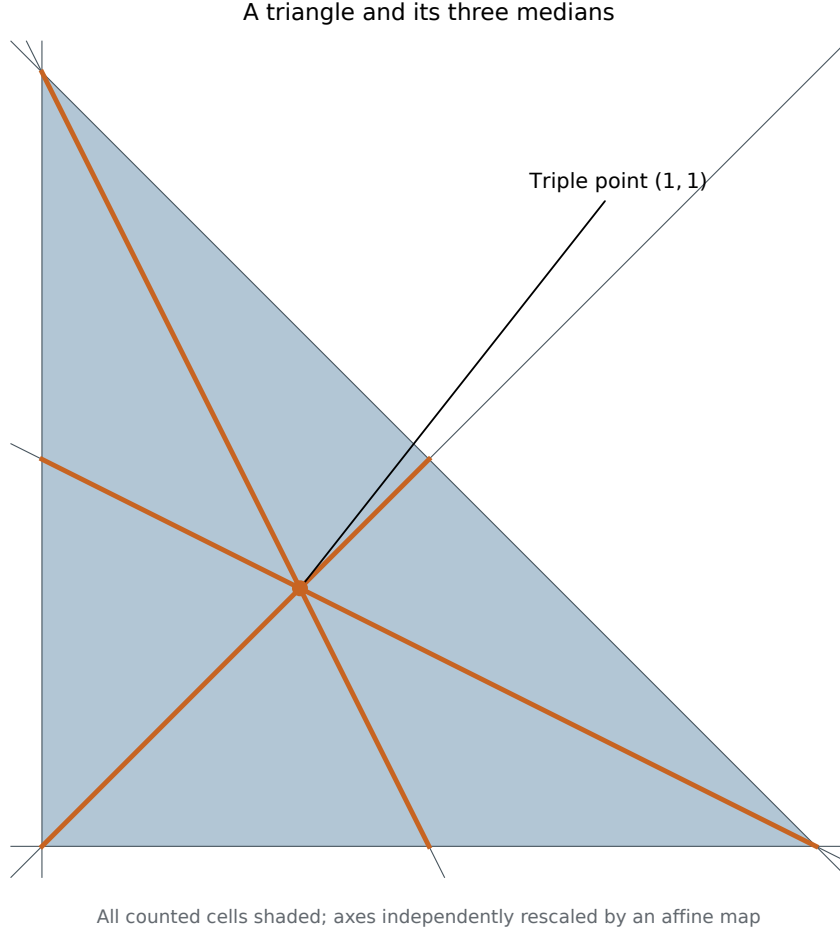
ProofAtlas reports a stronger restriction-specific result at  $n = 14$ , covering up to four finite multiple points [23]. Its proof was not independently reproduced here. Our two-point estimate covers all even orders and arbitrary core multiplicities, but neither that fact nor the table establishes priority for the restricted theorem.

## 8.6 A checked shared-fan example and the limits of smoothing

Consider the six lines

$$x = 0, \quad y = 0, \quad x + y = 3, \quad x - y = 0, \quad 2x + y = 3, \quad x + 2y = 3. \quad (74)$$

They are the three sides of a triangle and its three medians. There are six bounded triangular cells and four triple points. At  $v = (1, 1)$ , six distinct nonzero elementary radial segments are



**Figure 16:** The exact arrangement (74). Six triangular cells meet around the central triple point, and all six radial sides are shared. This refutes only a literal local incidence reading of the shared-pair statement in Clément–Bader’s Lemma 1; it does not refute their final upper theorem. The figure is newly generated from the integer equations.

shared sides, each belonging to two of those cells. Three have ordinary other endpoints and three have multiple other endpoints, giving  $d_1(v) = d_2(v) = 3$ ; globally  $D_1 = D_2 = 3$ .

The **SharedFan** module proves the real geometric facts: each designated triangle has nonempty uncut interior, the six interiors are pairwise disjoint, the radial segments are distinct and elementary, and each is a side of two distinct triangles. The proof uses ordinary kernel checking.

This matters when interpreting the Clément–Bader proof. Its local shared-pair sentence cannot literally bound all shared segments incident to a multiple point by two. An intended assignment of each shared side to one endpoint would be a different assertion requiring global justification. The example has only six triangles, below the reported seven-triangle upper bound at order six; it is therefore not a counterexample to that upper theorem. The present paper leaves that distinction explicit and uses the separate  $D_1, D_2$  accounting instead.

Nor can one justify a simple upper bound merely by perturbing every multiple point. For a retained two-triple-point Maiorana 14:54 witness, the four local generic resolution types have triangle counts 52, 52, 53, 52. These counts were checked with exact arithmetic; they are finite experimental evidence, not a new Lean theorem classifying all perturbations. They demonstrate why a universally nondecreasing smoothing argument needs more than the existence of a small generic perturbation. Our global upper argument never assumes such monotonicity.

## 8.7 Actual arrangement extraction and simple upper theorems

The formal inventory is now built from an actual finite injective family of certified triangles in  $n \geq 2$  pairwise nonparallel lines. It may be a proper subset of all triangular cells. The quantities  $U, D_1, D_2$  are defined relative to this selected family. The cell theorem proves interior disjointness, and the side-emptiness theorem proves that each side has consecutive endpoints on its supporting line.

An unordered endpoint pair represents one geometric side. At most two pairwise interior-disjoint nondegenerate triangles can share that side: three would put two on the same side of its support, forcing their interiors to overlap near an interior base point. Counting triangle-side incidences therefore gives  $3T = E - U + D_1 + D_2$ . Pair counting proves  $\sum r(r-1)t_r = n(n-1)$  and the actual consecutive-edge inventory gives  $E = n(n-2) - S$ . Thus

$$\delta = S + U - D_1 - D_2 \quad (75)$$

is proved from geometry by `certificate_defect_identity`; capacity, incidence and edge cardinality are conclusions of the proof, not assumed numerical identities.

**Theorem 8.12** (Actual clean-line charging and simple bounds). *Let  $n \geq 4$  be even. For any pairwise nonparallel real arrangement and finite injective certified triangle family,*

$$n - h \leq 2U + D_1. \quad (76)$$

*For every simple arrangement with  $T$  certified triangular cells,*

$$3T \leq n(n-2) \quad (n \geq 2), \quad 6T \leq n(2n-5) \quad (n \geq 4 \text{ even}). \quad (77)$$

*Proof.* The parity proof of Lemma 8.4 can also be organized as an odd-cardinality matching contradiction, which is the formal route. A clean line has exactly  $n-1$  distinct crossings. One common open half-plane contains a further crossing on every transverse support. Choose in that half-plane the consecutive bounded segment incident to each clean-line crossing. Existence is obtained from the nearest actual intersection, with the bounded-side and uniqueness lemmas handling both signs.

If every chosen edge had degree one in the certified triangle family, its unique triangle would have its base on the clean line: at an ordinary crossing a triangle must use one of the two incident supports. Consecutive edge uniqueness and interior disjointness then make these bases a disjoint two-element cover of the clean-line crossings. Such a cover has even cardinality, contradicting  $n-1$  odd. Hence some incident bounded edge has degree zero or two. A degree-two edge has a multiple other endpoint by the shared-side lemma, so it is counted by  $D_1$ .

Choose one resulting charge for each clean line. A fixed ordinary endpoint of an edge determines at most one clean transverse line. An unused edge has at most two such endpoints and receives at most two charges; a  $D_1$  edge has exactly one ordinary endpoint and receives at most one. This proves (76). The extraction, choice and fiber bounds are all in `certificate_clean_line_budget`, using `UpperCleanHalfplane`, `UpperCleanEdges`, `UpperCleanPairing` and `UpperCleanCover`. No charge-existence or pairing premise remains.

For a simple arrangement the core is empty, so  $S = D_1 = D_2 = h = 0$ . Equation (75) gives  $3T = n(n-2) - U$ , proving the first bound. In the even case  $n \leq 2U$ , so  $6T = 2n(n-2) - 2U \leq n(2n-5)$ . The `SimpleLowerBound` versions extract the injective finite family from its duplicate-free triangle list. These are `simple_lower_bound_upper` and `simple_lower_bound_even_upper`. Q.E.D.



The integer floor forms follow immediately. These classical inequalities retain their earlier attribution, including Blanc’s even parity bound. Their contribution here is a complete real-geometric formalization with standard logical axioms only. In particular, the retained simple 44 : 608 certificate attains  $\lfloor 44(88 - 5)/6 \rfloor = 608$ ; this proves simple optimality at that order, not unrestricted optimality. Theorem 8.12 does not justify applying the even polynomial to nonsimple witnesses such as 14 : 54.

## 8.8 Extremal fans and strengthened local restrictions

**Theorem 8.13** (Equality structure of a supplied cyclic fan). *Consider actual local sector data at an  $r$ -fold point,  $r \geq 3$ , satisfying the cyclic order, antipodality, triangle and ordinary-endpoint conditions of `UpperFan.Sectors`. If  $d_1 = 2r - 3$ , every sector is triangular and  $d_2 = 3$ . Consequently,*

$$d_2 \leq 2 \implies d_1 \leq 2r - 4, \quad 3d_1 + d_2 \leq 6r - 8 + 2e, \quad (78)$$

where  $e = 1$  when  $d_1 = 2r - 3$  and  $e = 0$  otherwise.

*Proof.* The no-long-run property from Lemma 8.6 holds for the selected ordinary-ended shared rays. If a sector were missing, both bounding rays would be absent from that selected set. The cyclic window count with these adjacent omissions strengthens its cardinality bound to  $2r - 4$ . Thus the extremal case has every sector present; all  $2r$  rays are then shared. Subtracting  $d_1 = 2r - 3$  gives  $d_2 = 3$ . If  $d_2 \leq 2$ , equality in the old bound is therefore impossible. In the nonextremal case,  $3d_1 + d_2 = 2d_1 + (d_1 + d_2) \leq 2(2r - 4) + 2r$ ; in the extremal case its value is  $6r - 6$ . These give (78). The exact proofs are `all_sectors_of_extremal`, `core_shared_card_eq_three_of_extremal` and `weighted_local_bound`. Q.E.D.

This strengthens the previous manuscript’s local hypothesis  $d_2 \leq 1$  to  $d_2 \leq 2$ . The theorem’s supplied local interface must still be extracted from every core in a whole arrangement before summation. Two further geometric restrictions are proved with their interfaces explicit. A supported extreme core cannot carry the full extremal fan when its core-ended rays are identified with points in the supporting closed half-plane (`UpperFanSupport` and `UpperCoreBoundary`). Also, two full extremal triple fans cannot be adjacent across the same matched elementary side with its two neighboring certified triangles (`UpperTripleFan`). The latter uses the actual coincident endpoints and opposing adjacent sectors; an arbitrary pair of abstract fan records need not supply that matching.

## 8.9 Core budgets: completed inputs and the precise conditional step

Actual extraction now proves

$$D_2 + h \leq I, \quad D_2 \leq \binom{q}{2}, \quad 2S - 7I + 15q \geq 0. \quad (79)$$

The first inequality counts consecutive core vertices on each support; the second maps a core-to-core elementary edge injectively to its two endpoints; the last sums  $2r(r - 2) - 7r + 15 = (r - 3)(2r - 5) \geq 0$  over core multiplicities  $r \geq 3$ . They are proved in `UpperCoreLineIncidence` and `UpperCoreCombinatorics`.

If a globally extracted family supplies the local bounds in (78), summation gives  $3D_1 + 2D_2 \leq 6I - 8q + 2e$ , where  $e$  counts exceptional full fans. Combining this with the actual defect and charging identities proves

$$2\delta \geq n + 2S - 7I + 8q - 2e + D_2. \quad (80)$$



This is `weighted_defect_with_core_edges`. The fan-summation premise is explicit; the equation is not yet a general Lean theorem on every nonsimple arrangement.

For at most three core points each core has at most two possible core neighbors. Once the missing ray/edge identification supplies  $D_1 \leq 2I - 4q$ , the actual inputs above and  $D_2 \leq 3$  imply, at  $n = 2m$ ,

$$\delta \geq m - 6, \quad T \leq \lfloor m(4m - 5)/3 \rfloor + 2. \quad (81)$$

For completeness, the algebra is short. Twice the defect and charging give  $2\delta \geq 2m + 2S - h - 3D_1 - 2D_2$ . Substitute  $h \leq I - D_2$  and  $D_1 \leq 2I - 4q$  to get  $2\delta \geq 2m + 2S - 7I + 12q - D_2 \geq 2m - 3q - D_2 \geq 2m - 12$ . Substitute  $\delta = 2m(2m - 2) - 3T$  and take the integer floor. `at_most_three_multiple_points` checks these inequalities while retaining the needed fan premise.

## 8.10 Formal status and the remaining global bridge

**Table 8:** Current upper-proof boundary. All completed sources use standard logical axioms; conditional implications retain the displayed premises.

| Component                      | Status in the October release                                                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Triangle-side geometry         | Actual elementary sides, at-most-two capacity and certified-cell disjointness completed.                                     |
| Edge inventory and core        | Actual multiplicity identity, consecutive-edge cardinality, shared endpoint classification and defect identity completed.    |
| Clean-line charging            | Actual bounded charge existence, odd-crossing parity contradiction, choice and finite charge fibers completed.               |
| Simple upper bounds            | Both classical polynomial inequalities and their floor forms completed from actual simple witnesses.                         |
| Local fan structure            | No long run, equality structure, supported-core and matched triple-fan obstructions completed for explicit local interfaces. |
| Core incidence                 | Actual $D_2 + h \leq I$ , $D_2 \leq \binom{q}{2}$ and nonnegative multiplicity surplus completed.                            |
| Global fan extraction          | Canonical cyclic ray order, actual multiplicity, endpoint/edge correspondence, and shared-triangle matching remain open.     |
| Restricted core upper formulas | Ordinary geometric arguments or conditional Lean arithmetic; no unqualified whole-arrangement theorem is claimed.            |
| Parallel classes               | General elementary accounting is included on paper; formal charging retains the nonparallel hypothesis.                      |

The older `CleanLineBudget` contains implications with symbolic parallel-pair terms. These do not prove the missing parallel charging lemma. Likewise, the exact checks on 597 test arrangements and the imported gallery/ Maiorana witnesses support and debug the accounting, but do not replace its geometric proofs. Completing global fan extraction would certify the restricted conclusions with their intended hypotheses. A new unrestricted numerical upper bound requires further mathematical control of the entire low-multiplicity core.

## 9 Exact certificates, external reproduction, and unsuccessful searches

The experimental program serves two purposes: producing concrete coordinate witnesses for lower bounds, and identifying which proposed general arguments survive exact checks. These purposes require different standards of evidence. A successful coordinate witness proves an existential lower bound at its specified order. A failed search ordinarily proves neither an upper bound nor the nonexistence of a better construction. This section records both positive and negative outcomes without merging their logical scopes.

### 9.1 From a numerical proposal to an exact geometric witness

The stored coordinates use the convention  $a_ix + b_iy = c_i$ , with rational or integer coefficients. Each line is normalized to a primitive integer triple. Pair intersections are computed as homogeneous integer triples, normalized with a positive final coordinate. Consequently, collinearity, concurrence, parallelism, and comparisons along a supporting line can all be tested without floating-point tolerances.

Two independently implemented counters are used. The first sorts the distinct intersection vertices on each line and constructs the graph of elementary bounded segments; a three-cycle supported by three different lines is a triangular cell. The second enumerates support triples and tests whether every other line has a common weak sign at their three vertices. Nonzero area and the distinction between open interiors and boundaries are checked explicitly. Where a complete independent verification is reported, both counters agree on the triangle set, not merely on a rounded area or a visual estimate. The Lean geometric soundness theorem provides a third, logically separate link from accepted coordinate predicates to real affine triangles.

Numerical optimization is used only to propose coordinates or rank candidate chambers. A proposal is rationalized and recounted before promotion. In the phase/chart experiments, progressively finer integer scales were accepted only when the entire exact triangle set and simplicity were retained. Such a finite acceptance does not prove that an arbitrary rounding of an irrational construction is valid. The real all-order theorem for  $G$  has its own determinant, trigonometric, and counting proof.

The released catalog contains 138 distinct promoted coordinate identities. Several identities can occur at the same order and triangle count, and an identity may have several repository source paths. Thus 138 is a catalog size, not a count of new records or independent discoveries. The complete catalog in Table 13 includes simple versus nonsimple status, provenance collections, and coordinate-hash prefixes; its machine-readable companion records full hashes, source paths, and theorem modules. The finite Lean certificates use native evaluation where indicated. This trust boundary is broader than that of the universal construction, whose proof uses only Lean’s standard logical axioms.

### 9.2 Finite improvements relative to the retained collection

The phase and affine-chart work raised the preceding saved values at eleven orders:

| $n$      | 39  | 47  | 48  | 51  | 53  | 54  | 55  | 59   | 60   | 99   | 195   |
|----------|-----|-----|-----|-----|-----|-----|-----|------|------|------|-------|
| previous | 469 | 690 | 720 | 817 | 884 | 918 | 954 | 1102 | 1140 | 3169 | 12481 |
| current  | 470 | 691 | 721 | 818 | 885 | 919 | 955 | 1103 | 1141 | 3170 | 12482 |

Each row is an improvement over the project’s preceding retained coordinate catalog. The underlying trigonometric family is due to Füredi–Palásti [10]; these comparisons do not establish

numerical first priority. The general formula  $G$  now explains all eleven displayed counts. The first projective search reached 47 : 691 after 11,264 proposals; the subsequent exposed-cap description replaced that search by a direct construction and an all-order proof.

The earlier OEIS-attributed values 28 : 238, 30 : 275, and 34 : 357 remain independently certified [18, 30]. The drawings in this paper render the currently retained exact witnesses, obtained by exterior extension of retained odd-order inputs. They are not asserted to be literal reproductions of the March diagrams. Likewise, the historical inventory in Table 14 preserves earlier outputs without upgrading every original derivation to a completed general theorem.

The current strongest saved simple and classical coordinate counts for every  $3 \leq n \leq 60$  appear in Table 12. An absent entry in the explicit OEIS table is recorded as a dash. Absence from that table is not evidence that a value is absent from the literature; conversely, a coordinate certificate does not establish optimality without an upper theorem covering the same arrangement class.

### 9.3 Independent reproduction of external constructions

The current Parpalak–Utkin coordinate gallery [19] supplied exact witnesses at

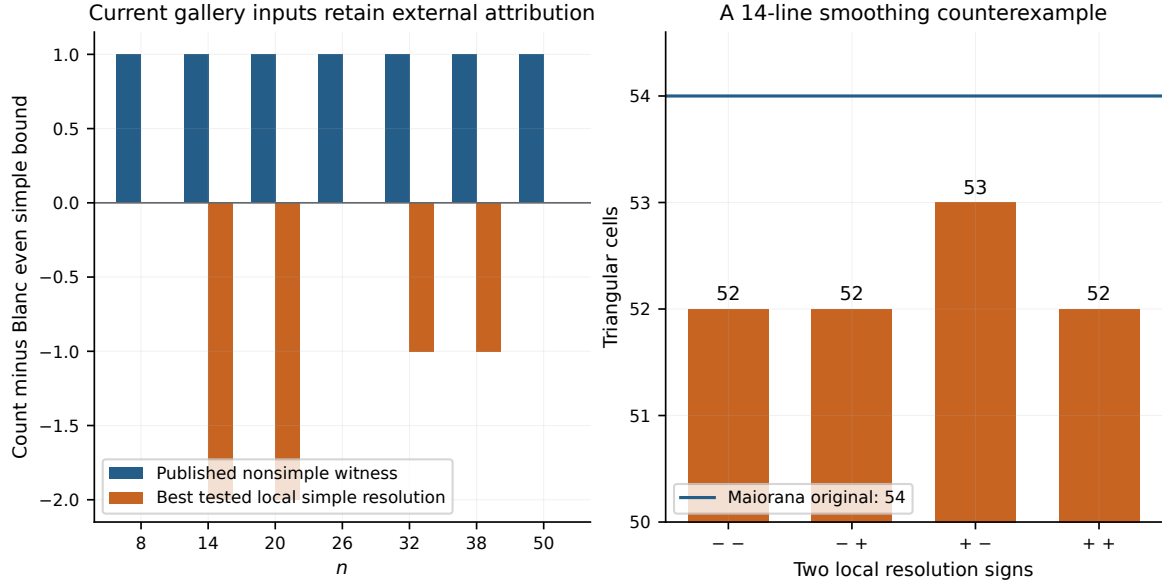
$$\begin{array}{cccccc} 8 : 15, & 14 : 54, & 20 : 117, & 26 : 204, & 32 : 315, \\ 36 : 402, & 38 : 450, & 42 : 553, & 46 : 667, & 50 : 792. \end{array}$$

Both independent counters reproduced their triangle sets. These witnesses retain their source attribution; in particular, the eight-line numerical value has older discovery priority than the coordinate gallery. Their addition to this repository improves its verified coverage, not the discoverer’s claim to the numbers. The inspected gallery snapshot is commit `feae4f5571a988a3ed45bd26a36259ebb7d7d3d2`.

All fifteen of Andrea Maiorana’s published 14 : 54 coordinate configurations [17] were also independently checked. Fourteen have exactly two triple points, and one has four; none has a parallel pair. The source snapshot is commit `e47c7cfd9661e54d29befb83118bf83d5f804028`. Its CC BY 4.0 licence, original coordinate files, and source attribution are retained. Added wrappers record counts and verification metadata. This reproduction imports no claim that a simple-arrangement upper theorem applies to these nonsimple witnesses.

These examples also expose a concrete limitation of smoothing. In Maiorana’s first example, the triple supports are  $\{0, 4, 10\}$  and  $\{0, 11, 12\}$ . Independently shifting the offsets of lines 4 and 11 by  $\pm 1/1000$  realizes all four local resolution choices while preserving every originally nonzero determinant sign. The resulting triangle counts are 52, 52, 53, 52, all below 54. Thus the proposed nondecreasing local smoothing strategy fails even when the local sign choices are independently realizable by straight lines. The argument identifying these sign choices with all sufficiently small local resolutions is an ordinary finite sign-stability argument, not a completed Lean theorem.

For the gallery’s collinear-core examples at orders 8, 14, 20, 26, 32, 38, 50, all 132 recorded local resolutions were realized by exact private-line offsets. Their respective best simple counts were 14, 51, 114, 203, 313, 448, 791. In particular, a universal claim that resolving such a core loses at most one triangle is also false. Figure 17 separates these finite obstructions from any general upper-bound assertion.



**Figure 17:** Exact local-resolution tests on attributed external constructions. Left: Parpalak–Utkin gallery witnesses and their best tested simple local resolutions, measured relative to Blanc’s even simple-arrangement polynomial; the eight-line value has older priority. Right: all four local resolutions of Maiorana’s two-triple-point 14 : 54 example. The original arrangements are nonsimple, so exceeding the simple polynomial is consistent. These are smoothing obstructions, not upper bounds for the classical problem.

#### 9.4 Projective charts, algebraic curves, and chamber searches

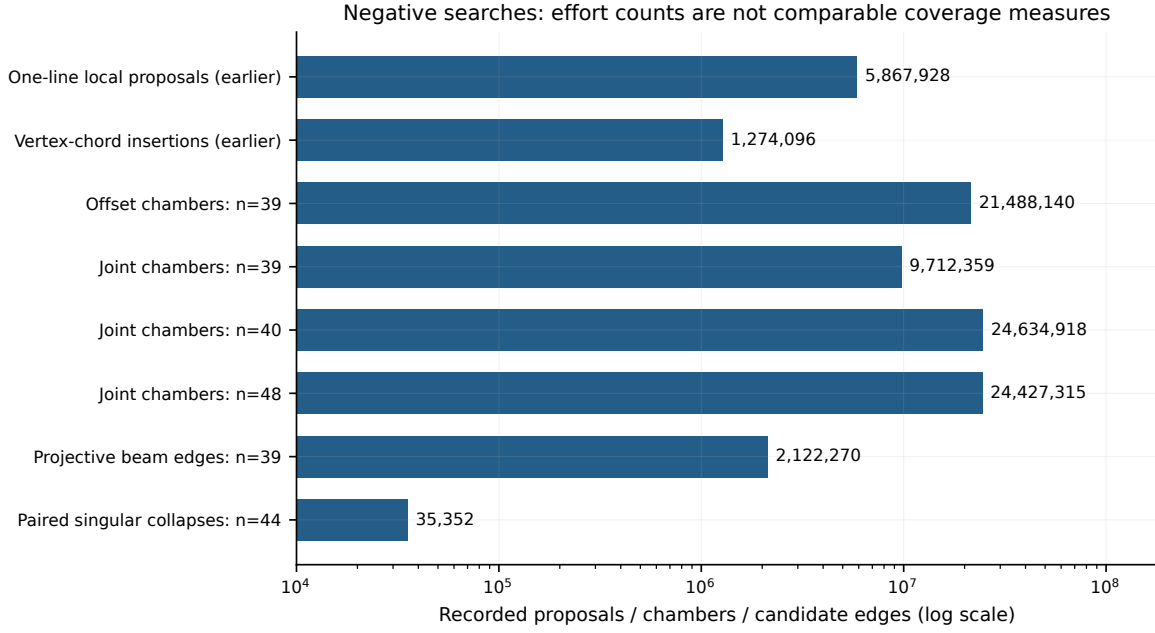
Projective geometry provides a useful global neighborhood: changing the line at infinity moves all lines together and may bound formerly unbounded triangular faces. It can also lose bounded cells. For the exact saved 81 : 2132 and 161 : 8532 hybrid arrangements, the projective triangular-face census already equals their bounded-cell count; a chart change alone cannot increase those particular counts.

Exact rational linear-arithmetic searches give sharper restrictions for some fixed configurations. The nine-line half-phase rational arrangement has 21 projective triangular faces, but the recorded QF\_LRA solver result rules out a chart retaining all 21. Similar fixed-seed tests find no gain for the tested half-phase arrangements at 14, 20, 26, 44, and no chart retaining all 471 projective triangles of the saved 39 : 470 witness. For a nonuniform cubic 48-line seed with 724 projective triangles, 2,903 exact linear-arithmetic checks rule out 722 bounded triangles within the tested chart-retention formulation. These are exact-solver results about specified rational seeds. Their UNSAT certificates have not been independently checked in Lean, and they are not global upper bounds at those orders.

The algebraic-curve direction tested both nodal-cubic angle perturbations and torsion cosets on real elliptic curves. A larger projective face count did not automatically produce a better affine construction. For the better two-real-component elliptic examples, the projective, original affine, and best found affine counts were respectively

| $n$               | 42  | 48  | 54  | 60   |
|-------------------|-----|-----|-----|------|
| projective        | 550 | 724 | 922 | 1144 |
| original affine   | 538 | 710 | 907 | 1127 |
| best found affine | 539 | 712 | 908 | 1129 |

All fall below the corresponding retained affine bounds. The chart scans at 54 and 60 had 90-



**Figure 18:** Selected unsuccessful searches, with their recorded units. Proposal, chamber, and candidate-edge totals describe different search neighborhoods. They do not measure exhaustive coverage, justify an upper bound, or rank the mathematical promise of the methods. All proposed improvements would require subsequent exact geometric verification.

second limits. Numerical ranking and bounded running time preclude a global nonexistence inference.

Oriented-matroid mutations and kinetic chamber sweeps addressed a different limitation: moving inside a fixed projective type cannot create new projective triangular faces. The mutation searches changed determinant signs, using linear-programming realization where appropriate. Kinetic sweeps varied many offsets, or both slopes and offsets, and visited concurrency events along selected rays. Their event counts are not counts of all realizable order types, and unsuccessful floating rays were discarded. Figure 18 records effort without treating heterogeneous search units as comparable coverage measures.

The final 39-line projective beam examined 2,122,270 candidate edges through 4,559 states without exceeding 470 affine or 471 projective triangles, even at its combinatorial scoring stage. The 48-line LP walk accepted 471 mutations out of 511 attempts and retained 721 affine and 724 projective triangles. At 44 lines, singular-boundary searches included 35,352 paired collapses; the retained beam checkpoint had 771 states and 446,422 candidate edges, without exceeding 608. An unfinished round was discarded. These results motivate a change of neighborhood or a structural argument rather than repeating the same local searches.

Other bounded approaches included Fourier deformations (2,687,600 proposals in one 39-line run), partial deletions from BBL constructions, and constrained tangent-grid transfer by LP/MILP. The tested optimal 23-line chart transfers did not admit a positive-margin realization on the desired grid; a constrained near-optimal search gave 144 triangles, below a separately known compatible 145. None of these observations proves nonexistence of another compatible seed or a different successful deletion pattern.

## 9.5 A proved algebraic obstruction to tangent-grid transfer

The October search connects oriented-matroid sign constraints, linear programming duality and exact algebraic number arithmetic. For the reciprocal-coordinate lines  $x - v_i y = a_i$ , a prescribed determinant sign becomes a strict linear inequality in the unknown reciprocal slopes. Write

$$D_{ijk} = (a_j - a_k)v_i + (a_k - a_i)v_j + (a_i - a_j)v_k. \quad (82)$$

Once the intercepts are fixed, feasibility is a linear sign problem, even though the intercepts themselves are algebraic.

**Lemma 9.1** (Positive-dependence obstruction). *Let  $A_i \in \mathbb{R}^d$  and let  $\lambda_i \geq 0$ , with at least one  $\lambda_i > 0$ , satisfy  $\sum_i \lambda_i A_i = 0$ . There is no  $v \in \mathbb{R}^d$  with  $A_i \cdot v > 0$  for every  $i$ .*

*Proof.* Taking the scalar product with  $v$  makes the left side a positive weighted sum, but the assumed vector identity makes it zero. This classical strict alternative is formalized by `homogeneous_infeasible`. The formal theorem also provides an affine form for nonhomogeneous strict systems. Q.E.D.

Set  $t = \tan(\pi/20)$ . Its exact algebraic specification is

$$p(t) = t^4 - 4t^3 - 14t^2 - 4t + 1 = 0, \quad \frac{3}{20} < t < \frac{17}{100}. \quad (83)$$

The quintuple-angle identity with  $\tan(5\pi/20) = 1$  gives  $(t - 1)p(t) = 0$ ; the interval excludes  $t = 1$ . Monotonicity on the interval identifies the intended root. `BBLTangentAlgebra` proves the quartic, root interval and uniqueness; `BBLTangentValues` proves that its explicit cubic power-basis expressions are the actual nine values  $\tan(k\pi/20)$  for  $1 \leq k \leq 9$ . Thus replacing trigonometric numbers by algebraic expressions does not insert a numerical guess.

Use the actual 20-intercept grid

$$a_j = \begin{cases} -\tan((9-j)\pi/20), & 0 \leq j < 9, \\ -\varepsilon, & j = 9, \\ \varepsilon, & j = 10, \\ \tan((j-10)\pi/20), & 11 \leq j < 20. \end{cases}$$

The following ten prescribed signs are impossible.

**Theorem 9.2** (One exact tangent-grid obstruction). *For every real  $\varepsilon$  and every real vector  $(v_0, \dots, v_{19})$ , at least one row of Table 9 fails the strict inequality  $s_i D_{j_i k_i \ell_i} > 0$ .*

*Proof.* For each row define  $A_i$  to be the coefficient vector of  $s_i D$  in (82). All ten displayed multipliers are positive throughout  $3/20 < t < 17/100$ , by rational polynomial inequalities. Using the proved power-basis expressions for the tangents, expand  $\sum_i \lambda_i(t) A_i$  one column at a time and reduce modulo  $p(t)$ . Every column is zero. The two central columns vanish because none of the ten supports uses indices 9 or 10. Lemma 9.1 gives a contradiction. The exact source contains the polynomial multiple of  $p$  for every nontrivial column, so the computation is reproducible without a floating-point linear solver. The declarations `actual_grid_infeasible` and `actual_grid_orientation_impossible` prove this for the actual tangent grid using only standard logical axioms. Q.E.D.

| Support triple | $s_i$ | Positive multiplier $\lambda_i(t)$ |
|----------------|-------|------------------------------------|
| (0, 6, 15)     | -1    | $-20t^3 + 120t^2 + 300t - 40$      |
| (0, 7, 13)     | +1    | $-8t^3 + 24t^2 + 96t + 8$          |
| (0, 8, 14)     | -1    | $-45t^3 + 205t^2 + 555t - 35$      |
| (0, 13, 19)    | +1    | $-16t^3 - 6t^2 + 40t + 2$          |
| (3, 6, 8)      | +1    | $20t^3 - 100t^2 - 220t + 100$      |
| (3, 11, 15)    | -1    | $20t^3 - 120t^2 - 220t + 200$      |
| (3, 13, 17)    | +1    | $20t^3 - 150t^2 - 300t + 130$      |
| (7, 11, 13)    | +1    | $-60t^3 + 240t^2 + 860t + 320$     |
| (8, 14, 17)    | -1    | $-105t^3 + 455t^2 + 1275t - 45$    |
| (8, 15, 19)    | -1    | $36t^3 - 188t^2 - 452t + 124$      |

**Table 9:** An explicit positive linear-dependence certificate. Indices are zero-based intercept indices. The rows use eleven distinct intercepts and avoid the two exceptional central indices. Each column of the weighted signed coefficient matrix vanishes modulo  $p(t)$ .

This excludes one explicit orientation system with arbitrary reciprocal slopes and every real central parameter. It does not classify all 21-line arrangements. An independent exact interval Gaussian-elimination check also proves persistence under independent perturbations of radius  $10^{-3}$  on the eleven used intercepts. That robustness calculation is external exact evidence, not part of the Lean theorem above.

## 9.6 The full supplied affine-class obstruction census

Parpalak–Utkin’s enumeration separates projective and Euclidean classes [20]. This distinction changed the scope of the search. The first checker verified 366 exact polynomial certificates for eighteen supplied projective representatives on  $0 < \varepsilon < 1/25000$ . A projective representative alone does not exhaust its affine cases: a choice of infinity support matters. That initial census was therefore insufficient for a complete affine claim.

The corrected checker expands all 236 supplied Euclidean classes and all 21 choices of distinguished support. It validates 3,765 exact polynomial dependencies and 15,060 reflected or reoriented variants, covering all  $236 \cdot 21 = 4,956$  class/support cases with no missing case, for

$$0 < \varepsilon < 1/2,000,000. \quad (84)$$

Every certificate is a positive linear contradiction of the form above, so it excludes arbitrary, even parameter-dependent, reciprocal slopes. The small positive interval is a uniform bound for this finite corpus; it is not the unrestricted all-real parameter assertion proved for the single ten-sign representative.

The logical dependencies remain separate. The completeness of the classification is supplied by Parpalak–Utkin and its dataset. The present independent checker validates the listed exact certificates, the class/support coverage and the explicit symmetries. It does not rerun their full enumeration inside Lean. The resulting exclusion of a perfect 21-line seed on this prescribed saturated grid and interval is therefore computer-assisted mathematics with an external classification input. Only Theorem 9.2 is a complete representative Lean obstruction. Other grids, larger seeds, nonsimple arrangements and other construction methods remain unrestricted by this census.



| Search                                     | Result                                                       | Limitation                                                 |
|--------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------|
| Deletions from a 49-line seed              | Best retained values 48 : 721, 47 : 677, 46 : 637, 45 : 598. | The tested deletion neighborhood only.                     |
| 57-line MILP deletions                     | Incumbents 56 : 990, 55 : 939, 54 : 887, 53 : 837.           | Timed runs; incumbents are not proved maxima.              |
| All 9,180 tested chords of a 17-line seed  | Best 18-line output has 93 triangles.                        | Chords of one seed.                                        |
| 407,253 vertex-pair insertions at 43 lines | Best 44-line output has 607 triangles, below retained 608.   | The stated insertion neighborhood.                         |
| Singular 16-to-18 insertion                | Best recorded count 89.                                      | No global exclusion.                                       |
| 39-pseudoline straightening                | Exact realizations with 353 and 358 triangles.               | Neither a record nor a proof of nonstretchability.         |
| Perfect 21-line grid fit                   | Exact affine-class obstruction census.                       | The prescribed grid, interval and supplied classification. |
| 49-line grid fit                           | Uniform 767-cell certificate and full visible boundary.      | Exact external certificate; full Lean base pending.        |

**Table 10:** The added research directions and their actual conclusions. The 49-line positive result is compared with its known BBL numerical family in Section 7.9.

## 9.7 New bounded searches and their exact outcomes

The October work also tested finite neighborhoods selected to change features that the earlier chart-only searches preserved. None produced a new numerical lower-bound record. The exact positive and negative results are summarized below; completed enumeration of a listed neighborhood does not imply enumeration of all arrangements.

The methodological connection to recent automated mathematical search is a separation of proposal, evaluation and proof, also emphasized by Georgiev, Gómez-Serrano, Tao and Wagner [12]. Here a numerical optimizer proposes a coordinate or dual vector; exact arithmetic certifies the finite claim; the symbolic construction or obstruction explains which statements hold uniformly. Their paper is methodological context, not evidence that Kobon has already been solved. The author’s other construction projects motivated preserving an iterable invariant and exploiting symmetry orbits; none of their unrelated theorems is imported as a Kobon lemma.

## 9.8 Reproduction and presentation audit

The research registers preserve source snapshots, experiment scripts, seeds, parameters, checkpoints, and exactness limits [31]. The paper’s figures are original vector renderings, not copied source figures. The generator `paper/scripts/generate_figures.py` reads the research data without modifying it and exports PDF figures, PNG previews, LaTeX tables, and machine-readable rendering data. The figure manifest records input-file hashes and the mathematical status of each caption. Geometric views recompute their exact triangle lists before converting coordinates to floating point for display. Cropped detail panels and independent affine axis rescaling are explicitly labeled; lengths and angles are not inferred from those views.

The appropriate experimental conclusion is therefore selective. Exact coordinates have strengthened the retained lower-bound envelope and revealed specific failed construction rules. The successful universal advance comes from replacing a numerical chart search by a geometric exposure proof and an all-order count. The remaining negative searches narrow useful next



steps, but they do not solve the general optimization problem.

## 10 Formal verification and reproducibility

### 10.1 The verified snapshot

The mathematical snapshot used by this paper is repository commit `2d69a71e32d3eaa59399b0e5bc61720ec3b00ac8`. It pins Lean 4.31.0 and Mathlib commit `fabf563a7c95a166b8d7b6efca11c8b4dc9d911f`. The local release audit completed on 2 October 2026. The independent [GitHub Actions run 37056093885](#) passed at the pinned revision; the preceding mathematical release run 37055143316 also passed. These are reproducibility statements about a particular collection of declarations and coordinates, not an assertion that every conjecture discussed in this paper has been formalized.

The audit built 270 targets and recorded hashes for all 375 active Lean source files. It inspected 5,241 theorem declarations: 4,405 use only the accepted standard logical axioms, while 836 also depend on explicitly recorded native evaluation. These are counts of theorem declarations, including auxiliary and generated declarations, not counts of independent mathematical contributions. The coordinate audit checked 138 distinct coordinate identities with two exact counters; 104 of the identities also satisfy the simplicity check.

### 10.2 Three proof and computation boundaries

**General symbolic proofs.** The arbitrary-order construction, the generic closed BBL recursion, the actual simple upper bounds, the representative grid obstruction and the local fan geometry use the usual Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. No unproved geometric construction is installed as an axiom. Their conclusions are not inferred from a finite range of successful tests. Lean and Mathlib provide the real analysis, finite combinatorics, and algebra used in these proofs [5, 29].

**Finite certificates.** Large coordinate checks use `native_decide`, which trusts Lean’s native compiler and runtime in addition to the ordinary logical kernel interface. The general certificate soundness theorem is proved symbolically; native evaluation verifies the finite integer sign conditions and list properties. The audit reports these dependencies rather than describing all certificate computation as kernel reduction. Small examples and many formula comparisons also have direct kernel-reduction proofs.

**Computational and paper-level evidence.** Exact rational searches, interval calculations, ordinary geometric arguments, and the experimental large witnesses are retained with separate status. They are not imported as axioms into the all-order theorem. In particular, the incomplete 642-line coordinate recount and the unfinished full real-parameter 33- and 49-line seed drafts are not active certificate results. The general recursive theorem now independently proves the 642-line existential bound. The full 21-line affine obstruction corpus is exact external evidence with a supplied classification input; the representative ten-sign obstruction is inside Lean. Conditional upper-core arithmetic is distinguished from the completed actual incidence and clean-line charging theorems.

### 10.3 What the audit rejects

The active-source scan strips comments and strings and rejects proof placeholders, admissions, custom axiom declarations, and unsafe definitions. The additional Lean audit recursively collects the axioms used by every theorem in the Kobon namespace and rejects unapproved dependencies. It permits and records the standard logical axioms and the generated native-evaluation dependencies just described.

The archived March file is deliberately outside the active library. Preserving it is evidence of the development history, not a way to conceal its eight placeholders inside an otherwise successful build. Likewise, an unfinished draft is kept under the research directory with an explicit status note rather than omitted silently from an allegedly complete theorem library.

### 10.4 Theorem-level correspondence

Table 11 gives the principal correspondence between the paper and the Lean development. The public repository contains more detailed logs and dependency information; the table is intended to make each major mathematical claim traceable without requiring a reader to inspect thousands of auxiliary declarations.

**Table 11:** Principal mathematical claims and formal sources. Names refer to the frozen mathematical snapshot.

| Mathematical claim                    | Formal source                                                                              |
|---------------------------------------|--------------------------------------------------------------------------------------------|
| Exact certificate soundness           | <code>Geometry.validate_sound; Simple.validate_simple_sound</code>                         |
| Certificates are nonoverlapping cells | <code>Cells.interior_eq_signCell; Cells.distinct_interiors_disjoint</code>                 |
| Classical affine construction         | <code>FurediPalasti.lower_bound; FurediPalastiCount.ordered_bound</code>                   |
| Shifted repeated-index gain           | <code>ShiftedCount.ordered_bound_plus; ShiftedFurediPalasti.simple_lower_bound</code>      |
| Actual projective transformation      | <code>Projective.oriented_transform; Projective.triangle_preserved</code>                  |
| One exposed cap                       | <code>FurediPalastiWrap.one_cap_simple_lower_bound</code>                                  |
| Two exposed caps                      | <code>FurediPalastiTwoCaps.two_cap_simple_lower_bound</code>                               |
| Universal $G$ and exact gain          | <code>Universal.baseline_sound; Universal.baseline_improvement</code>                      |
| Finite-enhanced all-order bound       | <code>Universal.all_n; AllN.dominates_every_saved_certificate</code>                       |
| Unconditional 49-seed target          | <code>AllN.from_49_unconditional; Universal.dominates_49_target</code>                     |
| Exterior addition from visible pairs  | <code>Exterior.extension</code>                                                            |
| Real compatible seed family           | <code>SeedFamily; Parametric; TangentBounds</code>                                         |
| Geometric one-step BBL doubling       | <code>BBLDoubling.doubling</code>                                                          |
| Closed seed and uniform recursion     | <code>BBLRecursiveSeed.seed_step; BBLInfinite.uniform_iterate</code>                       |
| Visible-boundary recursion            | <code>BBLEvenStep.canonical_visible_step; BBLEvenInfinite.infinite_even_family</code>      |
| Concrete odd and even families        | <code>BBLVerifiedFamilies.eleven_odd_family; BBLVerifiedFamilies.eleven_even_family</code> |

| Mathematical claim                          | Formal source                                                                                                                    |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Current all-order envelope $H$              | <code>RecursiveEnvelope.all_n; RecursiveEnvelope.strict_previous_tail</code>                                                     |
| Actual simple optimality windows            | <code>BBLFamilyOptimality.odd_window; BBLFamilyOptimality.even_window</code>                                                     |
| Next-grid saturation component              | <code>BBLNextSaturation.next_saturated</code>                                                                                    |
| Actual segment and defect inventory         | <code>UpperEdgeInventory.certificate_defect_identity</code>                                                                      |
| Actual clean-line charging                  | <code>UpperCleanCharging.certificate_clean_line_budget</code>                                                                    |
| Actual simple upper inequalities            | <code>UpperSimpleOptimality.simple_lower_bound_upper; UpperEvenSimpleOptimality.simple_lower_bound_even_upper</code>             |
| Extremal fan and matched triple obstruction | <code>UpperFan.Sectors.weighted_local_bound; UpperTripleFan.extremal_triple_fans_not_adjacent</code>                             |
| Actual core incidence                       | <code>UpperCoreLineIncidence.core_line_budget; UpperCoreCombinatorics.core_surplus</code>                                        |
| Conditional three-core bound                | <code>UpperCoreBudget.at_most_three_multiple_points</code>                                                                       |
| Tangent algebra and actual obstruction      | <code>BBLTangentAlgebra.tan_quartic; BBLTangentValues.value_eq_tan; BBLGridObstruction.actual_grid_orientation_impossible</code> |
| 49-grid analytic enclosures                 | <code>BBLTangent48Bounds</code> ; finite uniform seed check still incomplete                                                     |
| Local fan geometry and budgets              | <code>SharedFan; FanGeometry; FanCount; CyclicFan; CleanLineBudget</code>                                                        |
| Preserved historical finite bounds          | <code>Results.earlier_NNN; Results.best_classical_NNN; Results.best_simple_NNN</code>                                            |

The exterior theorem assumes a valid visible-pair list, not an unproved assertion that enough visible pairs always exist. The BBL theorem assumes the stated tangent-grid and saturation conditions, not its own desired conclusion. The concrete eleven-derived family discharges its entire seed and visibility interface. Conversely, the upper-core arithmetic still accepts the global fan inequalities as hypotheses: actual segment incidence, clean-line charging and core incidence are now completed, but cyclic fan extraction and its summation are not. Inspecting hypotheses is as important as inspecting the absence of placeholders.

## 10.5 Reproducing proofs, coordinates, and figures

After cloning the repository and installing the pinned Lean toolchain, the principal checks are:

```
lake exe cache get
python scripts/verify_lean.py --jobs 2
python scripts/verify_coordinates.py
python scripts/verify_seed_interval.py
python scripts/evidence_manifest.py --check
python manuscripts/2026-10-03/build.py --edition long
```

The finite envelope can be regenerated from the certificate index using `scripts/generate_all_n.py`. This generator emits theorem references; Lean checks the resulting proposition independently of the script's intentions. The generated source is not a trusted oracle for geometry.

All sixteen scientific figures retained from the initial manuscript are generated by `paper/scripts/generate_figures.py`. The script reads the saved formulas, coordinates, and experiment records. For geometric renderings it fixes the triangle combinatorics using exact rational arithmetic before converting coordinates to floating point for plotting. Vector PDF is the publication format; PNG images are previews. Rational proxies of the trigonometric examples are explicitly identified as finite illustrations, not substituted for the symbolic theorem.

The files `paper/figure_audit.md` and `paper/generated/figure-manifest.json` record data sources, hashes, and figure-specific claim status. The finite tables are generated from the same coordinate index used by verification. This makes a plot an inspectable derivative of evidence rather than an unsupported visual argument. The updated edition adds two vector figures for the recursive envelope and the uniform 49-line seed, with a separate figure manifest. The paper-specific build instructions accompany the LaTeX source. The shared proof map links every main result to immutable source locations and marks ordinary deductions, exact external evidence and excluded drafts. The earlier 60-page and 14-page editions remain preserved in the repository; this new edition does not rewrite their frozen verification manifests.

**Computer and AI assistance.** The development and preparation of this manuscript used computer algebra, exact arithmetic, automated search, Lean, and AI assistance for research, code, and drafting. AI-generated assertions are not treated as mathematical evidence. The reported conclusions are tied to explicit proofs, exact witnesses, or separately scoped computational records. Alejandro Zarzuelo Urdiales is the sole named author of this manuscript; cited construction authors retain their mathematical attribution.

## 11 Consequences, limitations, and directions for further work

### 11.1 What the program has established

The strongest completed total lower-bound function in this project is  $H$  from Theorem 7.7. It retains the all-order construction  $G$ , all 138 finite coordinate identities, and the fully recursive odd and even families. The affine phase/chart result changes the constant term of a standard all-order baseline. The recursive family has a one-triangle simple optimality window and improves the previous verified repository envelope at infinitely many orders. Off the sparse family it leaves the preceding general guarantee intact.

The significant proof advance is geometric. Every finite-depth iteration produces a real straight-line arrangement with the next seed's full grid, triangle, orientation and visible-boundary invariant. The concrete base discharges those hypotheses; an arithmetic recurrence alone would not. The simple upper halves are also derived from actual arrangements, not assumed polynomial benchmarks. This supplies a reusable formal construction and comparison framework. It does not establish first numerical priority for the inherited BBL family.

The earlier bounds at 28, 30 and 34 lines survive through exact finite certificates independently of the incomplete March proof. The actual exterior theorem and the wedge-resource obstruction explain both why individual successful steps exist and why unrestricted repetition of full gains is not justified by that method. The all-order numerical target beginning at 49 remains independently proved, but it is not a nested chain of successive extensions from an arbitrary optimal seed. The unrestricted successor assertion remains open in this development.

For upper bounds, the actual edge inventory, core incidence and clean-line charge are complete. The local fan equality structure and matched triple-fan obstruction identify restrictions on the remaining nonsimple core. The explicit algebraic grid obstruction provides a rigorous

search limitation. The larger affine-class corpus and the uniform 49-line seed are exact external computations with stated input and verification boundaries. Neither an incomplete seed check nor a published classification input is silently promoted to an unconditional Lean theorem.

## 11.2 Complete concrete seed interfaces before changing the recurrence

The generic odd and visible recursion is now complete. The next seed work is therefore local: finish the uniform 33- or 49-line finite base checks, link the resulting actual lines to the seed record, and invoke the existing induction. The 49-line tangent enclosures and the isolated graph/arithmetic checks already pass, but the full base computation was stopped before completion. Decomposing that computation into independently checked chunks is a proof-engineering task; its completion would not make the known BBL numerical series new.

A new numerical improvement requires a smaller initial defect, a source of extra triangles beyond the conserved count, or a different invariant. The invariant  $q^2 - 1 - 3T$  is preserved by pure doubling. The perfect 21-line obstruction applies to its specified tangent grid and small parameter interval, so it motivates testing different intercept families rather than declaring all perfect compatible seeds impossible. Partial doubling and interpolation at intermediate orders remain independent construction questions.

## 11.3 The remaining global core interfaces

Two interfaces isolate the unfinished upper-bound proof. First choose a linear functional nonzero on every radial direction at a core, orient one representative of each incident line positively, and sort the resulting slopes. Appending the opposite representatives should supply actual cyclic order, positive consecutive determinants and antipodality. Second, match rays to actual shared elementary edges. On a bounded ray choose the nearest intersection vertex; on an unbounded ray use a noncentral auxiliary point. Requiring an actual vertex on every ray would be a false assumption. Certified triangular sectors must then be associated with their actual certificate indices and endpoint pairs.

The required incidence equivalences are

$$\sum_v d_1(v) = D_1, \quad \sum_v d_2(v) = 2D_2.$$

They must identify the centers injectively and use actual intersection multiplicities. The present abstract local fan records alone do not supply those facts. For a shared edge joining centers  $v, w$ , the two adjacent certified triangles must also produce the matching needed by `UpperTripleFan`, after a proved cyclic reindexing. This is why the new local restrictions do not yet imply an unrestricted upper bound.

## 11.4 A conditional graph deduction for the next proof stage

The following deduction is a research plan, not an additional Lean result of the release. At a triple fan write  $u_i$  for the radial vectors. Positive consecutive determinants and antipodality  $u_3 = -\lambda u_0$  with  $\lambda > 0$  give

$$\det(u_0, u_1) > 0, \quad \det(u_1, u_2) > 0, \quad \det(u_0, u_2) = \det(u_2, u_3)/\lambda > 0.$$

Thus the first three radial directions are pairwise noncollinear and their opposites give six distinct radial points. Once formalized, this will connect three exceptional shared-ray indices to three distinct neighboring core points.

Form the finite simple graph whose vertices are actual core points and whose edges are shared elementary segments with two core endpoints. Let  $X$  be its exceptional triple cores, with  $|X| = e$ . Suppose the missing ray/edge correspondence proves degree three at every vertex of  $X$ , and the matched triple-fan theorem proves  $X$  independent. Counting incidences at  $X$  then gives

$$3e \leq D_2, \quad D_2 - 2e \geq e, \quad D_2 - 2e \geq D_2/3. \quad (85)$$

No planarity theorem is needed: each edge meeting  $X$  has exactly one endpoint there. The distinct-neighbor and matching assumptions remain essential.

If every exceptional fan is triple, substituting in the already checked conditional budget (80) yields

$$2\delta \geq n + 2S - 7I + 8q + e, \quad (86)$$

$$6\delta \geq 3n + 6S - 21I + 24q + D_2. \quad (87)$$

If all core multiplicities are three,  $I = S = 3q$ , so the right sides become  $n - 7q + e$  and  $3n - 21q + D_2$ . These replace a potentially negative exceptional correction by a nonnegative term. They are consequences of the exact correction  $D_2 - 2e$ , not stronger estimates when that original term is already retained. The negative core-count term remains; no new unrestricted numerical upper bound follows from this calculation alone. These deductions and all unproved interfaces are recorded explicitly in [the next-proof plan](#).

## 11.5 Numerical novelty and future comparison

Several published special families are stronger than a compact uniform baseline at their orders; taking maxima or padding already produces other all-order functions. Accordingly, neither validity for every  $n$  nor omission from an explicit OEIS table establishes a numerical state of the art. The present claims are a proved affine refinement relative to the quoted formula, completed recursive real geometry, a stronger verified repository envelope, and precise structural/certificate results. A first-discovery claim would require a separate exhaustive priority audit.

Future comparisons should inspect implicit consequences of earlier families and extensions as well as printed tables. In particular, the 49-line branch is already covered numerically by BBL, and the 33-line branch by Forge–Ramírez Alfonsín. The author’s earlier recorded contribution, an independently supplied coordinate certificate and the first publication of a numerical inequality need not have the same provenance. Maintaining those distinctions preserves the valid progress without overstating the solution of the problem.

**Data and code availability.** The complete LaTeX source, bibliography, original vector figures, figure-generation scripts, proof-to-claim correspondence, exact finite catalog, negative experiments and verification records accompany the public repository [31]. This comprehensive edition preserves the earlier research while updating its proof status. A separate journal-length companion presents the completed contributions with links to the same immutable proof revision. Neither publication of a repository nor preparation of these manuscripts asserts external peer review.

## References

- [1] Nicolas Bartholdi, Jérémy Blanc, and Sébastien Loisel. On simple arrangements of lines and pseudo-lines in  $\mathbb{P}^2$  and  $\mathbb{R}^2$  with the maximum number of triangles. In *Surveys on*

- Discrete and Computational Geometry*, volume 453 of *Contemporary Mathematics*, pages 105–116. American Mathematical Society, 2008. doi: 10.1090/conm/453/08797. URL <https://arxiv.org/abs/0706.0723>. Preprint first posted in 2007.
- [2] J  r  my Blanc. The best polynomial bounds for the number of triangles in a simple arrangement of  $n$  pseudo-lines. *Geombinatorics*, 21(1):5–14, 2011. URL <https://arxiv.org/abs/0801.2845>. Preprint first posted in 2008.
  - [3] Gilles Cl  ment and Johannes Bader. Tighter upper bound for the number of Kobon triangles, December 2007. URL <https://oeis.org/A006066/a006066.pdf>. Draft dated 21 December 2007.
  - [4] G  bor Dam  sdi, Leonardo Mart  nez-Sandoval, D  niel T. Nagy, and Zolt  n L  r  nt Nagy. Triangle areas in line arrangements. *Discrete Mathematics*, 343:112105, 2020. doi: 10.1016/j.disc.2020.112105. URL <https://real.mtak.hu/114794/7/1-s2.0-S0012365X20302910-main.pdf>.
  - [5] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction—CADE 28*, Lecture Notes in Computer Science, pages 625–635. Springer, 2021. doi: 10.1007/978-3-030-79876-5\_37.
  - [6] Jeff Erickson. New lower bounds for convex hull problems in odd dimensions. *SIAM Journal on Computing*, 28(4):1198–1214, 1999. doi: 10.1137/S0097539797315410. URL <https://jeffe.cs.illinois.edu/pubs/pdf/convex.pdf>.
  - [7] Stefan Felsner and Klaus Kriegel. Triangles in Euclidean arrangements. *Discrete & Computational Geometry*, 22(3):429–438, 1999. doi: 10.1007/PL00009471. URL <https://page.math.tu-berlin.de/~felsner/Paper/tri.pdf>.
  - [8] David Forge and Jorge Luis Ram  rez Alfons  n. Straight line arrangements in the real projective plane. *Discrete & Computational Geometry*, 20(2):155–161, 1998. doi: 10.1007/PL00009373.
  - [9] Kobon Fujimura. *The Tokyo Puzzles*. Charles Scribner’s Sons, New York, 1978. ISBN 0684155362. Edited by Martin Gardner; translated by Fumie Adachi.
  - [10] Zolt  n F  redi and Ilona Pal  sti. Arrangements of lines with a large number of triangles. *Proceedings of the American Mathematical Society*, 92(4):561–566, 1984. doi: 10.1090/S0002-9939-1984-0760946-2. Also indexed by JSTOR DOI 10.2307/2045427.
  - [11] Martin Gardner. *Wheels, Life, and Other Mathematical Amusements*. W. H. Freeman, New York, 1983. Kobon discussion cited in Cl  ment–Bader at pp. 170–171 and 178.
  - [12] Bogdan Georgiev, Javier G  mez-Serrano, Terence Tao, and Adam Zsolt Wagner. Mathematical exploration and discovery at scale, 2025. URL <https://arxiv.org/abs/2511.02864v3>. Version 3, 22 December 2025; cited for methodology, not as a solution of Kobon.
  - [13] Branko Gr  nbaum. *Arrangements and Spreads*, volume 10 of *CBMS Regional Conference Series in Mathematics*. American Mathematical Society, Providence, RI, 1972. doi: 10.1090/cbms/010.
  - [14] HandWiki contributors. Kobon triangle problem, 2026. URL [https://handwiki.org/wiki/Kobon\\_triangle\\_problem](https://handwiki.org/wiki/Kobon_triangle_problem). Consulted 22 September 2026; a secondary record of the extension claim, not its proof.



- [15] Heiko Harborth. Some simple arrangements of pseudolines with a maximum number of triangles. *Annals of the New York Academy of Sciences*, 440(1):31–33, 1985. doi: 10.1111/j.1749-6632.1985.tb14536.x.
- [16] Ke Liang, Youming Liu, and Yurui Zhang. On the symmetries of optimal simple Kobon arrangements for  $n = 28$ . Zenodo preprint, version 4, July 2026. URL <https://zenodo.org/records/21181834>. Version-specific record, 3 July 2026; not an unrestricted exact-value theorem.
- [17] Andrea Maiorana. Exact fourteen-line Kobon configurations with 54 triangles. Public coordinate repository, 2026. URL [https://github.com/rufio72/kobon\\_triangles\\_k14/tree/e47c7cfd9661e54d29befb83118bf83d5f804028](https://github.com/rufio72/kobon_triangles_k14/tree/e47c7cfd9661e54d29befb83118bf83d5f804028). Descriptive title; commit e47c7cfd9661e54d29befb83118bf83d5f804028, 12 August 2026; CC BY 4.0.
- [18] OEIS Foundation Inc. Entry A006066: Kobon triangles. The On-Line Encyclopedia of Integer Sequences, 2026. URL <https://oeis.org/A006066>. Consulted 22 September 2026; revision 245 and its attribution comments audited.
- [19] Roman Parpalak and Denis Utkin. The  $18 \cdot 2^t + 1$  triangle-maximal series of straight lines, 2026. URL <https://arxiv.org/abs/2604.22035>. Version 1, 23 April 2026.
- [20] Roman Parpalak and Denis Utkin. Enumeration and classification of triangle-maximal pseudoline arrangements, 2026. URL <https://arxiv.org/abs/2607.29236v1>. Version 1, 31 July 2026; classification is external input to the exact grid-obstruction corpus.
- [21] Roman Parpalak and Denis Utkin. The 8–14–26 even series. Public research draft and construction repository, 2026. URL <https://github.com/parpalak/kobon-even-draft/blob/4da100acd30d9fb2c5b2a1fa3fb9ce848a2e52a5/8-14-26-even-series.md>. Snapshot at commit 4da100acd30d9fb2c5b2a1fa3fb9ce848a2e52a5; descriptive repository title.
- [22] Roman Parpalak and Denis Utkin. Kobon solutions: coordinate gallery. Public data repository, 2026. URL <https://github.com/ud1/kobon-solutions/tree/feae4f5571a988a3ed45bd26a36259ebb7d7d3d2>. Snapshot at commit feae4f5571a988a3ed45bd26a36259ebb7d7d3d2, 21 September 2026; original individual credits retained.
- [23] ProofAtlas. Kobon triangle problem: collaboration research page, 2026. URL <https://www.proofatlas.ai/collaboration/kobon-triangle-problem/>. Consulted 22 September 2026; reported restricted results were not independently replayed.
- [24] G. B. Purdy. Triangles in arrangements of lines. *Discrete Mathematics*, 25(2):157–163, 1979. doi: 10.1016/0012-365X(79)90018-9.
- [25] George Purdy. Triangles in arrangements of lines. II. *Proceedings of the American Mathematical Society*, 79(1):77–81, 1980. doi: 10.1090/S0002-9939-1980-0560588-4.
- [26] Jean-Pierre Roudneff. The maximum number of triangles in arrangements of pseudolines. *Journal of Combinatorial Theory, Series B*, 66(1):44–74, 1996. doi: 10.1006/jctb.1996.0006.
- [27] Pavlo Savchuk. Constructing optimal Kobon triangle arrangements via table encoding, SAT solving, and heuristic straightening, 2025. URL <https://arxiv.org/abs/2507.07951>. Version 1 inspected.

- [28] Thomas O. Strommer. Triangles in arrangements of lines. *Journal of Combinatorial Theory, Series A*, 23(3):314–320, 1977. doi: 10.1016/0097-3165(77)90022-X.
- [29] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pages 367–381. ACM, 2020. doi: 10.1145/3372885.3373824. URL <https://leanprover-community.github.io/papers/mathlib-paper.pdf>.
- [30] Alejandro Zarzuelo Urdiales. New lower bounds for even Kobon numbers. Archivara preprint, March 2026. URL <https://archivara.org/paper/48b411c9-0e03-4592-931e-179b9a1c2312>. Original March result; the archived general extension proof was incomplete.
- [31] Alejandro Zarzuelo Urdiales. Verified Kobon constructions: research and Lean proof repository. Source code, exact certificates, and research records, 2026. URL <https://github.com/alejandrozukobon-proof>. Research state reviewed through 3 October 2026; proof snapshot 2d69a71e32d3eaa59399b0e5bc61720ec3b00ac8.

## A Complete saved finite table, orders 3–60

The entries below are the strongest saved coordinate lower bounds in the frozen release. They are separated by arrangement class. The classical column may use multiple intersections; the simple column may not. The OEIS column is a dated explicit-table comparison, not an exhaustive priority audit. The upper columns reproduce attributed benchmark values from the September comparison. The simple polynomial upper bounds now also have actual geometric Lean proofs; the unrestricted reported Clément–Bader values retain their external-source status. A difference between a lower and upper entry represents an unresolved gap within the stated scope, not a proof that intermediate values are unattainable.

**Table 12:** Saved finite lower bounds, separated by arrangement class, and attributed comparison upper values. OEIS is the explicit table snapshot of 20 September 2026; a dash means omitted. Upper values retain their external attribution; the simple polynomial bounds are also formalized in this release.

| $n$ | $G(n)$ | Saved simple | Saved classical | OEIS lower | Blanc simple | $U_T$ |
|-----|--------|--------------|-----------------|------------|--------------|-------|
| 3   | 1      | 1            | 1               | 1          | 1            | 1     |
| 4   | 2      | 2            | 2               | 2          | 2            | 2     |
| 5   | 5      | 5            | 5               | 5          | 5            | 5     |
| 6   | 7      | 7            | 7               | 7          | 7            | 8     |
| 7   | 11     | 11           | 11              | 11         | 11           | 11    |
| 8   | 14     | 14           | 15              | 15         | 14           | 16    |
| 9   | 20     | 21           | 21              | 21         | 21           | 21    |
| 10  | 24     | 25           | 25              | 25         | 25           | 26    |
| 11  | 31     | 32           | 32              | 32         | 33           | 33    |
| 12  | 37     | 37           | 38              | 38         | 38           | 40    |
| 13  | 45     | 47           | 47              | 47         | 47           | 47    |
| 14  | 52     | 53           | 54              | 54         | 53           | 56    |
| 15  | 62     | 65           | 65              | 65         | 65           | 65    |
| 16  | 70     | 72           | 72              | 72         | 72           | 74    |
| 17  | 81     | 85           | 85              | 85         | 85           | 85    |
| 18  | 91     | 93           | 93              | 93         | 93           | 96    |
| 19  | 103    | 107          | 107             | 107        | 107          | 107   |
| 20  | 114    | 116          | 117             | 117        | 116          | 120   |
| 21  | 128    | 133          | 133             | 133        | 133          | 133   |
| 22  | 140    | 143          | 143             | 143        | 143          | 146   |
| 23  | 155    | 161          | 161             | 161        | 161          | 161   |
| 24  | 169    | 172          | 172             | 172        | 172          | 176   |
| 25  | 185    | 191          | 191             | 191        | 191          | 191   |
| 26  | 200    | 203          | 204             | 204        | 203          | 208   |
| 27  | 218    | 225          | 225             | 225        | 225          | 225   |
| 28  | 234    | 238          | 238             | 238        | 238          | 242   |
| 29  | 253    | 261          | 261             | 261        | 261          | 261   |
| 30  | 271    | 275          | 275             | 275        | 275          | 280   |
| 31  | 291    | 299          | 299             | 299        | 299          | 299   |
| 32  | 310    | 314          | 315             | 315        | 314          | 320   |
| 33  | 332    | 341          | 341             | 341        | 341          | 341   |
| 34  | 352    | 357          | 357             | 357        | 357          | 362   |
| 35  | 375    | 385          | 385             | 385        | 385          | 385   |
| 36  | 397    | 402          | 402             | 402        | 402          | 408   |
| 37  | 421    | 431          | 431             | 431        | 431          | 431   |
| 38  | 444    | 449          | 450             | 450        | 449          | 456   |
| 39  | 470    | 470          | 470             | –          | 481          | 481   |
| 40  | 494    | 494          | 494             | –          | 500          | 506   |

*Continued from previous page*

| $n$ | $G(n)$ | Saved simple | Saved classical | OEIS lower | Blanc simple | $U_T$ |
|-----|--------|--------------|-----------------|------------|--------------|-------|
| 41  | 521    | 533          | 533             | 533        | 533          | 533   |
| 42  | 547    | 553          | 553             | 553        | 553          | 560   |
| 43  | 575    | 587          | 587             | 587        | 587          | 587   |
| 44  | 602    | 608          | 608             | —          | 608          | 616   |
| 45  | 632    | 645          | 645             | 645        | 645          | 645   |
| 46  | 660    | 667          | 667             | 667        | 667          | 674   |
| 47  | 691    | 691          | 691             | —          | 705          | 705   |
| 48  | 721    | 721          | 721             | —          | 728          | 736   |
| 49  | 753    | 767          | 767             | 767        | 767          | 767   |
| 50  | 784    | 791          | 792             | 792        | 791          | 800   |
| 51  | 818    | 818          | 818             | —          | 833          | 833   |
| 52  | 850    | 850          | 850             | —          | 858          | 866   |
| 53  | 885    | 885          | 885             | —          | 901          | 901   |
| 54  | 919    | 919          | 919             | —          | 927          | 936   |
| 55  | 955    | 955          | 955             | —          | 971          | 971   |
| 56  | 990    | 990          | 990             | —          | 998          | 1008  |
| 57  | 1028   | 1045         | 1045            | —          | 1045         | 1045  |
| 58  | 1064   | 1073         | 1073            | —          | 1073         | 1082  |
| 59  | 1103   | 1103         | 1103            | —          | 1121         | 1121  |
| 60  | 1141   | 1141         | 1141            | —          | 1150         | 1160  |

## B The complete 138-certificate catalog

Each row identifies a finite arrangement and its certified triangle count. A hash distinguishes coordinate identities even when  $n$  and  $T$  coincide. The complete coordinates, support triples, source paths, original metadata, and Lean theorem names are machine-readable in the accompanying catalog. Printing hundreds of large rational coefficients would obscure rather than improve the mathematical presentation; the hash-indexed files provide their exact reproducible specification.

The catalog contains 138 distinct promoted coordinate identities, not 138 new numerical records. S denotes a simple arrangement; M denotes a verified arrangement with multiple intersections. Every row has a classical lower-bound theorem; S rows also have simplicity certificates. The hash is a prefix of the canonical coordinate SHA-256 identity, not of the source file. Full source paths, source-file hashes, theorem modules, and complete identities are supplied in `paper/generated/certificate-catalog.json`.

Source codes identify repository collections, not discovery authors: FT, retained finite table (mixed published and derived sources); EX, exact extension; HI, historical work inventory; HY, compatible-seed/hybrid work; SC, current session (phase/chart or classical dyadic construction, distinguished in the JSON metadata); MA, Andrea Maiorana’s external certificates; PU, Parpalak–Utkin gallery inputs; KR, earlier research collection; OT, other documented source. Published constructions retain their original attribution. Native-evaluation trust boundaries are discussed in the text.

**Table 13:** Complete promoted coordinate catalog. Repeated orders and counts can have distinct coordinate witnesses; no numerical priority follows from inclusion.

| ID | $n$ | $T$ | Class | Collections | Coordinate identity |
|----|-----|-----|-------|-------------|---------------------|
| 1  | 3   | 1   | S     | FT          | 05c82de62571        |
| 2  | 4   | 2   | S     | FT          | 0e541bd5bbe7        |

*Continued from previous page*

| ID | $n$ | $T$  | Class | Collections | Coordinate identity |
|----|-----|------|-------|-------------|---------------------|
| 3  | 5   | 5    | S     | FT          | 948a0bdf2dd3        |
| 4  | 6   | 7    | S     | FT,EX,HI    | 6ea9ebaa7bf2        |
| 5  | 7   | 11   | S     | FT          | e6b9a626a163        |
| 6  | 8   | 14   | S     | FT          | 2a63f4d9308d        |
| 7  | 9   | 21   | S     | FT          | 6ebbed82d063        |
| 8  | 10  | 25   | S     | FT          | b767c427fa30        |
| 9  | 11  | 32   | S     | FT,HY       | b955e7c9923a        |
| 10 | 12  | 38   | M     | FT          | 38f69f380097        |
| 11 | 13  | 47   | S     | FT          | ddec1f788550        |
| 12 | 14  | 53   | S     | FT          | 45157b2f20fe        |
| 13 | 15  | 65   | S     | FT          | 4d3bf3b46db5        |
| 14 | 16  | 72   | S     | FT          | 391cf79afe24        |
| 15 | 17  | 85   | S     | FT          | 7cf3c061a447        |
| 16 | 18  | 93   | S     | FT          | edb6246a6de7        |
| 17 | 19  | 107  | S     | FT          | a2f32bdbcd1e        |
| 18 | 20  | 116  | S     | FT,EX,HI    | adc03eed024d        |
| 19 | 21  | 133  | S     | FT          | 36bae754f198        |
| 20 | 22  | 143  | S     | FT          | 54396c646a7b        |
| 21 | 23  | 161  | S     | FT          | 969ad7d3adaf        |
| 22 | 24  | 172  | S     | FT          | e63e906b9903        |
| 23 | 25  | 191  | S     | FT          | b568bfffb5a19       |
| 24 | 26  | 203  | S     | FT          | e61b423c7c3d        |
| 25 | 27  | 225  | S     | FT          | 7d6ae9a04c46        |
| 26 | 28  | 238  | S     | FT          | 56ef2acb9edd        |
| 27 | 29  | 261  | S     | FT          | d4805c25a121        |
| 28 | 30  | 275  | S     | FT          | dcfe649a79cf        |
| 29 | 31  | 299  | S     | FT          | 70f23b9463cb        |
| 30 | 32  | 314  | S     | FT          | 9ab78b00c233        |
| 31 | 33  | 341  | S     | FT          | 95f3b770119e        |
| 32 | 34  | 357  | S     | FT          | fbde0a760c6e        |
| 33 | 35  | 385  | S     | FT          | 9ee99af6b980        |
| 34 | 36  | 402  | S     | FT          | 3fc01b7d2a1c        |
| 35 | 37  | 431  | S     | FT          | 11f0bde7bac3        |
| 36 | 38  | 449  | S     | FT          | 46ad5a32af83        |
| 37 | 39  | 469  | M     | FT,HI       | cf24d7d33c00        |
| 38 | 40  | 494  | S     | FT          | 785374ca691b        |
| 39 | 41  | 533  | S     | FT          | 01e2ef66f8fd        |
| 40 | 42  | 553  | S     | FT          | 2bdc9d67365a        |
| 41 | 43  | 587  | S     | FT          | 861d8be27dee        |
| 42 | 44  | 608  | S     | FT,HI       | b5277675e356        |
| 43 | 45  | 645  | S     | FT          | 3dc6e882e9b8        |
| 44 | 46  | 667  | S     | FT,PU       | 32b8ca2f231d        |
| 45 | 47  | 690  | S     | FT          | d8af27fc13d1        |
| 46 | 48  | 720  | S     | FT          | 5e9767e73b31        |
| 47 | 49  | 767  | S     | FT          | 957e6f151a57        |
| 48 | 50  | 791  | S     | FT          | 2ea22b558bf8        |
| 49 | 51  | 817  | M     | FT,HI,KR    | bd215f67ab10        |
| 50 | 52  | 850  | S     | FT          | 4b0d823a1fa8        |
| 51 | 53  | 884  | S     | FT          | 723d7eff8d45        |
| 52 | 54  | 918  | S     | FT          | 915abb8ed17c        |
| 53 | 55  | 954  | S     | FT          | ea7675d9b8a3        |
| 54 | 56  | 990  | S     | FT          | de8a415c9b29        |
| 55 | 57  | 1045 | S     | FT          | 9a1d337c75ba        |
| 56 | 58  | 1073 | S     | FT          | d988db552530        |

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| ID  | $n$ | $T$   | Class | Collections | Coordinate identity |
|-----|-----|-------|-------|-------------|---------------------|
| 57  | 59  | 1102  | S     | FT          | fd3b4280376e        |
| 58  | 60  | 1140  | S     | FT          | 16af6d0b260e        |
| 59  | 12  | 37    | S     | FT, HY      | 8c718a334b7b        |
| 60  | 39  | 468   | S     | FT          | 5c5b0b26ee22        |
| 61  | 51  | 816   | S     | FT          | 6fa6b7d91d28        |
| 62  | 5   | 3     | S     | EX          | 6265d6a7c557        |
| 63  | 6   | 4     | S     | EX          | 7e060e9a0ac1        |
| 64  | 19  | 107   | S     | EX          | 5d2a7588c1e2        |
| 65  | 21  | 126   | S     | EX          | f18756b72791        |
| 66  | 5   | 5     | S     | EX, HI      | dd0748b7cdfb        |
| 67  | 7   | 10    | S     | EX, HI      | 4b5bb6e99d62        |
| 68  | 21  | 132   | S     | HY          | 3d8bae2fa729        |
| 69  | 22  | 142   | S     | HY          | e683945e66cd        |
| 70  | 41  | 532   | S     | HY          | 6d48a3411f41        |
| 71  | 42  | 552   | S     | HY          | 7b29a6bd1191        |
| 72  | 81  | 2132  | S     | HY          | 7bc6b303744f        |
| 73  | 82  | 2172  | S     | HY          | d75c9101c8a5        |
| 74  | 161 | 8532  | S     | HY          | 230691d59eee        |
| 75  | 162 | 8612  | S     | HY          | 1512e6966074        |
| 76  | 9   | 19    | M     | HI          | e16410b887ee        |
| 77  | 15  | 61    | M     | HI          | 7c38d54f9e52        |
| 78  | 21  | 127   | M     | HI          | b8dcff224cdb        |
| 79  | 27  | 217   | M     | HI          | 6670a2b14284        |
| 80  | 33  | 331   | M     | HI          | 25fd8f0c3311        |
| 81  | 48  | 715   | M     | HI          | ba6f5baae76d        |
| 82  | 99  | 3169  | M     | HI, KR      | 4fef38d87d0c        |
| 83  | 195 | 12481 | M     | HI, KR      | cd3497dbf846        |
| 84  | 195 | 12480 | S     | HI          | 1b9230fb738a        |
| 85  | 44  | 602   | S     | HI          | 0b0250bf6213        |
| 86  | 99  | 3168  | S     | HI          | a0ea30825a73        |
| 87  | 39  | 470   | S     | SC          | 30caa6f7f894        |
| 88  | 47  | 691   | S     | SC          | fac44af18c55        |
| 89  | 48  | 721   | S     | SC          | 2fb5ef0df4cf        |
| 90  | 51  | 818   | S     | SC          | c18e1baf25f5        |
| 91  | 53  | 885   | S     | SC          | 6bc5b14d0b68        |
| 92  | 54  | 919   | S     | SC          | a254c1b4f05b        |
| 93  | 55  | 955   | S     | SC          | 202625a39085        |
| 94  | 59  | 1103  | S     | SC          | ba482375103b        |
| 95  | 60  | 1141  | S     | SC          | 9bc5096cc3fd        |
| 96  | 65  | 1365  | S     | SC          | b1dcbeb4ff39        |
| 97  | 66  | 1397  | S     | SC          | 327ecbb9b7ba        |
| 98  | 99  | 3170  | S     | SC          | ea507e2de137        |
| 99  | 129 | 5461  | S     | SC          | 6dae0012aa49        |
| 100 | 130 | 5525  | S     | SC          | e0559b54aee5        |
| 101 | 195 | 12482 | S     | SC          | 5413dd7b60ee        |
| 102 | 14  | 54    | M     | MA          | d47aea63afc1        |
| 103 | 14  | 54    | M     | MA          | 18bbcff4f433        |
| 104 | 14  | 54    | M     | MA          | 78f750c071a5        |
| 105 | 14  | 54    | M     | MA          | 912da8ec9bc7        |
| 106 | 14  | 54    | M     | MA          | 9134999072a9        |
| 107 | 14  | 54    | M     | MA          | 9c871e103a46        |
| 108 | 14  | 54    | M     | MA          | b3dcae0bc1d8        |
| 109 | 14  | 54    | M     | MA          | 9219cdaa6f90        |
| 110 | 14  | 54    | M     | MA          | d2babc1b9431        |

*Continued from previous page*

| ID  | $n$ | $T$ | Class | Collections | Coordinate identity |
|-----|-----|-----|-------|-------------|---------------------|
| 111 | 14  | 54  | M     | MA          | 2b34ca22dc89        |
| 112 | 14  | 54  | M     | MA          | fa427b3d27ef        |
| 113 | 14  | 54  | M     | MA          | 1ec1fd5cb2ee        |
| 114 | 14  | 54  | M     | MA          | 3e5d182f8408        |
| 115 | 14  | 54  | M     | MA          | 17f8c0524eb1        |
| 116 | 14  | 54  | M     | MA          | c472f97237f1        |
| 117 | 14  | 52  | S     | MA          | 2eb83dc242a1        |
| 118 | 14  | 53  | S     | MA          | 5e6068f5003e        |
| 119 | 14  | 52  | S     | MA          | b3f4749f6dee        |
| 120 | 14  | 52  | S     | MA          | c6615430ae3d        |
| 121 | 8   | 14  | S     | PU          | ebff58687458        |
| 122 | 14  | 51  | S     | PU          | d67e7a6b3b08        |
| 123 | 20  | 114 | S     | PU          | bc312fc85ea5        |
| 124 | 26  | 203 | S     | PU          | 5452b4ffa1a9        |
| 125 | 32  | 313 | S     | PU          | 018f9fc6b73f        |
| 126 | 38  | 448 | S     | PU          | 7d59f504aba8        |
| 127 | 50  | 791 | S     | PU          | 6bde86a3dcdb        |
| 128 | 8   | 15  | M     | PU          | a138081033c3        |
| 129 | 14  | 54  | M     | PU          | 105765145381        |
| 130 | 20  | 117 | M     | PU          | 382171ac62cc        |
| 131 | 26  | 204 | M     | PU          | f71379c4db03        |
| 132 | 32  | 315 | M     | PU          | 5914f2bb88d1        |
| 133 | 36  | 402 | S     | PU          | ce43a1077597        |
| 134 | 38  | 450 | M     | PU          | 8ba2bdb14e85        |
| 135 | 42  | 553 | S     | PU          | 9849e255d5da        |
| 136 | 50  | 792 | M     | PU          | 3d40cb582616        |
| 137 | 19  | 107 | S     | OT          | 6e8ca6f56a85        |
| 138 | 6   | 6   | M     | OT          | af8f599eb8a1        |

## C Historical inventory of the author's outputs

This table preserves the earlier research inventory, including bounds later superseded by  $G$ , by a stronger retained certificate, or by known literature constructions. Its original labels describe how a number entered that inventory. They do not retroactively establish a missing general geometric argument. The active result aliases independently verify the retained numerical inequalities. Published inputs and older matching results keep their original attribution.

This is a historical numerical inventory, not a list of original discoveries. B means an elementary base case; C means an explicit coordinate construction in that inventory; E means the then-evaluated defect-extension consequence (called G in the historical file, renamed here to avoid confusion with the present function  $G$ ). The numerical inequalities are now retained independently by the compiled `Results.earlier_NNN` aliases. This does not formalize all premises of the former general defect argument or establish priority. The right columns display current, separately proved bounds.

**Table 14:** Earlier work inventory compared with the present unconditional formula and saved certificate envelope.

| $n$ | Earlier output | Historical basis | Current $G(n)$ | Best saved classical |
|-----|----------------|------------------|----------------|----------------------|
| 3   | 1              | B                | 1              | 1                    |
| 4   | 2              | E                | 2              | 2                    |



*Continued from previous page*

| $n$ | Earlier output | Historical basis | Current $G(n)$ | Best saved classical |
|-----|----------------|------------------|----------------|----------------------|
| 5   | 5              | B/C              | 5              | 5                    |
| 6   | 7              | C                | 7              | 7                    |
| 7   | 10             | C                | 11             | 11                   |
| 8   | 14             | E                | 14             | 15                   |
| 9   | 19             | C                | 20             | 21                   |
| 10  | 25             | E                | 24             | 25                   |
| 11  | 28             | E                | 31             | 32                   |
| 12  | 36             | E                | 37             | 38                   |
| 13  | 39             | E                | 45             | 47                   |
| 14  | 53             | E                | 52             | 54                   |
| 15  | 61             | C                | 62             | 65                   |
| 16  | 72             | E                | 70             | 72                   |
| 17  | 76             | E                | 81             | 85                   |
| 18  | 93             | E                | 91             | 93                   |
| 19  | 98             | E                | 103            | 107                  |
| 20  | 116            | C                | 114            | 117                  |
| 21  | 127            | C                | 128            | 133                  |
| 22  | 143            | E                | 140            | 143                  |
| 23  | 149            | E                | 155            | 161                  |
| 24  | 172            | E                | 169            | 172                  |
| 25  | 178            | E                | 185            | 191                  |
| 26  | 203            | E                | 200            | 204                  |
| 27  | 217            | C                | 218            | 225                  |
| 28  | 238            | E                | 234            | 238                  |
| 29  | 245            | E                | 253            | 261                  |
| 30  | 275            | E                | 271            | 275                  |
| 31  | 283            | E                | 291            | 299                  |
| 32  | 314            | E                | 310            | 315                  |
| 33  | 331            | C                | 332            | 341                  |
| 34  | 357            | E                | 352            | 357                  |
| 35  | 366            | E                | 375            | 385                  |
| 36  | 402            | E                | 397            | 402                  |
| 37  | 411            | E                | 421            | 431                  |
| 38  | 449            | E                | 444            | 450                  |
| 39  | 469            | C                | 470            | 470                  |
| 40  | 470            | E                | 494            | 494                  |
| 41  | 496            | E                | 521            | 533                  |
| 42  | 553            | E                | 547            | 553                  |
| 43  | 564            | E                | 575            | 587                  |
| 44  | 608            | C                | 602            | 608                  |
| 45  | 618            | E                | 632            | 645                  |
| 46  | 667            | E                | 660            | 667                  |
| 47  | 679            | E                | 691            | 691                  |
| 48  | 715            | C                | 721            | 721                  |
| 49  | 722            | E                | 753            | 767                  |
| 50  | 791            | E                | 784            | 792                  |
| 51  | 817            | C                | 818            | 818                  |
| 52  | 818            | E                | 850            | 850                  |
| 53  | 852            | E                | 885            | 885                  |
| 54  | 886            | E                | 919            | 919                  |
| 55  | 920            | E                | 955            | 955                  |
| 56  | 956            | E                | 990            | 990                  |
| 57  | 992            | E                | 1028           | 1045                 |
| 58  | 1073           | E                | 1064           | 1073                 |

*Continued from previous page*

| $n$ | Earlier output | Historical basis | Current $G(n)$ | Best saved classical |
|-----|----------------|------------------|----------------|----------------------|
| 59  | 1088           | E                | 1103           | 1103                 |
| 60  | 1104           | E                | 1141           | 1141                 |
| 99  | 3169           | C                | 3170           | 3170                 |
| 195 | 12481          | C                | 12482          | 12482                |

## D Evidence navigation and excluded drafts

The following stable repository paths locate the complete evidence. This appendix is a navigation aid, not an additional source of axioms.

| Path or collection                                                                     | Contents and role                                                                                                                                                                       |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>archive/2026-03-original/</code>                                                 | Original March sources, preserved as historical evidence; incomplete and excluded from the active build.                                                                                |
| <code>Kobon/</code>                                                                    | Active Lean geometry, counts, transformations, certificates and formal proof audit.                                                                                                     |
| <code>research/kobon-extension/</code>                                                 | Both-parity extension manuscript, boundary-defect analysis and examples.                                                                                                                |
| <code>research/successor-formalization/</code>                                         | Recurrence status, 49-seed numerical target, and exterior-resource obstruction.                                                                                                         |
| <code>research/kobon-hybrid/</code>                                                    | Compatible eleven-line seed, interval argument and sparse family manuscript.                                                                                                            |
| <code>research/kobon-research/</code>                                                  | Additional-line family draft and finite witnesses.                                                                                                                                      |
| <code>research/kobon-own-results/</code>                                               | Historical inventory, candidate comparisons and priority audit.                                                                                                                         |
| <code>research/all-n-formalization/</code>                                             | Earlier and current all-order formulas, finite enhancement table and interpretation.                                                                                                    |
| <code>research/six-hour-2026-09-21/</code>                                             | Three-hour research release, source audits, local upper arguments, family components and cumulative review. The directory keeps its original allocation name.                           |
| <code>experiments/2026-09-20/ and<br/>experiments/2026-09-21/<br/>verification/</code> | Frozen search scripts, parameters, exact outputs, failed approaches and interrupted-run status. Build logs, theorem audit, coordinate recounts, source hashes and the interval summary. |
| <code>evidence/sources.json</code>                                                     | Bibliographic roles, versions, immutable commits and provenance.                                                                                                                        |
| <code>paper/</code>                                                                    | Preserved September comprehensive and journal manuscripts, with their corrected rendering and frozen proof manifests.                                                                   |
| <code>research/three-hour-2026-10-02/</code>                                           | Closed odd/even recursion, exact obstruction corpus, actual upper extraction, new searches, seed drafts and continuation plan.                                                          |
| <code>manuscripts/2026-10-03/</code>                                                   | This revised comprehensive edition, journal companion, updated vector figures, shared proof map and reproduction instructions.                                                          |

The unfinished full real-parameter 33- and 49-line seed checks are separately labelled and excluded from the active library. The large 321-, 322-, and 641-line coordinate files remain outside the 138 Lean-promoted finite identities; the interrupted direct coordinate recount at 642 lines is still incomplete. Existence at all four orders now follows independently from the general verified family. None of those coordinate files or seed drafts is an assumption of the all-order theorems.

## E Release deltas and reproducible research boundaries

**Table 16:** Changes from the preserved September mathematical release.

| Topic                                      | Previous status                                              | Current status                                                                                                                 |
|--------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| All-order affine $G$<br>Finite coordinates | Complete Lean proof.<br>138 identities, 104 simple.          | Unchanged and retained.<br>Same catalog and attributions;<br>no new numerical record added.                                    |
| BBL output invariant                       | One-step geometry;<br>recursive integration open.            | Closed grid, seed and<br>visible-boundary records; all<br>finite depths proved.                                                |
| Eleven-derived families                    | Finite checkpoints and<br>published-theorem<br>consequences. | Unconditional actual odd/even<br>Lean existence.                                                                               |
| All-order envelope                         | $L = \max(G, C)$ .                                           | $H = \max(L, R)$ ; strict<br>repository gains from every<br>321/322 family pair onward.                                        |
| Simple upper comparisons                   | Polynomial benchmarks.                                       | Actual general simple upper<br>proofs and one-triangle family<br>windows.                                                      |
| Global upper extraction                    | Local fans and conditional<br>arithmetic.                    | Actual edge inventory, core<br>incidence and clean-line<br>charging complete; cyclic fan<br>extraction still open.             |
| Perfect 21-grid transfer                   | Bounded failed searches.                                     | One explicit Lean obstruction;<br>complete exact corpus for<br>supplied affine classes, with<br>external classification input. |
| 49 compatible seed                         | Research target.                                             | Uniform exact external<br>certificate; tangent enclosures in<br>Lean; full base/family drafts<br>pending.                      |
| Active Lean audit                          | 220 targets, 325 sources,<br>4,227 theorems.                 | 270 targets, 375 sources, 5,241<br>theorems; 4,405 standard-axiom<br>and 836 native-dependent<br>declarations.                 |
| March universal extension                  | Incomplete archived<br>argument.                             | Still unproved; finite numerical<br>consequences independently<br>supported.                                                   |

The [verification record](#) and [proof index](#) distinguish five exact replays and three isolated Lean probes from the unfinished full 49-line seed check.